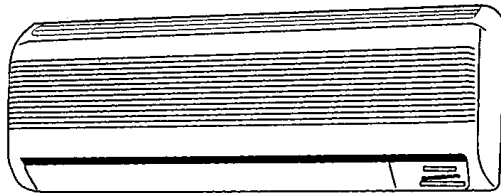
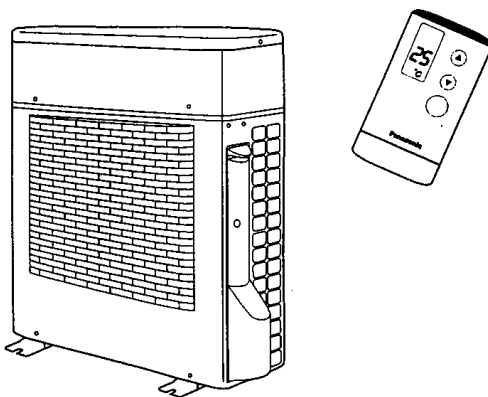


Service Manual

Room Air Conditioners



CS-VG120KE
CU-VG120KE



WARNING

This service information is designed for experienced repair technicians only and is not designed for use by the general public. It does not contain warnings or cautions to advise non-technical individuals of potential dangers in attempting to service a product. Products powered by electricity should be serviced or repaired only by experienced professional technicians. Any attempt to service or repair the product or products dealt with in this service information by anyone else could result in serious injury or death.

IMPORTANT SAFETY NOTICE

There are special components used in this equipment which are important for safety. These parts are marked by  in the Schematic Diagrams, Circuit Board Diagrams, Exploded Views and Replacement Parts List. It is essential that these critical parts should be replaced with manufacturer's specified parts to prevent shock, fire or other hazards. Do not modify the original design without permission of manufacturer.

Panasonic

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1 Features

• Product

- Powerful Mode for quick cool/heat
- Compressor operating frequency control to maintain desired room temperature
- Automatic Restart after power failure
- Battery weak indication mark at remote control
- Washable front panel
- Power Monitor Display

• Serviceability

- Self diagnosis
- Test Run at both Cooling and Heating rated frequency

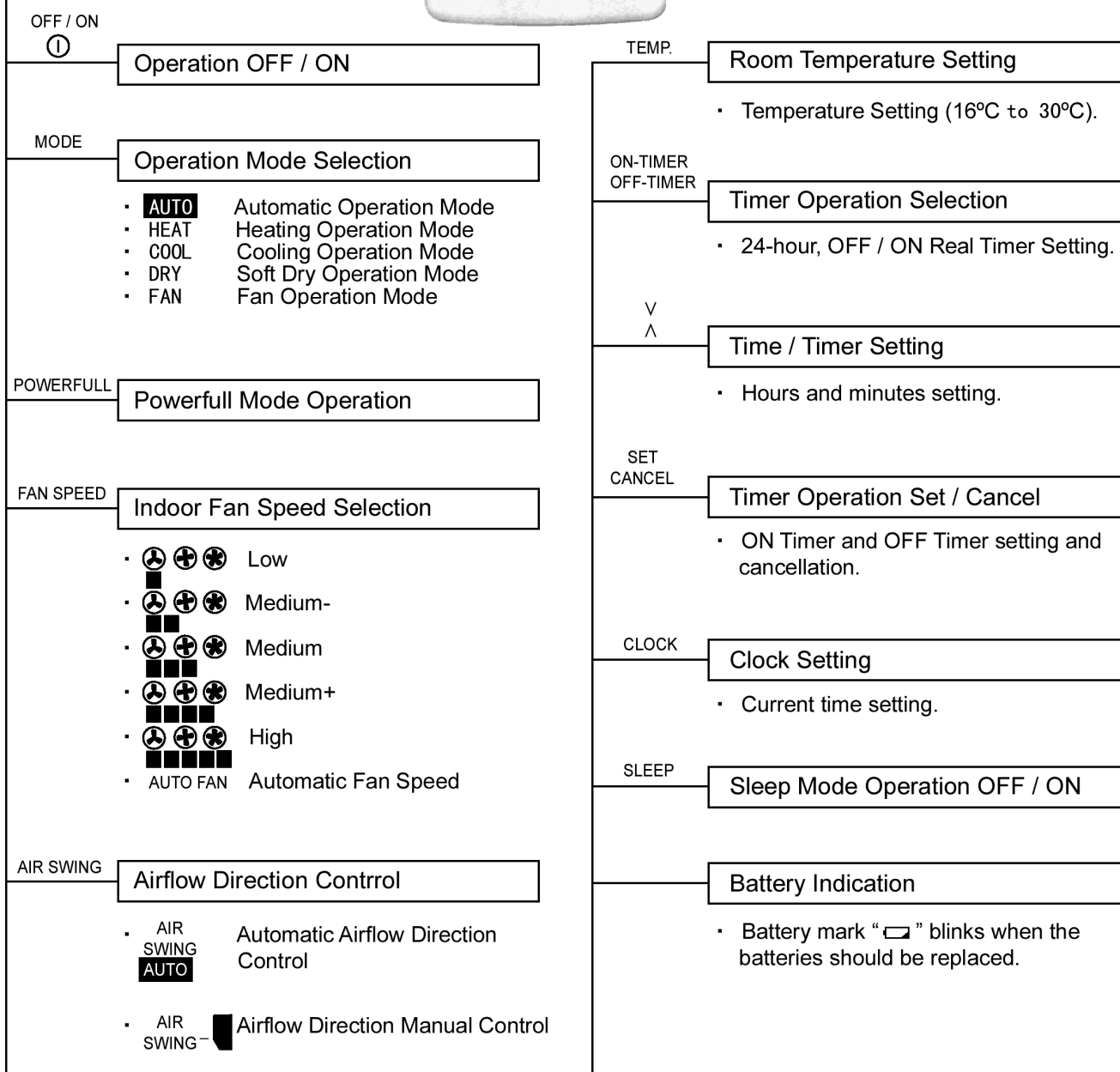
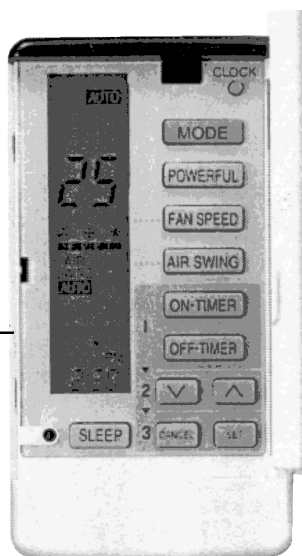
• Quality Improvement

- High power supply voltage protection
- Low power supply voltage protection

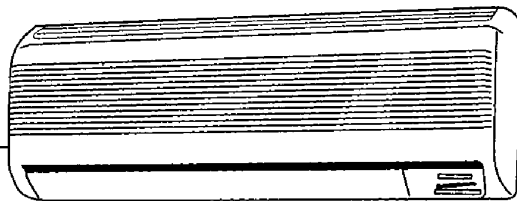
- Gas leakage detection

2 Functions

2.1. REMOTE CONTROL



2.2. INDOOR UNIT



FOR ALL OPERATIONS

POWER ①

Power Switch OFF / ON ①

AUTO
OFF / ON

Automatic Operation Switch

- Used when the remote control cannot be used.

Remote Control Signal Recieving
Sound Control OFF / ON ※

- It can be controlled by pressing Automatic Operation Switch for 10 seconds.

TEST RUN
OFF / ON

Test Run Operation

- Used when test running or servicing.
- Compressor operation: rated frequency
Cooling: 73Hz
Heating: 92Hz

Operation Indication Lamps

- POWER (Red) - Lights up in operation, blinks in Automatic Operation judging and Hot Start operation.
- TIMER (Orange) - Lights up in Timer Setting. Blinks in Self Diagnosis Control.
- SLEEP (Orange) - Lights up in Sleep Mode Operation.

Power Monitor Display ※

- Lights up during compressor operation.

Operation Mode

- Automatic, Heating, Cooling, Soft Dry and Fan Operation.

Automatic Restart Control ※

- Operation is restarted after power failure at previous setting mode.

Sleep Operation Mode ※

Timer Operation ※

Powerful Mode ※

- For quick cooling or heating

Indoor Fan Speed Control ※

Airflow Direction Control ※

COOLING/SOFT DRY OPERATION

Room Temperature Control ※

Operation Mode	Compressor Operation Frequency
Cooling	9~86 Hz
Soft Dry	9~29 Hz
Heating	9~124 Hz

Temperature Shift ※

Self Diagnosis ※

Low Pressure Control ※
(Gas Leakage Detection)High Power Supply Voltage
Protection ※

Indoor Power Relay Control ※

Deodorizing Control ※

Sensible Heat Control ※

Anti-Fog Discharge Control ※

Anti-Dew Formation Control ※

Anti-Freezing Control ※

Quiet Operation ※

HEATING OPERATION

Anti-Cold Draft Control ※

Hot Start ※

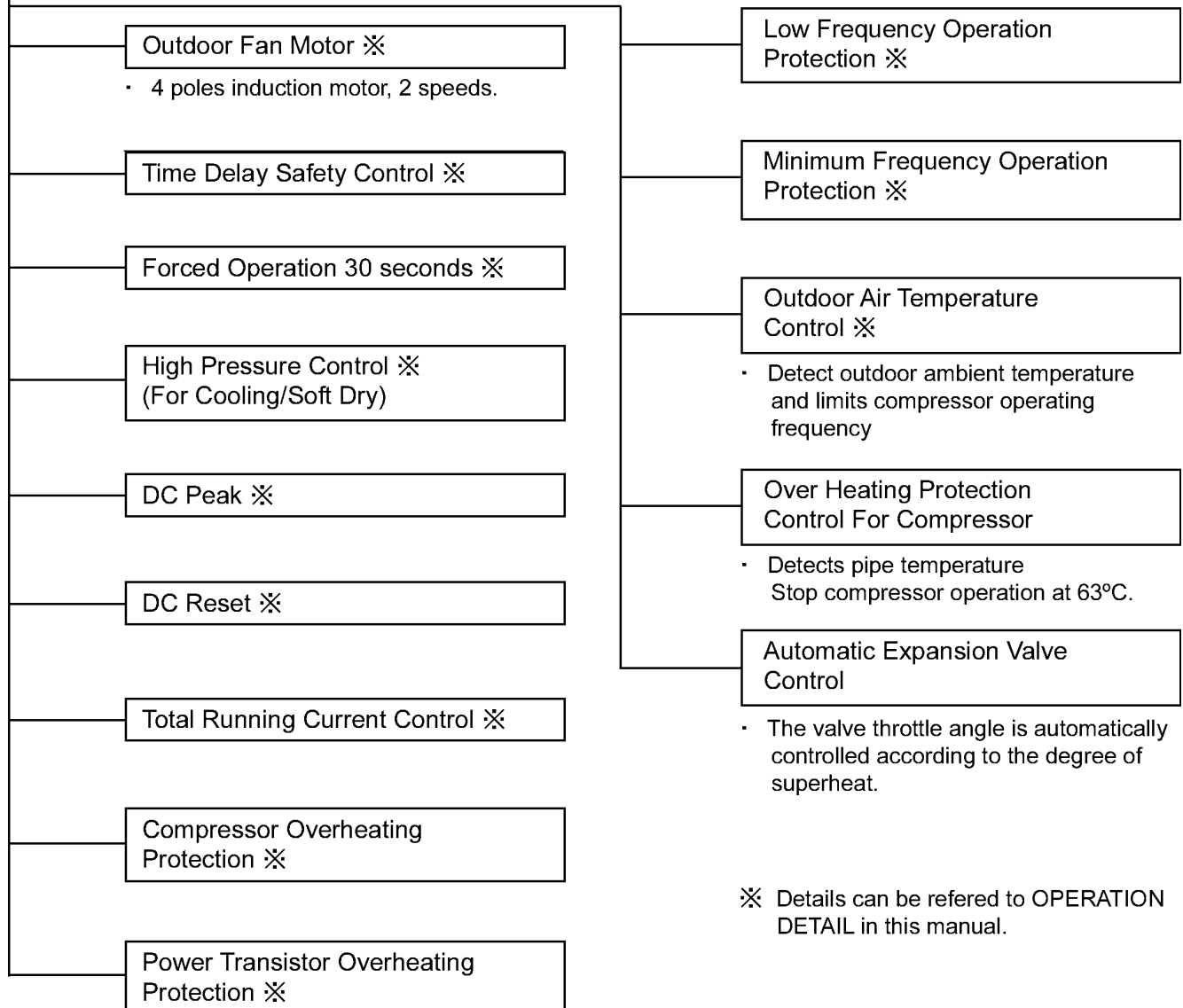
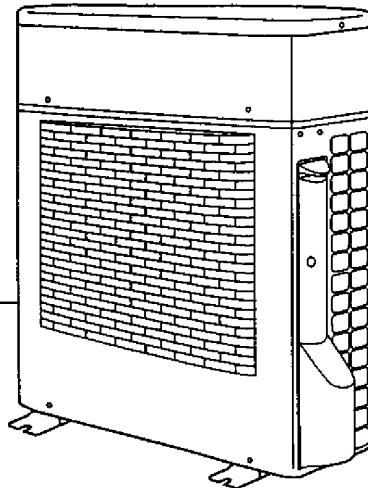
Intake Air Temperature Control ※

High Pressure Control ※

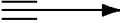
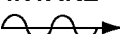
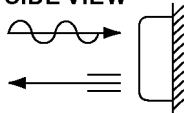
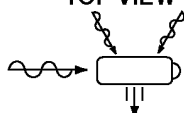
AUTOMATIC OPERATION

FAN OPERATION

2.3. OUTDOOR UNIT



3 Product Specifications

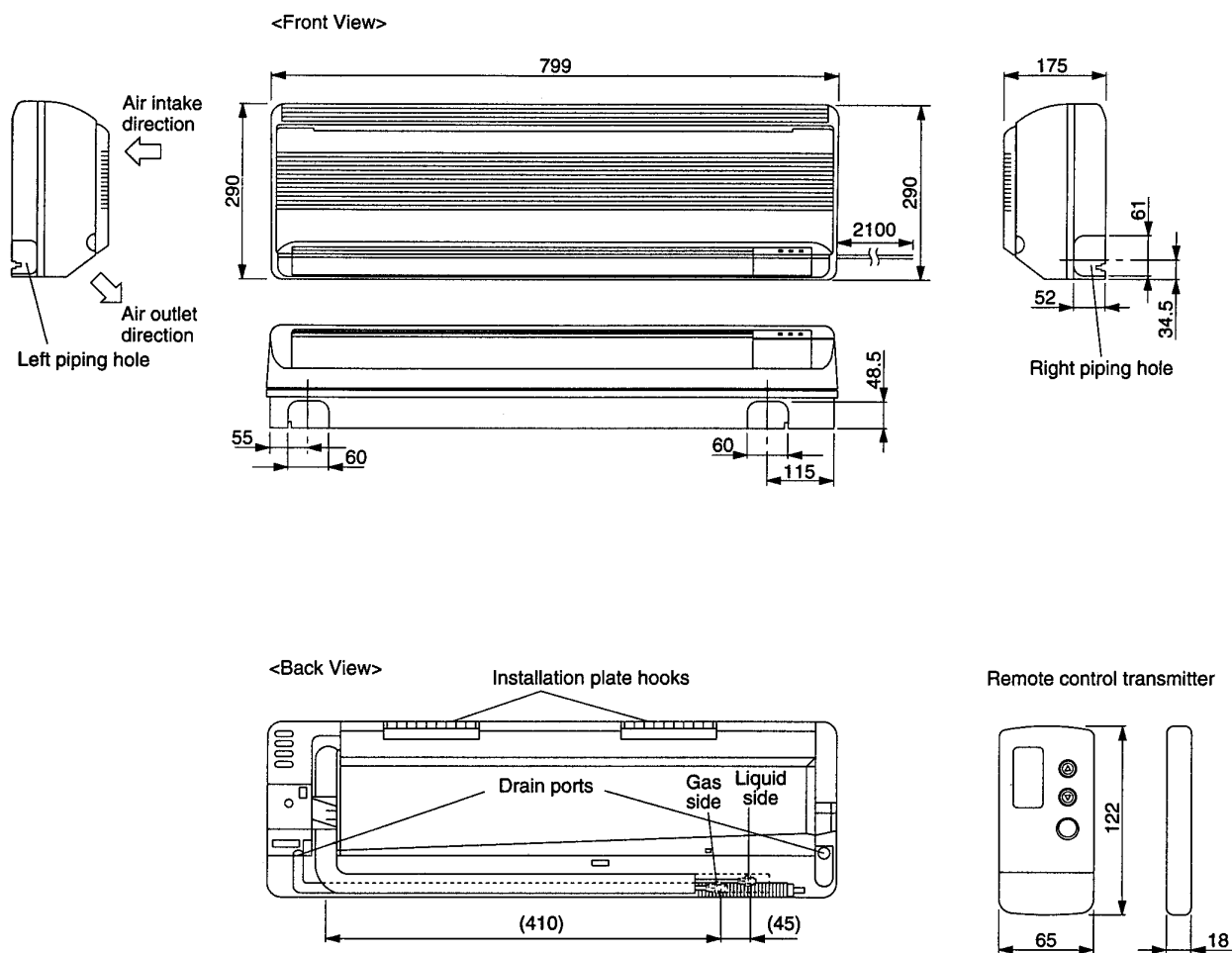
		Unit	CS-VG120KE	CU-VG120KE
Cooling Capacity		kW Btu/h	3.45(0.90 ~ 3.85) 11,800 (3,100 ~ 13,100)	
Heating Capacity		kW Btu/h	4.8 (0.78 ~ 6.30) 14,300 (2,700 ~ 20,500)	
Moisture Removal		ℓ /h Pint/h	2.0 4.2	
Power Source		Phase V Cycle	Single 230 50	
Airflow Method		OUTLET  INTAKE 	SIDE VIEW 	TOP VIEW 
Air Volume	Indoor Air(Lo)	m³ / min(cfm)	Cooling ; 7.4(261) Heating ; 7.6(268)	-
	Indoor Air(Me)	m³ / min(cfm)	Cooling ; 8.1(286) Heating ; 8.3(293)	-
	Indoor Air(Hi)	m³ / min(cfm)	Cooling ; 9.3(328) Heating ; 9.8(346)	-
	Outdoor Air	m³ / min(cfm)	-	Hi; 24.5(865) Low; 21.5(759)
Noise Level		dB(A)	Cooling ; High 41,Low 37 heating ; High 41,Low 37	Cooling ; 48 Heating ; 51
Electrical Data	Input	kW	Cooling ; 1.91(0.27 ~ 1.46) Heating ; 1.6(0.25 ~ 2.00)	
	Running Current	A	Cooling ; 5.4 (max.6.9) Heating ; 7.0 (max.9.0)	
	COP		Cooling ; 2.9 Heating ; 3.0	
	Starting Current	A	5.9	
Piping Connection Port (Flare piping)		inch inch	G ; Half Union 1/2" L ; Half Union 1/4"	G ; 3-Way valve 1/2" L ; 2-Way valve 1/4"
Pipe Size (Flare piping)		inch inch	G(gas side) ; 1/2" L(liquid side) ; 1/4"	G(gas side) ; 1/2" L(liquid side) ; 1/4"
Drain Hose	Inner diameter	mm	12	-
	Length	m	0.7	-
Power Cord length			2.1 m	-
Number of core-wire			3 core wires x 1.5mm²	-
Dimensions	Height	inch(mm)	11 - 7/16(290)	25 - 25/32(655)
	Width	inch(mm)	31 - 15/32(799)	21 - 1/4(540)
	Depth	inch(mm)	6 - 29/32(175)	9 - 23/32(247)
Net Weight		lb(kg)	18(8.0)	90(41)
Compressor	Type		-	Scroll (1 cylinder)
	Motor Type		-	Brushless(4-poles)
	Rated Output	W	-	650
Air Circulation	Type		Cross-flow Fan	Propeller Fan
	Material		AS+Glass Fiber 30 %	AES+Glass Fiber 12 %
	Motor Type		Transiator(4-poles)	Induction(4-poles)
	Input	W	-	71
	rated Output	W	20	30
	Fan Speed	Low	rpm 1,200	1,025
		Medium	rpm 1,310	-
		High(heating)	rpm 1,500(1,550)	1,165
Heat Exchange	description		Evaporator	Condenser
	Tube material		Copper	Copper
	Fine material		Aluminium	Aluminium
	Fine type		Slit Fin	Corrugated Fin
	Row/Stage		(Plate fin configuration,forced draft) 2/12 2/24	
	FPI		21	17
	Size(W x H x L)		mm 600 x 252 x 25.4	673 x 504 x 18.2 615 x 504 x 18.2
Refrigerant Control Device			-	Expansion Valve
refrigeration Oil		(c.c)	-	RB88A(290)
Refrigerant(R410A)		g(oz)	-	850(30)
Thermostat			Electronic Control	-
Protection Device				Electronic Control

		Unit	CS-VG120KE	CU-VG120KE
Air Filter	Material style		P.P Honeycomb	-
Capacity Control			Expansion valve	
Fan motor Capacitor		μF,VAC	-	1.8μF,400VAC

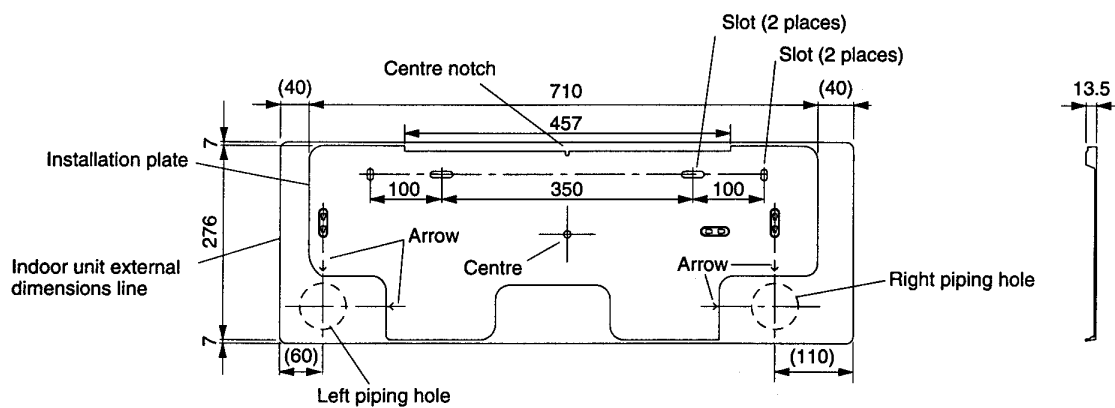
- Specifications are subject to change without notice for further improvement.

4 Dimensions

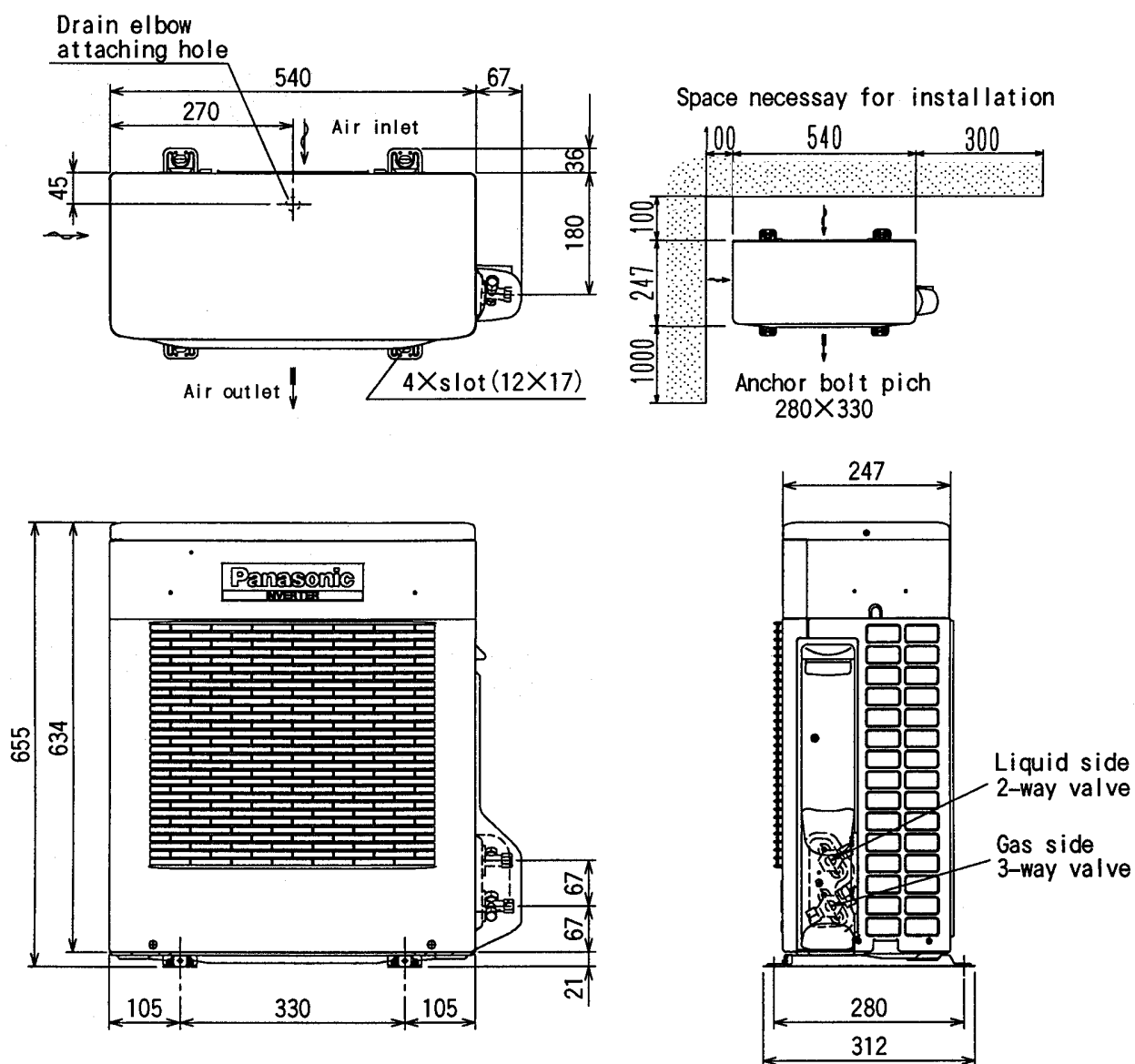
4.1. CS-VG120KE (INDOOR UNIT)



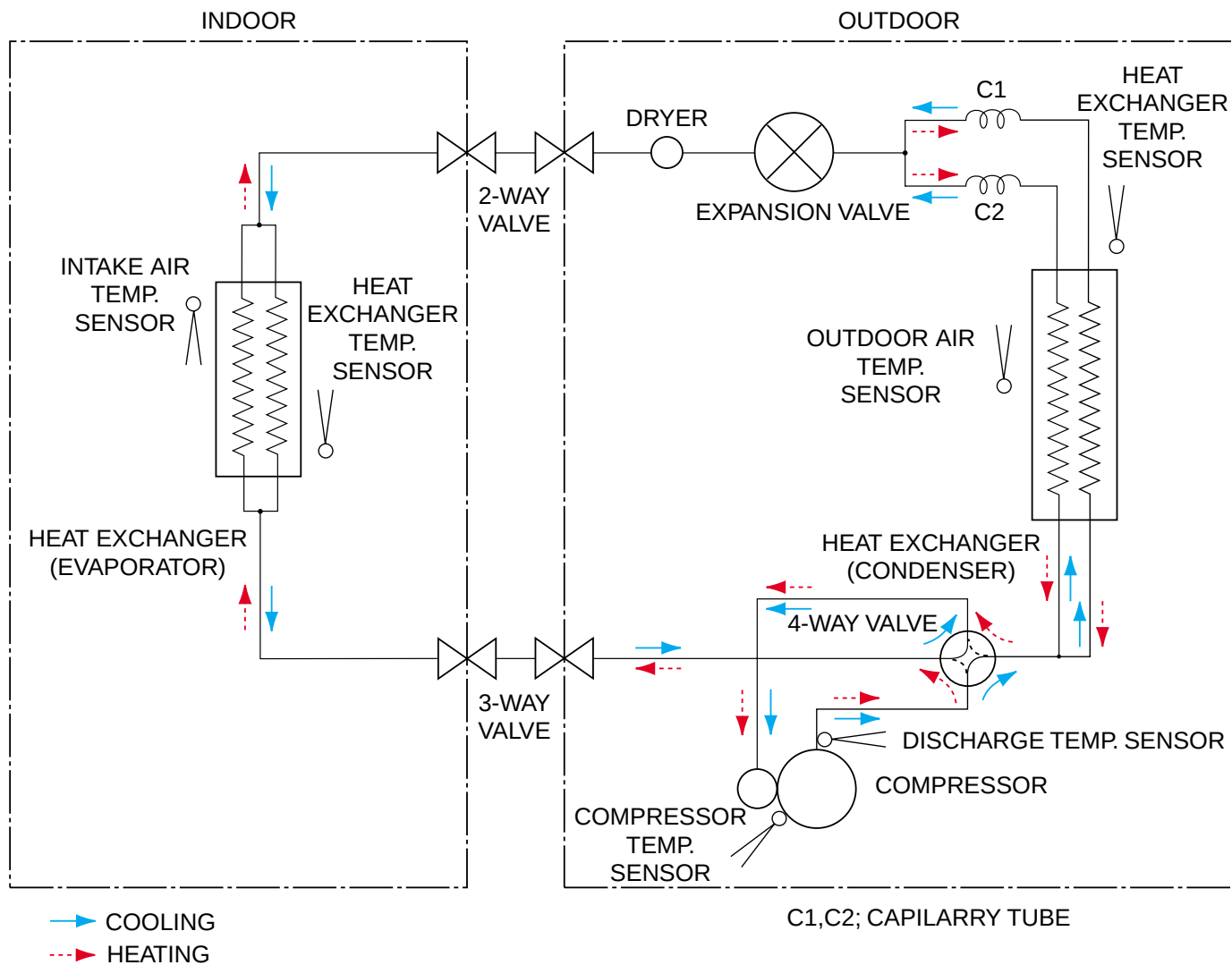
Relative position between the indoor unit and the installation plate <Front View>



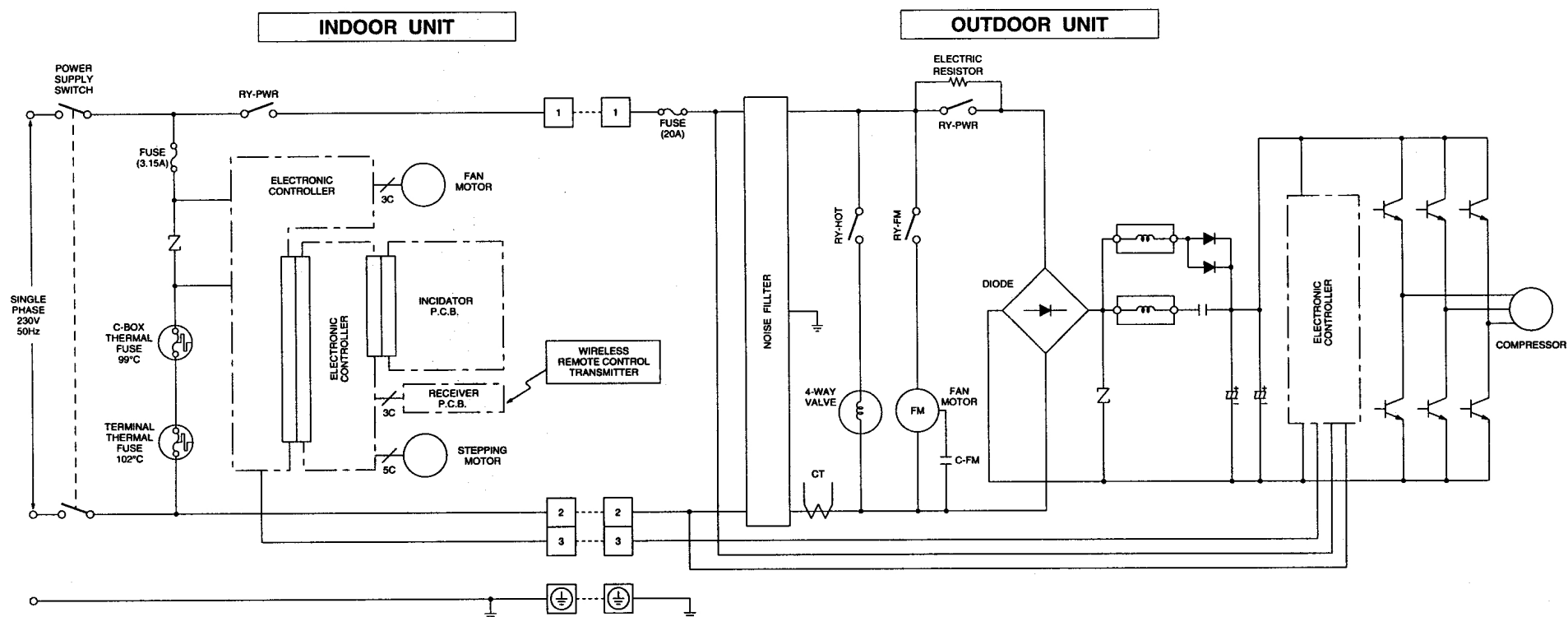
4.2. CU-VG120KE (OUTDOOR UNIT)



5 Refrigeration Cycle Diagram



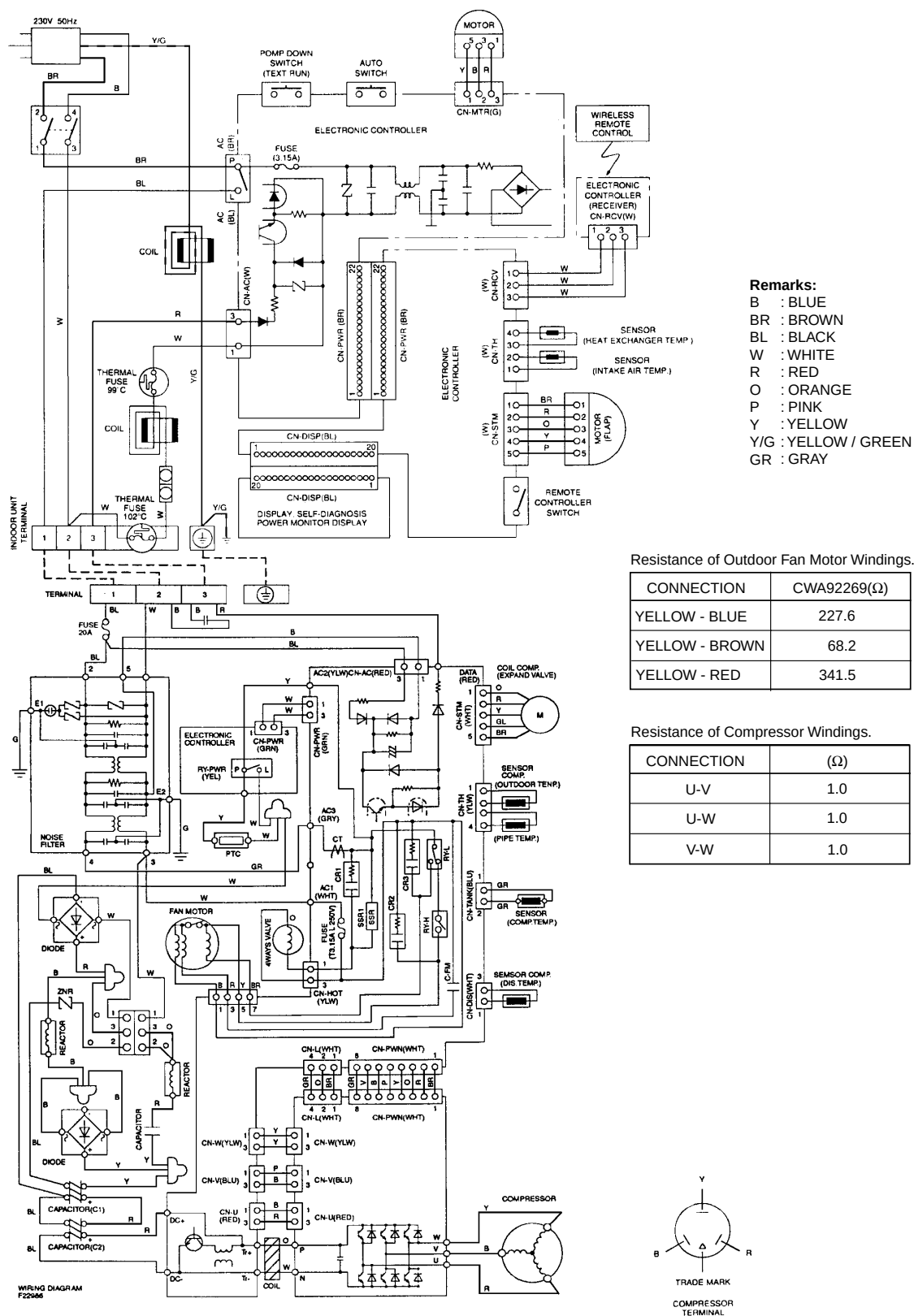
6 Block Diagram



✕ Indicates the electronic control unit.

✕ "C" Indicates the number of core wires. (Example: 5C=5 core wires).

7 Wiring Diagram



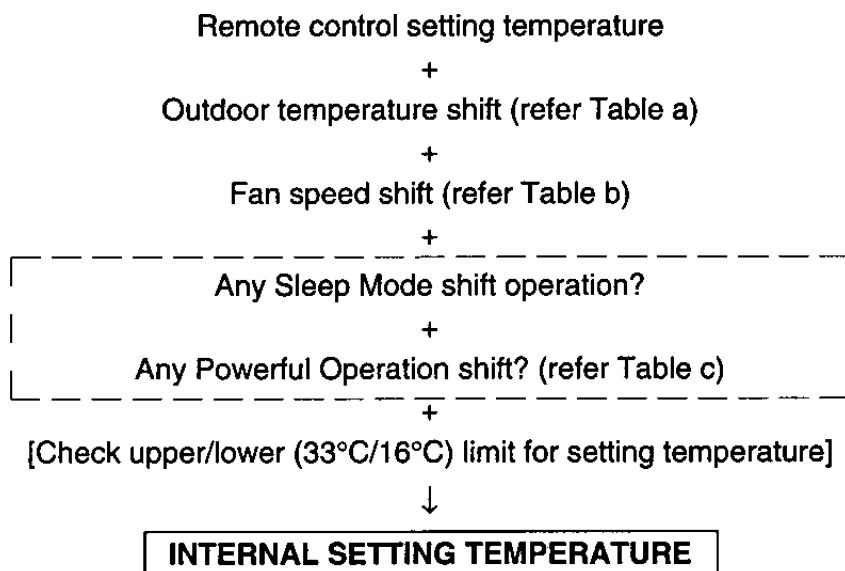
8 Operation Details (FUNCTIONS)

8.1. TEMPERATURE SHIFT

Once the operation starts, the remote control setting temperature will be shifted internally based on the setting fan speed and outdoor air temperature. In addition, if Sleep Mode or Powerful Mode are set, the temperature shift will be carried out.

Setting of internal Setting Temperature

The internal setting temperature can be decided as follows:



8.1.1. Table a

Setting Temperature Shift based on outdoor air temperature.

1. Cooling, Soft Dry

	Shift amount
38°C ≤ Outdoor air temperature	−0.5°C
30°C ≤ Outdoor air temperature < 38°C	0.0°C
23°C ≤ Outdoor air temperature < 30°C	+0.5°C
Outdoor air temperature < 23°C	+1.0°C

2. Heating

	Shift amount
21°C ≤ Outdoor air temperature	−1.5°C
17°C ≤ Outdoor air temperature < 21°C	−1.5°C
13°C ≤ Outdoor air temperature < 17°C	−1.0°C
9°C ≤ Outdoor air temperature < 13°C	−1.0°C
5°C ≤ Outdoor air temperature < 9°C	0.0°C
1°C ≤ Outdoor air temperature < 5°C	+1.0°C
−3°C ≤ Outdoor air temperature < 1°C	+1.0°C
Outdoor air temperature < −3°C	+1.5°C

8.1.2. Table b

Setting Temperature Shift based on fan speed.

Remote control setting fan speed	Cooling	Dry	Heating
Lo	+2.0°C	+2.5°C	+1.0°C
Me-, Me, Me+, Auto fan speed	+2.0°C		+1.0°C
Hi	+2.0°C		+0.5°C

8.1.3. Table c

Powerful Mode Shift

	Cooling	Dry	Heating
Powerful	−4.0°C	−3.0°C	+6.0°C

8.2. COOLING OPERATION

8.2.1. Room Temperature Control

1. When the remote control setting temperature is less than 24°C.

Cooling		Compressor Operation Frequency (Hz)	
Intake air temp. – Internal setting temp.(°C)	+1.5	73 (Fc) or 86 (Fc max)	73 (Fc) or 86 (Fc max)
		55	70
	+1.0		
		45	67
	+0.5		
Internal setting temp.	0	37	62
		29	37
	-0.5		
Compressor OFF temp.	-1.0	17	17
		Comp OFF	Comp OFF
Outdoor Air Temperature		Less than 38°C	38°C and above

2. When the remote control setting temperature is 24°C and above.

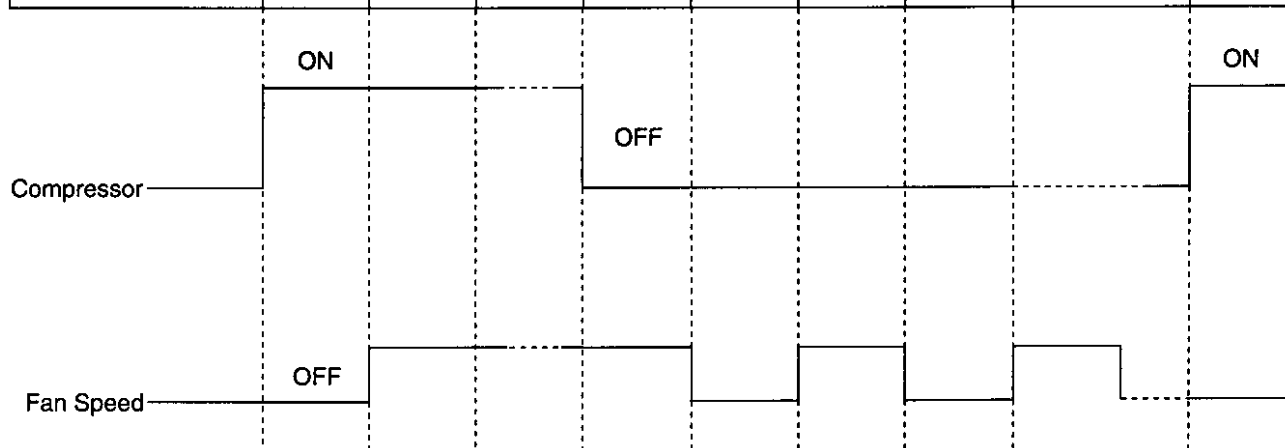
Cooling	Compressor Operation Frequency (Hz)		
Intake air temp. – Internal setting temp.(°C)	+1.5	73 (Fc) or 86 (Fc max)	73 (Fc) or 86 (Fc max)
	+1.0	55	55
	+0.5	45	45
Internal setting temp.	0	37	37
	–0.5	29	29
Compressor OFF temp.	–1.0	17	17
		Comp OFF	Comp OFF
Outdoor Air Temperature		Less than 38°C	38°C and above

- Compressor OFF temperature = Compressor ON temperature.
- The operation frequency can be changed every 30 seconds.
- 30 minutes from the start of the operation, the compressor is operating at Fc max Hz.
- The compressor stops when the intake air temperature reaches 1°C below internal setting temperature and continues for 3 minutes.
- When the compressor stops, it will not begin operation for 3 minutes. (Time Delay Safety Control)
- When the intake air temperature reaches the Compressor ON temperature, the Compressor starts operation.
- When the compressor stops, the outdoor fan motor stops 30 seconds later.

8.2.2. Deodorizing Control

- This control is available during automatic fan speed for Cooling and Soft Dry Operation. It is not available during anti-freezing control.

Deodorizing Status	1	2	3	4	5	6	7	6, 7, 6, 7,...	1
Compressor	ON			OFF					ON
Time (second)	40	50		30	60	30	60		40
Fan Speed	OFF	SLo	Auto	SLo	OFF	SLo	OFF		OFF



- When the compressor is in operation, the deodorizing status starts from 1 → 2 → 3.
- When the compressor stops operation, the deodorizing status starts from 4 → 5 → 6 → 7.
- If the compressor still stops operation after 3 minutes, the deodorizing status will start from 6.

8.2.3. Sensible Heat Control

- This control is to improve the feeling in high fan speed during low operation frequency. When the operation frequency is less than 37Hz, the fan speed will reduce. When the operation frequency is above 38Hz continuously for 5 minutes, the fan speed will resume to normal condition.

8.3. SOFT DRY OPERATION

8.3.1. Room Temperature Control

At the start of operation, cooling operation is running until the intake air temperature is 0.5°C higher than internal setting temperature, then the operation will shift to Soft Dry with indoor fan speed SLo.

Soft Dry		Compressor Operation Frequency (Hz)	
Intake air temp. – Internal setting temp.(°C)	+1.5	Cooling Operation	
	+1.0	29	
	+0.5	29	
	0	29	
	-0.5	29	
Internal setting temp.	-1.0	21	
Compressor OFF temp.	-1.0	Comp OFF	
Condition		When the temperature falls	When the temperature rises

- Compressor OFF temperature = Compressor ON temperature.
- The operation frequency changes every 30 seconds.
- 30 minutes from the start of the operation, the compressor is operating at Fc max Hz.
- The Compressor stops when the intake air temperature reaches 1°C below internal setting temperature and continues for 3 minutes.
- When the Compressor stops, it will not begin operation for 3 minutes. (Time Delay Safety Control)
- When the intake air temperature reaches the Compressor ON temperature, the Compressor starts operation immediately.
- When the Compressor stops, the outdoor fan motor stops 30 seconds later.

8.3.2. Deodorizing Control

- This control is available during automatic fan speed for Cooling and Soft Dry Operation. It is not available during anti-freezing control.

Deodorizing Status	1	2	3	4	5	6	7	6, 7, 6, 7,...	1	
Compressor	ON			OFF						ON
Time (second)	40	50		30	60	30	60		40	
Fan Speed	OFF	SLo	SLo	SLo	OFF	SLo	OFF		OFF	

Compressor

Fan Speed

- When the compressor is in operation, the deodorizing status starts from 1 → 2 → 3.
- When the compressor stops operation, the deodorizing status starts from 4 → 5 → 6 → 7.
- If the compressor still stops operation after 3 minutes, the deodorizing status will start from 6.

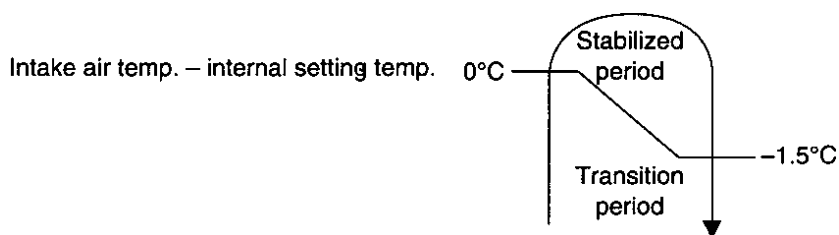
8.3.3. Sensible Heat Control

- This control is to improve the feeling in high fan speed and low operation frequency. When the operation frequency is less than 37Hz, the fan speed will reduce. When the operation frequency is above 38Hz continuously for 5 minutes, the fan speed will resume to normal condition. (During Cooling operation).

8.4. HEATING OPERATION

8.4.1. Room Temperature Control

- During heating operation, the room temperature control depends on intake air temperature and internal setting temperature. Basically it can be divided into 2 periods as shown below:



- When indoor fan speed is Medium or above.

Heating Operation	Compressor Operation Frequency (Hz)						
	Transition Period	Stabilized Period					
Remote Control Setting Temp.	16°C ~ 30°C	16°C ~ 20°C		21°C ~ 25°C		26°C ~ 30°C	
Intake air temperature – Internal setting temperature	Comp Off	Comp Off	Comp Off	Comp Off	Comp Off	Comp Off	Comp Off
Compressor OFF							
+1.5	17	17	17	17	17	17	17
+1.0	17	17	17	37	17	37	17
+0.5	21	21	21	45	21	45	21
0	70	45	45	55	45	55	45
-0.5	88	55	55	70	55	70	55
-1.0	88	70	70	88	70	88	70
-1.5	92 (Fh) or 124 (Fhmax)	92 (Fh) or 124 (Fhmax)	92 (Fh) or 124 (Fhmax)	92 (Fh) or 124 (Fhmax)	92 (Fh) or 124 (Fhmax)	92 (Fh) or 124 (Fhmax)	92 (Fh) or 124 (Fhmax)
Outdoor air temperature		Less than -1°C	-1°C and above	Less than -1°C	-1°C and above	Less than -1°C	-1°C and above

- When indoor fan speed is lower than Medium.

Heating Operation	Compressor Operation Frequency (Hz)						
	Transition Period	Stabilized Period					
Remote Control Setting Temp.	16°C ~ 30°C	16°C ~ 20°C		21°C ~ 25°C		26°C ~ 30°C	
Intake air temperature – Internal setting temperature	Comp Off	Comp Off	Comp Off	Comp Off	Comp Off	Comp Off	Comp Off
Compressor OFF							
+1.5	17	17	17	17	17	17	17
+1.0	17	29	29	37	29	37	29
+0.5	21	37	37	45	37	45	37
0	70	45	45	55	45	55	45
-0.5	88	55	55	70	55	70	55
-1.0	88	70	70	75	70	75	70
-1.5	92 (Fh) or 124 (Fhmax)	92 (Fh) or 124 (Fhmax)	92 (Fh) or 124 (Fhmax)	92 (Fh) or 124 (Fhmax)	92 (Fh) or 124 (Fhmax)	92 (Fh) or 124 (Fhmax)	92 (Fh) or 124 (Fhmax)
Outdoor air temperature		Less than -1°C	-1°C and above	Less than -1°C	-1°C and above	Less than -1°C	-1°C and above

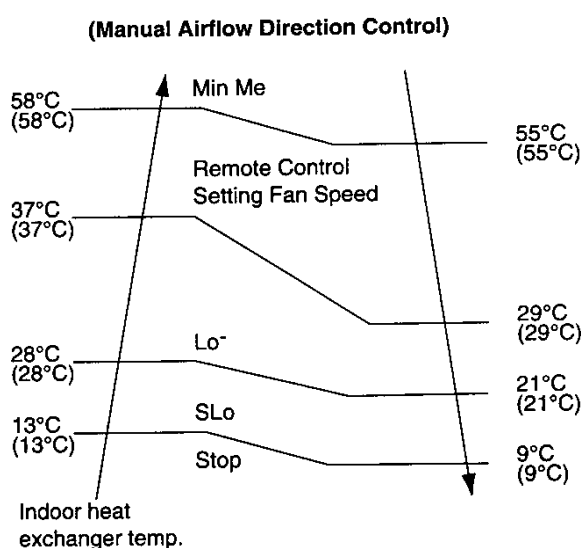
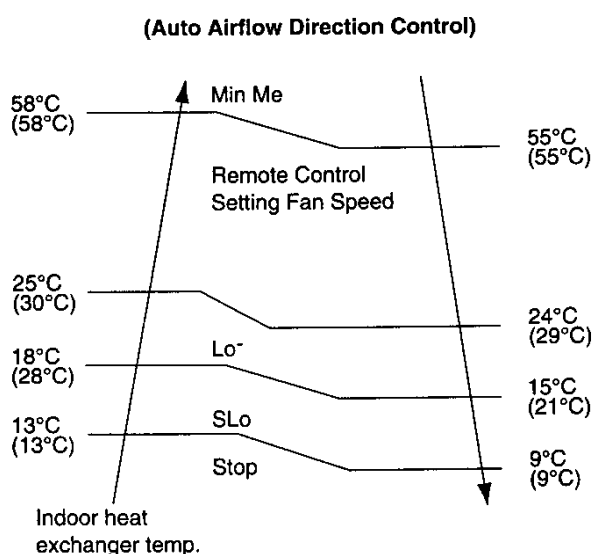
- Compressor OFF temperature = compressor ON temperature.
- The operation frequency changes every 30 seconds.
- When the difference of the intake air temperature and Internal setting temperature is -1.5°C or more, compressor will operate at Fh continuously for 3 minutes and then change over to Fhmax.
- The compressor stops when the intake air temperature reaches 1.5°C above internal setting temperature and continues for 3 minutes.
- When the compressor stops, it will not start operation for 3 minutes. (Time Delay Safety Control)
- When the intake air temperature decreases to the compressor ON temperature, the compressor starts immediately.
- When the compressor stops, the outdoor fan motor stops 30 seconds later.

8.4.2. Anti Cold Draft Control

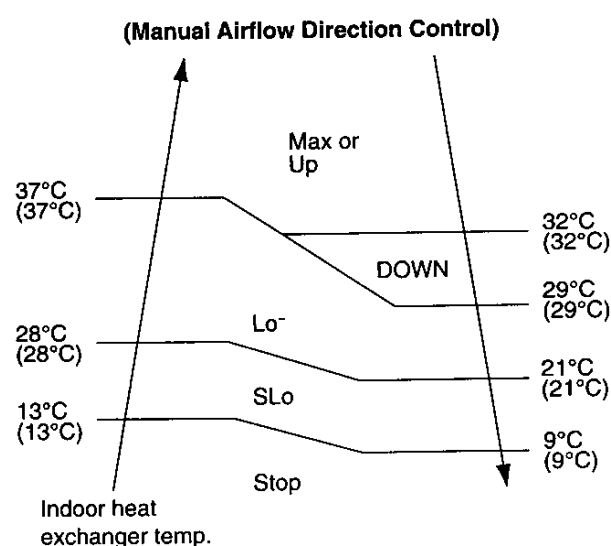
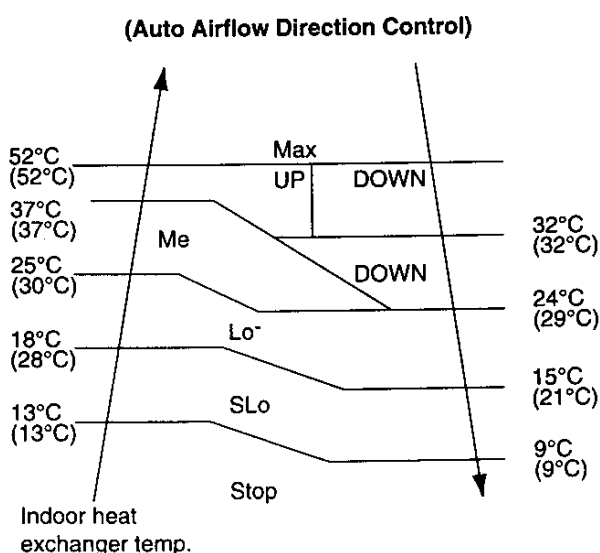
1. Indoor Fan Control

Indoor fan speed and airflow direction varies in accordance to the indoor heat exchanger temperature as shown below:

a. Manual Fan speed control



b. Auto Fan speed control



Note:

- UP means fan speed is increased by 1 rank.
- DOWN means fan speed is decreased by 1 rank.
- Max means fan speed is running at maximum auto fan speed.
- Temperature in () is indicating when powerful mode is selected.

2. Hot Start

- At the start of heating operation, the indoor fan stops and compressor operates at Fhmax frequency (124Hz). This is to heat up the indoor heat exchanger in order to avoid cold air discharged.
- Hot Start ends when
 - a. Indoor heat exchanger temperature reaches over 15°C
 - or
 - b. 4 minutes after heating operation starts
- After Hot Start operation, compressor operates at Fhmax (124Hz) for 2 minutes.

8.4.3. Deice Operation

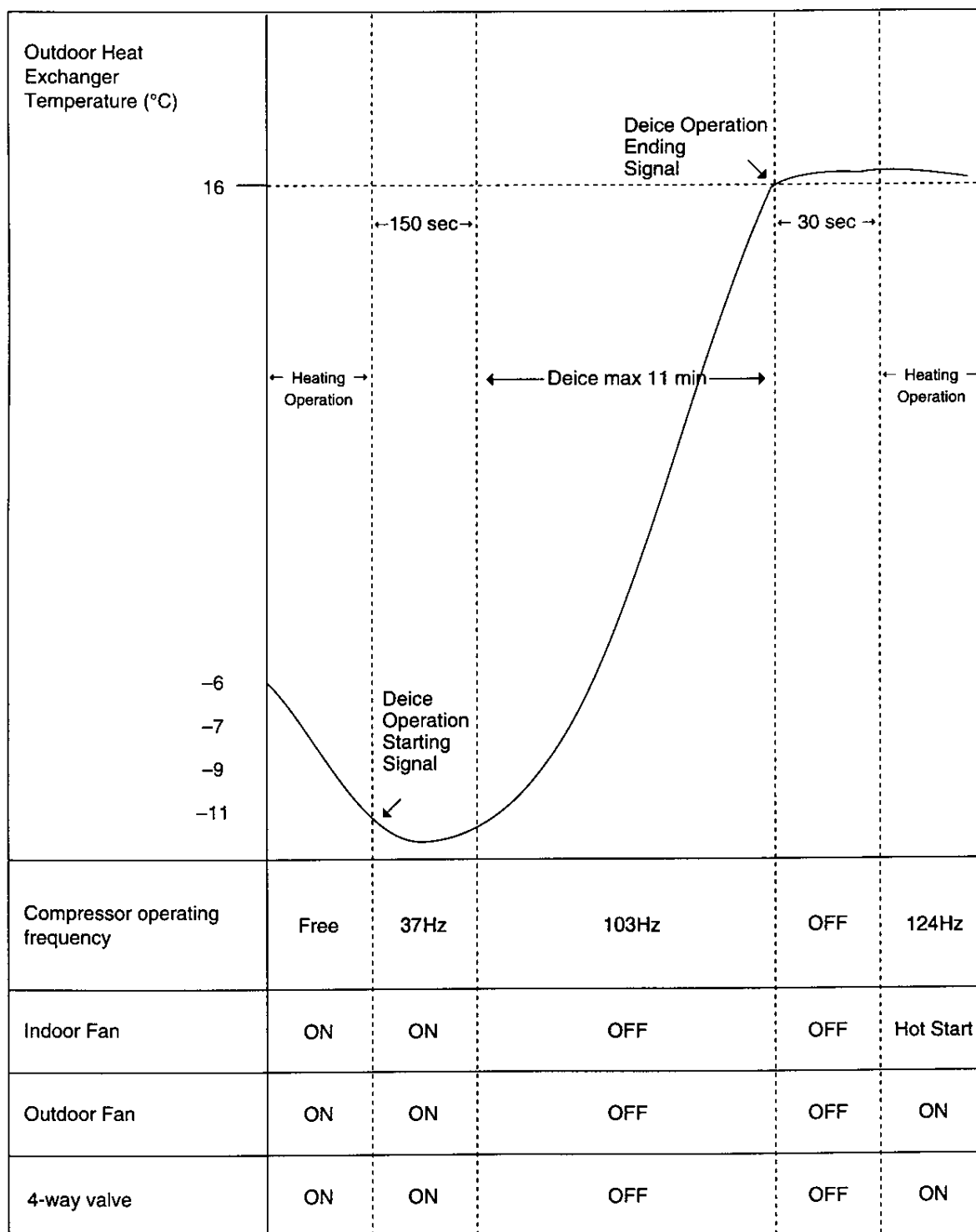
Deice operation occurs when the deice operation starting signal is generated. This happens when one of the following conditions occurs. However, the first deice operation will begin 1 hour after start of heating operation.

	Outdoor Heat Exchanger Temp. Th $\leq 3^{\circ}\text{C}$	Outdoor Heat Exchanger Temp. (Th)	Outdoor Temp. (To)
Case 1	120 minutes continuously	Th $\leq -6^{\circ}\text{C}$ for 3 min continuously	To $\geq -1^{\circ}\text{C}$
Case 2	80 minutes continuously	Th $\leq -7^{\circ}\text{C}$ for 3 min continuously	To $\geq -1^{\circ}\text{C}$
Case 3	40 minutes continuously	Th $\leq -9^{\circ}\text{C}$ for 3 min continuously	To $\geq -3^{\circ}\text{C}$
Case 4	40 minutes continuously	Th $\leq -11^{\circ}\text{C}$ for 3 min continuously	To $\geq -3^{\circ}\text{C}$

Note:

The above 4 cases are under compressor operating condition.

Deice Operation Time Chart



- Compressor frequency is set at 73Hz when the deice operation starting signal is generated.
- 150 seconds later deice operation starting signal is generated, indoor fan, outdoor fan, 4-way valve are turned off and compressor operates at 95Hz for 30 seconds. (Deice operation starts)
- During deice operation, the compressor operating frequency is set at 103Hz.
- Deice will end when the outdoor heat exchanger temperature rises to 16°C or after 11 minutes.

8.5. FAN OPERATION

This operation is to enable the fan operation without compressor running. Timer operation is valid for fan operation.

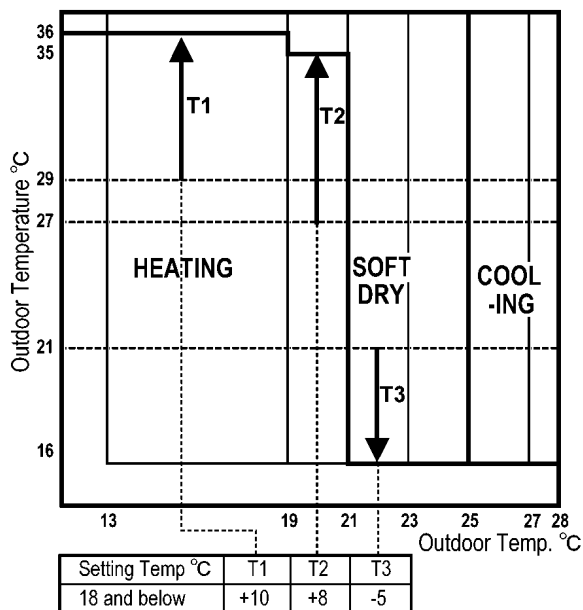
8.6. AUTOMATIC OPERATION

When the Automatic mode is selected, the operation mode is decided in accordance to remote control setting temperature, intake air temperature and outdoor air temperature.

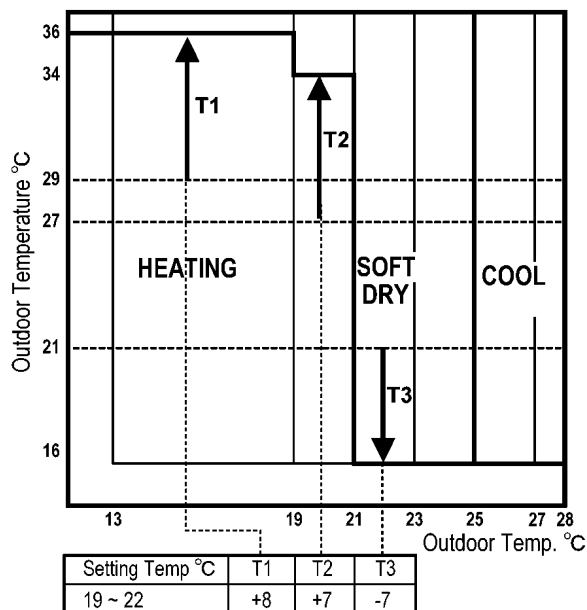
- During judging the operation mode, indoor fan is running at Lo-speed and outdoor fan at ON in order to sense the indoor intake air temperature and outdoor air temperature for 20 seconds. At this time, Power LED is blinking. After the operation mode is selected, Power LED light up.

There are 4 determination charts when automatic mode is selected.

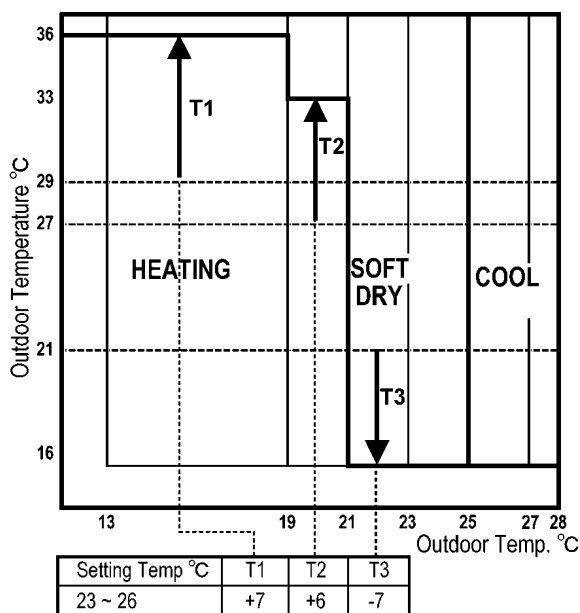
(A) When setting temperature of remote control is 18 °C and below



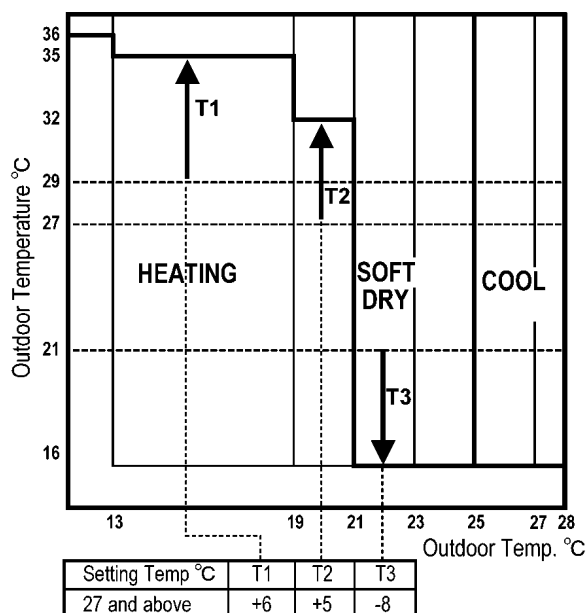
(B) When setting temperature of remote control is 19 °C ~ 22 °C



(C) When setting temperature of remote control is 23 °C ~ 26 °C

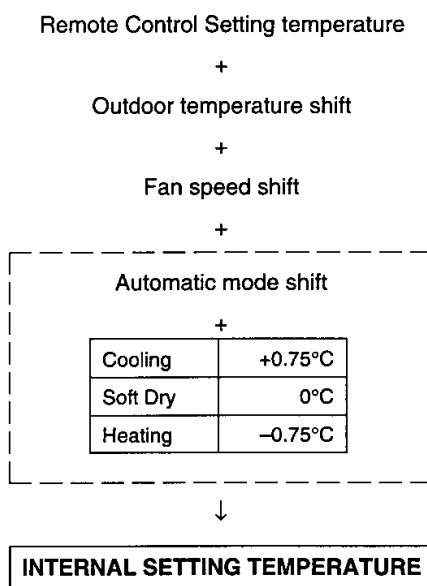


(D) When setting temperature of remote control is 27 °C and above



Setting temp °C	T1	T2	T3	Note: Base temperature for T1 is 29 °C Base temperature for T2 is 27 °C Base temperature for T3 is 21 °C
18 and below	+10	+8	-5	
19 - 22	+8	+7	-7	
23 - 26	+7	+6	-7	
27 and above	+6	+5	-8	

- When the operation mode is changed over, the value for t1, t2 and t3 are shifted as follows:
Cooling/Soft dry → Heating: -2°C
Heating → Cooling/Soft dry: +2°C
- When the indoor intake air temperature is lower than 16°C, heating operation is immediately started.
- When the outdoor air temperature is more than 25°C, and the intake air temperature is over 16°C, cooling operation is immediately started.
- The operation mode is judged every 30 minutes.
- When the operation mode (Heating, Cooling or Soft Dry) is decided, the internal setting temperature will shift as shown below:



8.7. INDOOR FAN SPEED CONTROL

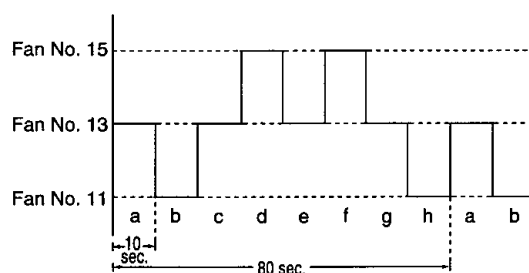
Fan speed no.	Voltage Supply to Fan Motor DC(V)	Cooling		Soft Dry	Remarks	Heating		Remarks
		Manual	Auto			MAnnual	Auto	
0		OFF	OFF	OFF		OFF	OFF	• hot Start Control
1								
2	20.4			SLO	• Deodorizing control • Sleep Mode			• Slo:Hot Start Control
3	21.7		Lo-		• Auto operation mode judgement • Quijet operation			• Lo-:Hot Start Control Sleep Mode Thermo OFF Anti Cold Draft Control
4	22.0							
5	22.2							
6	22.5				• Sensible Heat Control			
7	22.7	Lo			• Sensible Heat Control • On Timer preparatory operation + Auto Fan	Lo	Auto Fan (Min)	• On Timer preparatory operation + Auto Fan
8	22.7					Me-		
9	22.7							
10	23.2	Me-			• Sensible Heat Control • Sensible Heat Control + Auto Fan			
11	23.5		Auto Fan					• Auto Fan + Powerful
12	24.1				• Sensible Heat Control + Auto Fan			
13	24.3		Auto Fan		• Sensible Heat Control • Auto Fan + Powerful	Me		• Manual Fan + Powerful • On Timer preparatory operation + Manual Fan + Powerful • Temporary Operation
14	24.6	Me			• Manual Fan + Powerful • On Timer preparatory operation + Manual Fan + powerful • Sensible Heat Control + Auto Fan • Temporary Operation			
15	24.9		Auto Fan		• Auto Fan + Powerful		Auto Fan (Max.)	
16	25.4				• Sensible Heat Control			
17	26.0				• Auto Fan + Powerful			
18	26.5					Me+		
19	27.1	Me+			• Manual Fan + Powerful • On Timer preparatory operation + Manual Fan + Powerful			
20	27.4							
21	28.0							
22	28.4							
23	28.8							
24	29.4							
25	29.7							
26	30.0	Hi			• Manual Fan + Powerful • On Timer preparatory operation + Manual Fan + Powerful • Test Run			
27	30.9							• Auto Fan + Powerful
28	31.5				• SHi:Maximum Capacity Operation			
29	32.1							

Fan speed no.	Voltage Supply to Fan Motor DC(V)	Cooling		Soft Dry	Remarks	Heating		Remarks
		Manual	Auto			MAnnual	Auto	
30	32.6					SHI		• Manual Fan + Powerful • On Timer preparatory operation + Manual Fan + Powerful • Test Run
31	36.0							• SSHI:Maximum Capacity Operation

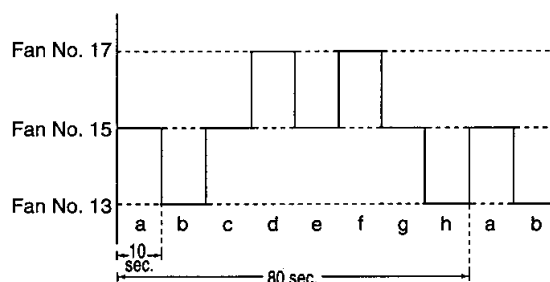
1. Cooling Automatic Fan Speed

The Automatic Fan Speed for cooling operation is shown as below:

a. When Automatic Fan Speed is selected



b. When Automatic Fan Speed and Powerful Mode are selected



Note:

The Fan Speed will change every 10 seconds and it will be repeated from a to h every 80 seconds.

2. Heating Automatic Fan Speed

The Automatic Fan Speed for heating operation is shown below:

- When Automatic Fan Speed is selected, the Fan speed will change every 10 seconds from Fan Speed No. 7 to No. 15 depending on indoor heat exchanger temperature. Each time the Fan Speed will move 1 rank up or down.
- When Automatic Fan Speed and Powerful Mode are selected, the Fan Speed will change for every 10 seconds from Fan Speed No. 11 to No. 27 depend on heat exchanger temperature. Each time the Fan Speed will move 1 rank up or down.

3. Cooling Operation at SHi Speed

During Cooling operation, Indoor Fan speed is set at SHi when the following conditions occur:

- Outside air temperature is 30°C or above
- Compressor operates at 73Hz and above
- Remote control setting fan speed is High
- Indoor intake air temperature is 24°C or above
- Remote control setting temperature is 16°C
- Within 30 minutes after start of operation.

4. Heating Operation at SSHi Speed

During Heating operation, Indoor Fan speed is set at SSHi when the following conditions occur:

- Heating operation for 2 hours or more
- When remote control setting fan speed is High
- Indoor intake air temperature is 17°C or above and less than 23°C
- Outdoor air temperature is 4°C or below
- Remote control setting temperature is 30°C
- Compressor operates at 92Hz and above
- Airflow Direction is set at Manual.

8.8. OUTDOOR FAN CONTROL

- Outdoor fan motor is controlled with 2 speeds. Fan is in operation when the compressor starts operation and stops 30 seconds after compressor stops operation.

(During 80 seconds, Fan speed is at low).

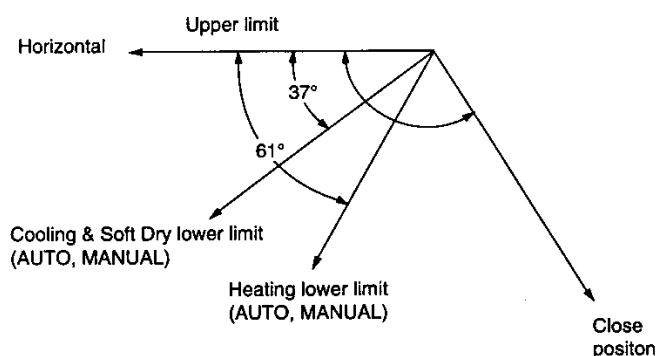
Fan Speed	Compressor Operating Frequency	
	Cooling-Soft Dry	Heating
High	Over 45Hz	Over 73Hz
Low	Below 45Hz	Below 73Hz

8.9. AIRFLOW DIRECTION

- Blade angle setting (Upper limit reference)

Operation				Blade angle				
				1	2	3	4	5
heating	Airflow direction auto	Indoor heat exchanger temperature	A	14°				
			B	61°				
			C	0°				
	Airflow direction manual	-	-	0°	18°	29.4°	45.1°	61°
Cooling	Airflow direction auto	-		0° - 37°				
		Anti-dew formation control		3° - 37°				
	Airflow direction manual	-		0°	8°	18°	28°	37°
		Anti-dew formation control		3°	11.5°	20°	28.5°	37°
Dry	Airflow direction auto	-		0° - 37°				
		Anti-dew formation control		3° - 37°				
	Airflow direction manual	-		0°	8°	18°	28°	37°
		Anti-dew formation control		3°	11.5°	20°	28.5°	37°
Stop				138°				

- Sitting angle

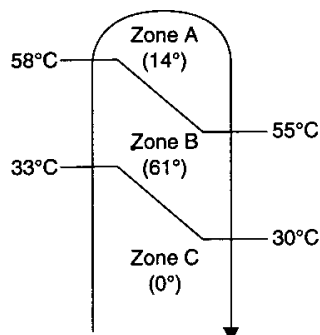


1. Airflow Direction Manual

By pressing the remote control airflow direction setting switch, the blade will move to the indicated angle (1, 2, 3, 4, 5) as shown in the table. When the remote control OFF/ON switch is pressed, the blade will move to the Close position.

2. Airflow Direction Auto

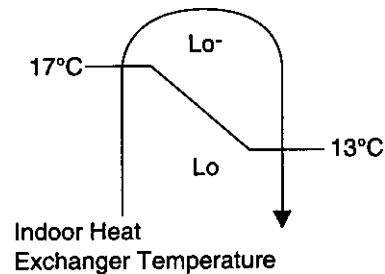
By setting the airflow direction to AUTO, the blade swings up and down from 0°-37° during Cooling and Soft Dry operation. During Heating operation, the blade angle will shift according to the indoor heat exchanger temperature as shown below:



8.10. QUIET OPERATION

The purpose of this control is to reduce indoor operating noise. Indoor fan speed is set to Lo- when the following conditions occur.

- Indoor fan speed is set at Low
- Indoor heat exchanger temperature rises to 17°C or above
- Compressor operates for 5 minutes or above
- Operation frequency is less than 45Hz

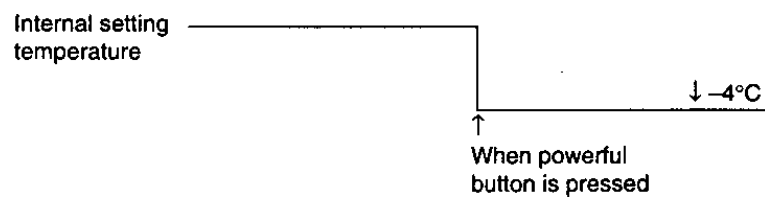


When the indoor heat exchanger temperature is decreased to 13°C or below, the control is cancelled and the indoor fan speed will resume from Lo- to Low.

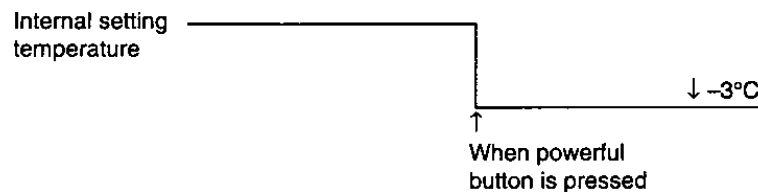
8.11. POWERFUL MODE OPERATION

When the powerful mode is selected, the internal setting temperature will shift to achieve the setting temperature quickly.

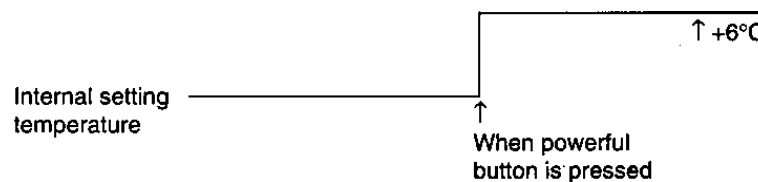
1. Cooling Operation



2. Soft Dry Operation



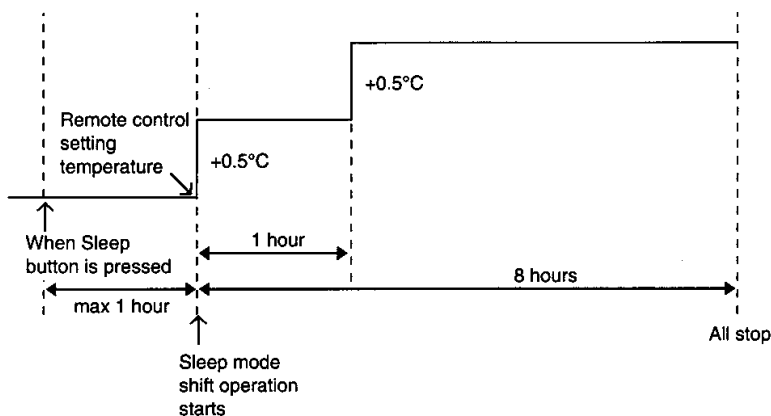
3. Heating Operation



8.12. SLEEP MODE OPERATION

1. Cooling Operation/Soft Dry Operation

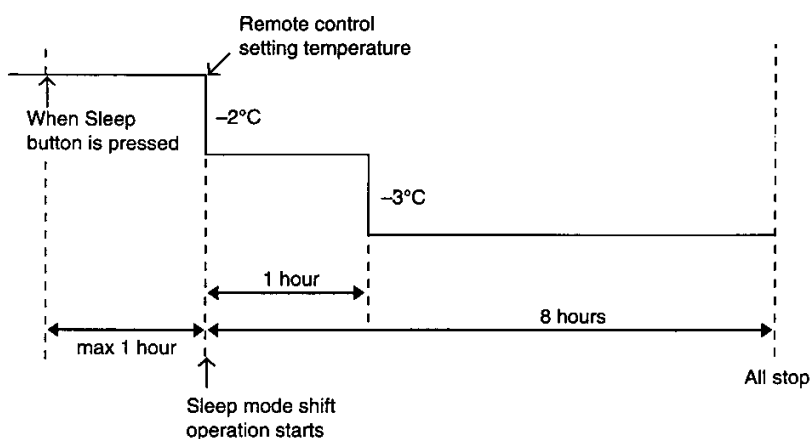
- When the sleep button is pressed, the remote control setting temperature will increase 0.5°C after 1 hour or when the remote control setting temperature is reached. After another hour, 0.5°C will be increased again.



- During sleep shift operation, indoor fan speed operates at SLo.

2. Heating Operation

- When the sleep button is pressed, the remote control setting temperature will decrease 2°C after 1 hour or when the remote control setting temperature is reached. After another hour, 3°C will be decreased again.

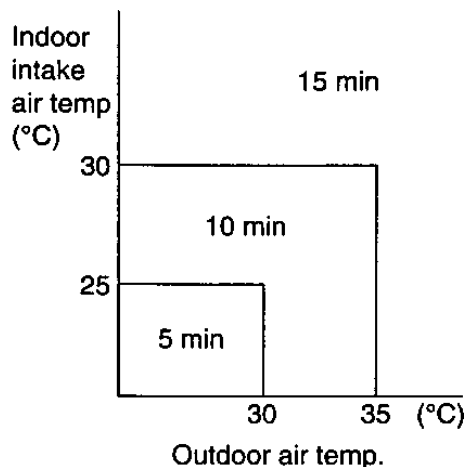


- During sleep shift operation, indoor fan speed operates at Lo-.

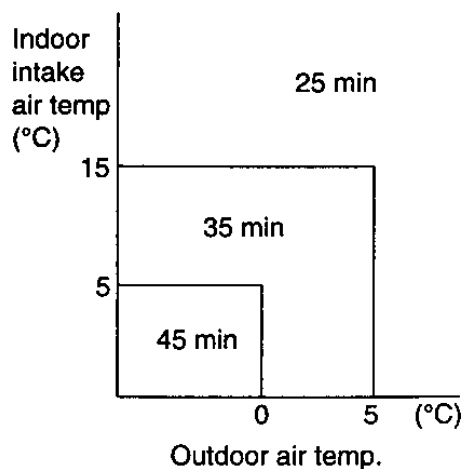
8.13. DELAY ON TIMER CONTROL

- When the Delay On Timer is set by using remote control, the unit will start to operate slightly earlier before the set time, so that the room will nearly reach the set temperature by the On Timer set time.
- 60 minutes before the set time, the indoor fan operates at SLo and outdoor fan operates for 20 seconds to sample the indoor intake air temperature and outdoor air temperature in order to determine the starting time for preparatory operation. (The Power LED blinks during sampling.)
- The time of the preparatory operation will start before the On Timer set time.

1. Cooling/Soft Dry



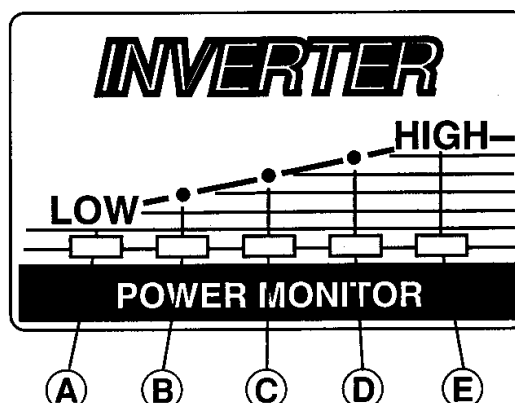
2. Heating



8.14. AUTO RESTART CONTROL

- If there is a power failure, operation will automatically be restarted when the power is resumed. It will start with the previous operation mode and airflow direction. (Time Delay Safety Control is valid)
- Auto Restart Control is not available when Timer or Sleep Mode is set.

8.15. POWER MONITOR DISPLAY



Power Monitor LED lights on when the compressor is in operation. The number of the LED lights on is in accordance to the compressor operating frequency.

Display	A	A, B	A, B, C	A, B, C, D	A, B, C, D, E
Cooling & Soft Dry Indication Frequency (Hz)	9-29	30-55	56-67	68-73	74-86
Heating Indication Frequency (Hz)	9-29	30-67	68-79	80-88	89-124

8.16. REMOTE CONTROL SIGNAL RECEIVING SOUND ON/OFF

- Press the AUTO button for 10 seconds or longer to switch off the signal receiving sound.
- Press the AUTO button for 10 seconds or longer again to switch on the signal receiving sound.

8.17. INDOOR POWER RELAY CONTROL

- The power relay turns on when one of the indoor LED lights on.
- The power relay turns off when all of the indoor LEDs lights off.
- When the air conditioner is stopped during operation, the power relay stays ON for 3 minutes.
- The power relay will turn off if a sudden power failure occurs for 0.5 second and below. The power relay will turn on again after 3 minutes. It will start with the previous operation condition before the power failure.

9 Operation Details (PROTECTION)

9.1. PROTECTION CONTROL FOR ALL OPERATIONS

1. Time Delay Safety Control

- The compressor is not restarted for 3 minutes after stop of compressor.

2. 30 Second Forced Operation

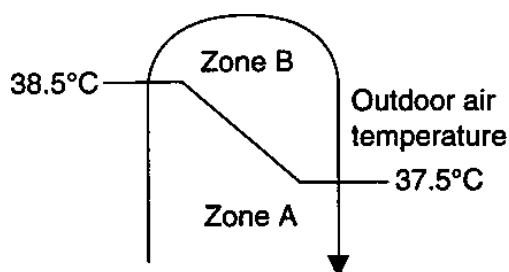
- Once the compressor is ON, it will not turn OFF for 30 seconds. However, it is turned off by remote control or Automatic switch.

3. Total Running Current Control

- When the outdoor unit total running current (AC) exceeds I1, the frequency is lowered by 1 rank. If I1 is not exceeded for 30 seconds, the frequency is highered by 1 rank at one time. If the outdoor unit total running current exceeds I2, the compressor is immediately stopped for 3 minutes.

< Cooling, Soft Dry set value >

	Zone A	Zone B
I1 (A)	5.9	5.0
I2 (A)	12.3	12.3



Note:

Zone A will be used 30 minutes after operation starts.

< Heating set value >

I1 (A)	8.4
I2 (A)	12.3

4. Power Transistor Overheating Prevention Control

- When the power transistor temperature rises to 110°C, the OLP goes into operation and compressor stops immediately. The compressor is restarted when the power transistor temperature decreases to 95°C after 3 minutes (Time Delay Safety Control).

5. Compressor Overheating Prevention Control

- When the temperature of compressor rises to 95°C, the frequency is reduced as shown in diagram below. When the temperature rises to 105° or above, the compressor stops. The compressor will start operating at low frequency when the temperature falls to 105°C and resume to normal condition when the temperature falls to 84°C.

Compressor Temp.(°C)	When temperature rises	When temperature falls
105	Comp OFF	Comp OFF
95	max 73Hz	max 73Hz
84	Free	Free
Condition	When temperature rises	When temperature falls

6. Low Pressure Control (Gas Leakage Detection)

- When the following conditions as shown in the below table occur, the compressor stops and restarts after 3 minutes. If this phenomenon is continuously occurring twice within 20 minutes, the air conditioner will stop operation and the Timer LED blinks. At this time, [F91] is displayed on the indoor unit.

Comp. Frequency $\otimes$	73Hz	92Hz
Total Running Current $\otimes$	≥ 1.51 A and ≤ 2.15 A	≥ 1.51 A and ≤ 2.30 A
Indoor Heat Exchanger Temp.	15 °C or above	30 °C or below
Operation	Cooling / Soft Dry	Heating

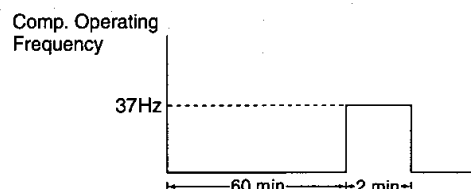
Note:

The above conditions are not valid during Deice operation.

- this conditions are continuous for 5 minutes.

7. Minimum Frequency Operation Protection

- When the compressor operates at less than 37Hz for 60 minutes, the operating frequency will increase to 37Hz for 2 minutes.

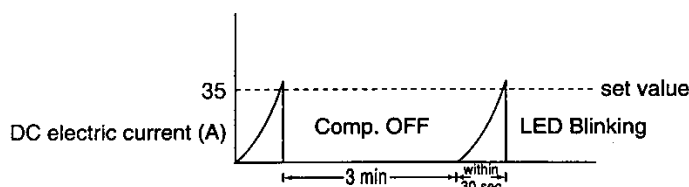
**8. Low Frequency Operation Protection**

- When the following conditions occur, the compressor will operate at minimum 37 Hz.

Indoor intake temp.(°C)	≥ 30 °C or ≤ 15 °C	≥ 28 °C or ≤ 19 °C
Outdoor air temp.(°C)	≥ 38 °C or ≤ 16 °C	≥ 24 °C or ≤ 4 °C
Indoor heat exchanger temp.(°C)	≤ 30 °C	≥ 30 °C
Operation	Cooling / Soft Dry	Heating

9. DC Peak Current Control

- When the electric current to the power transistor exceeds the set value, DC 35±4A, the compressor stops. The compressor restarts after 3 minutes.
- If within 30 seconds the set value exceeds again after start of the compressor, all indoor and outdoor relays will be turned off and all the LEDs will blink.

**10. DC Reset**

- When the voltage supply (DC) to power transistor is below the set value i.e, 178±5V, the compressor is stopped. The compressor will restart after 3 minutes.
- If within 30 seconds the voltage supply (DC) is below the set value again after start of the compressor, all indoor and outdoor relays will be turned off and all the LEDs will blink.

11. High Power Supply Voltage Protection

- When the voltage supply (AC) exceeds $295 \pm 15V(T_0)$, the air conditioner stops and restarts automatically when the voltage supply (AC) is below $(T_0-5)V$. However, waiting for 3 minutes is necessary for re-operation.

9.2. PROTECTION CONTROL FOR COOLING & SOFT DRY

1. Anti-Freezing Control

- When the temperature of the indoor heat exchanger becomes low, the compressor operating frequency is reduced and stopped when the temperature falls to lower than 2°C continuously for 6 minutes. This is to prevent freezing of indoor heat exchanger. When the temperature rises to 10°C or above, the compressor restarts with 3 minutes. The compressor operating frequency will resume to normal when the temperature reaches 14°C.
- Indoor fan speed will increase when the temperature falls and it will resume to original speed when the temperature increases to 14°C for 5 minutes continuously.

Indoor Heat Exchanger Temp.(°C)	14 12 10 6 4 2	Free Original	Original	Free	Original
				37Hz 17Hz	1 rank up
				OFF	
		Comp.	Indoor Fan	Comp.	Indoor Fan
		When the temp. falls		When the temp. rises	

Note:

The above phenomenon occurs when the fan is running at Me+ or below.

2. Anti-Dew Formation Control

- When the following conditions occur for 20 minutes continuously, anti-dew formation is controlled:
 - (1) Indoor intake air temperature is 24°C or above.
 - (2) Outdoor air temperature is less than 30°C.
 - (3) Remote control setting temperature is less than 25°C.
- During anti-dew formation control, compressor operates at 55Hz and vertical airflow direction blade moves down slightly (as shown in Airflow Direction Control). Indoor fan speed increases by 1 rank if it is set at Low or below or decreases by 2 ranks if it is set at Hi speed.
- This control is cancelled immediately when either condition 1-3 as written above is changed, or remote control setting temperature or fan speed is changed.

3. Anti-Fog Discharge Control

- The compressor operating frequency is regulated by outdoor air temperature and operation time to prevent fog discharged from indoor unit as shown in the table below.

		Compressor operating frequency (Hz)	
Operation Time, T (min)	0 ≤ T < 30	73	73
	30 ≤ T < 90	73	73
	90 ≤ T < 180	73	73
	T ≥ 180	73	73
Outdoor air temperature (°C)		less than 24°C	24°C and above

Note:

- Indoor fan is running at Me+ or below.
- After 420 minutes from the start of operation, the operation timer counting is restarted from "0".

4. High Pressure Control at Minimum Frequency Control

- When the outdoor air temperature increases to 38.6°C, compressor operating frequency reduces to minimum of 37Hz. Compressor operating frequency will resume to normal when outdoor air temperature decreases to 37.8°C.

5. Electrical Components Overheating Protection

- When outdoor air temperature is 38.5°C or above and the compressor operating frequency is 9Hz or above, total running current set value is reduced (refer to Total Running Current Control) and the operating frequency is reduced.
- The protection control is cancelled when the outdoor air temperature decreases to 37.5°C or compressor stops.

9.3. PROTECTION CONTROL FOR HEATING OPERATION

1. Intake Air Temperature Control

- When the intake air temperature is 10°C or above and remote control setting fan speed is less than Medium, the compressor operates at 92Hz.

2. Outdoor Air Temperature Control

- The compressor operating frequency is regulated in accordance to the outdoor air temperature as shown in the diagram below. This control will begin 50 seconds after the compressor starts.

Outdoor air Temperature (°C)	21	74Hz	
	19	82Hz	
	12		
	10	Free	
Condition		When temp. rises	When temp. falls

3. High Pressure Control

- The compressor operating frequency is regulated in accordance to the indoor heat exchanger temperature.

Indoor air exchanger temp.(°C)	59	OFF	
	53	45Hz	
	51		
	47	Min 45Hz	
	43		
	37	Free	
Condition		When temp. rises	When temp. falls

10 Installation And Servicing Air Conditioner Using R410A

10.1. OUTLINE

10.1.1. About R410A Refrigerant

1. Converting air conditioners to R410A

Since it was declared in 1974 that chlorofluorocarbons (CFC), hydro chlorofluorocarbons (HCFC) and other substances pose a destructive danger to the ozone layer in the earth's upper stratosphere (20 to 40 km above the earth), measures have been taken around the world to prevent this destruction.

The R22 refrigerant which has conventionally been used in ACs is an HCFC refrigerant and, therefore, possesses this ozone-destroying potential. International regulations (the Montreal Protocol on Ozone-Damaging Substances) and the domestic laws of various countries call for the early substitution of R22 by a refrigerant which will not harm the ozone layer.

· In ACs, the HFC refrigerant which has become the mainstream alternative is called R410A. Compared with R22, the pressure of R410A is approximately 1.6 times as high at the same refrigerant temperature, but the energy efficiency is about the same. Consisting of hydrogen (H), fluorine (F) and carbon (C), R410A is an HFC refrigerant. Another typical HFC refrigerant is R407C. While the energy efficiency of R407C is somewhat inferior to that of R410A, it offers the advantage of having pressure characteristics which are about the same as those of R22, and is used mainly in packaged ACs.

2. The characteristics of HFC (R410A) refrigerants

a. Chemical characteristics

The chemical characteristics of R410A are similar to those of R22 in that both are chemically stable, non-flammable refrigerants with low toxicity.

However, just like R22, the specific gravity of R410A gas is heavier than that of air. Because of this, it can cause an oxygen deficiency if it leaks into a closed room since it collects in the lower area of the room. It also generates toxic gas when it is directly exposed to a flame, so it must be used in a well ventilated environment where it will not collect.

Table 1 Physical comparison of R410A and R22

	R410A	R22
Composition (wt%)	R32/R125 (50/50)	R22 (100)
Boiling point (°C)	-51.4	-40.8
Vaporizing pressure (25°C)	1.56 Mpa (15.9 kgf/cm ²)	0.94 Mpa (9.6 kgf/cm ²)
Saturated vapor density	64.0 kg/m ³	44.4 kg/m ³
Flammability	Non-flammable	Non-flammable
Ozone-destroying point (ODP)	0	0.055
Global-warming point (GWP)	1730	1700

b. Compositional change (pseudo-azeotropic characteristics)

R410A is a pseudo-azeotropic mixture comprising the two components R32 and R125. Multi-component refrigerants with these chemical characteristics exhibit little compositional change even from phase changes due to vaporization (or condensation), which means that there is little change in the circulating refrigerant composition even when the refrigerant leaks from the gaseous section of the piping.

Accordingly, R410A can be handled in almost the same manner as the single-component refrigerant R22.

However, when charging, because there is a slight change in composition between the gas phase and the liquid phase inside a cylinder or other container, charging should basically begin with the liquid side.

c. Pressure characteristics

As seen in Table 2, the gas pressure of R410A is approximately 1.6 times as high as that of R22 at the same refrigerant temperature, which means that special R410A tools and materials with high-pressure specifications must be used for all refrigerant piping work and servicing.

Table2 Comparison of R410A and R22 saturated vapor density

Refrigerant Temperature(°C)	Unit:MPa	
	R410A	R22
-20	0.30	0.14
0	0.70	0.40
20	1.35	0.81
40	2.32	1.43
60	3.73	2.33
65	4.15	2.60

d. R410A refrigerating machine oil

Conventionally, mineral oil or a synthetic oil such as alkylbenzene has been used for R22 refrigerating machine oil. Because of the poor compatibility between R410A and conventional oils like mineral oil, however, there is a tendency for the refrigerating machine oil to collect in the refrigerating cycle. For this reason, polyester and other synthetic oils which have a high compatibility with R410A are used as refrigerating machine oil.

Because of the high hygroscopic property of synthetic oil, more care must be taken in its handling than was necessary with

conventional refrigerating machine oils. Also, these synthetic oils will degrade if mixed with mineral oil or alkylbenzene, causing clogging in capillary tubes or compressor malfunction. Do not mix them under any circumstances.

10.1.2. Safety Measures When Installing/Service Refrigerant Piping

Cause the gas pressure of R410A is approximately 1.6 times as high as that of R22, a mistake in installation or servicing could result in a major accident. It is essential that you use R410A tools and materials, and that you observe the following precautions to ensure safety.

1. Do not use any refrigerant other than R410A in ACs that have been used with R410A.
2. If any refrigerant gas leaks while you are working, ventilate the room. Toxic gas may be generated if refrigerant gas is exposed to a direct flame.
3. When installing or transferring an AC, do not allow any air or substance other than R410A to mix into the refrigeration cycle. If it does, the pressure in the refrigeration cycle can become abnormally high, possibly causing an explosion and/or injury
4. After finishing the installation, check to make sure there is no refrigerant gas leaking.
5. When installing or transferring an AC, follow the instructions in the installation instructions carefully. Incorrect installation can result in an abnormal refrigeration cycle or water leakage, electric shock, fire, etc.
6. Do not perform any alterations on the AC unit under any circumstances. Have all repair work done by a specialist. Incorrect repairs can result in an water leakage, electric shock, fire, etc.

10.2. TOOLS FOR INSTALLING/SERVICING REFRIGERANT PIPING

10.2.1. Necessary Tools

In order to prevent an R410A AC from mistakenly being charged with any other refrigerant, the diameter of the 3-way valve service port on the outdoor unit has been changed. Also, to increase its ability to withstand pressure, the opposing dimensions have been changed for the refrigerant pipe flaring size and flare nut. Accordingly, when installing or servicing refrigerant piping, you must have both the R410A and ordinary tools listed below.

Table 3 Tools for Installation, transferring or replacement

Type of work	Ordinary tools	R410A tools
Flaring	Flaring tool (clutch type), pipe cutter, reamer	Copper pipe gauge for clearance Adjustment, flaring tool (clutch type)*1)
Bending, connecting pipes	Torque wrench (nominal diameter 1/4, 3/8, 1/2) Fixed spanner (opposing sides 12 mm, 17 mm, 19 mm) Adjustable wrench, Spring bender	
Air purging	Vacuum pump Hexagonal wrench (opposing sides 4 mm)	Manifold gauge, charging hose, vacuum pump adaptor
Gas leak inspection	Gas leak inspection fluid or soapy water	Electric gas leak detector for HFC refrigerant*2)

*1). You can use the conventional (R22) flaring tool. If you need to buy a new tool, buy the R410A type.

*2). Use when it is necessary to detect small gas leaks.

For other installation work, you should have the usual tools, such as screwdrivers (+,-), a metal-cutting saw, an electric drill, a hole core drill (65 or 70 dia.), a tape measure, a level, a thermometer, a clamp meter, an insulation tester, a voltmeter, etc.

Table 4 Tools for servicing

Type of work	Ordinary tools	R410A tools
Refrigerant charging		Electronic scale for refrigerant charging Refrigerant cylinder Charging orifice and packing for refrigerant cylinder
Brazing (Replacing refrigerating cycle parts*1)	Nitrogen blow set (be sure to use nitrogen blowing for all brazing), and brazing machine	

*1). Always replace the dryer of the outdoor unit at the same time. The replacement dryer is wrapped in a vacuum pack. Replace it last among the refrigerating cycle parts. Start brazing as soon as you have opened the vacuum pack, and begin the vacuuming operation within 2 hours.

10.2.2. R410A Tools

1. Copper tube gauge for clearance adjustment
(used when flaring with the conventional flaring tool (clutch type))
 - This gauge makes it easy to set the clearance for the copper tube to 1.0-1.5 mm from the clamp bar of the flaring tool.

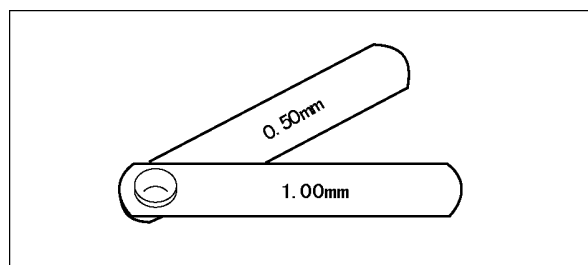


Fig. 1 Copper tube gauge for clearance adjustment

2. Flaring tool (clutch type)
 - In the R410A flaring tool, the receiving hole for the clamp bar is enlarged so the clearance from the clamp bar can be set to 0-0.5 mm, and the spring inside the tool is strengthened to increase the strength of the pipe-expanding torque. This flaring tool can also be used with R22 piping, so we recommend that you select it if you are buying a new flaring tool.

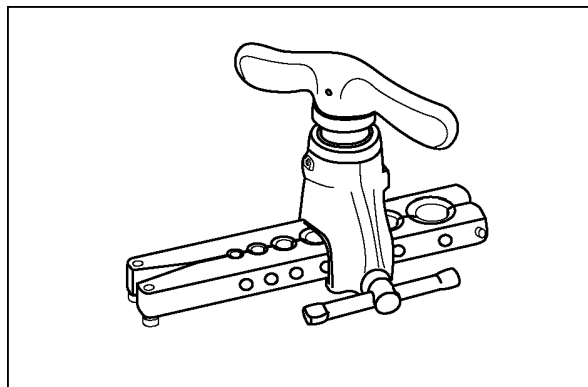


Fig. 2 Flaring tool (clutch type)

3. Torque wrenches

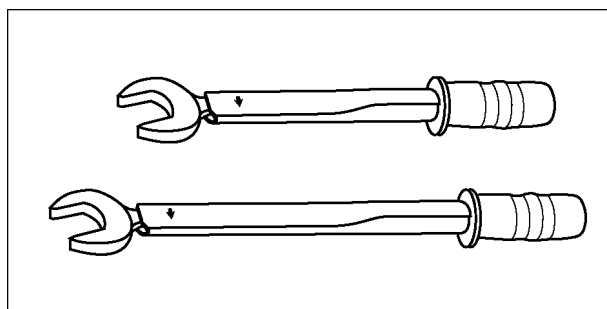


Fig. 3 Torque wrenches

Table 5

	Conventional wrenches	R410A wrenches
For 1/4 (opposite side x torque)	17mm X 18 N·m (180 kgf·cm)	17mm X 18 N·m (180 kgf·cm)
For 3/8 (opposite side x torque)	22mm X 42 N·m (420 kgf·cm)	22mm X 42 N·m (420 kgf·cm)
For 1/2 (opposite side x torque)	24mm X 55 N·m (550 kgf·cm)	26mm X 55 N·m (550 kgf·cm)

4. Manifold gauge
 - Because the pressure is higher for the R410A type, the conventional type cannot be used.

Table 6 Difference between R410A and conventional high/low-pressure gauges

	Conventional gauges	R410A gauges
High-pressure gauge (red)	-76 cmHg - 35 kgf/cm ³	-0.1 - 5.3 Mpa -76 cmHg - 53 kgf/cm ³
Low-pressure gauge (blue)	-76 cmHg - 17 kgf/cm ³	-0.1 - 3.8 Mpa -76 cmHg - 38 kgf/cm ³

- The shape of the manifold ports has been changed to prevent the possibility of mistakenly charging with another type of refrigerant.

Table 7 Difference between R410A and conventional manifold port size

	Conventional gauges	R410A gauges
Port size	7/16 UNF 20 threads	1/2 UNF 20 threads

5. Charging hose

- The pressure resistance of the charging hose has been raised to match the higher pressure of R410A. The hose material has also been changed to suit HFC use, and the size of the fitting has been changed to match the manifold ports.

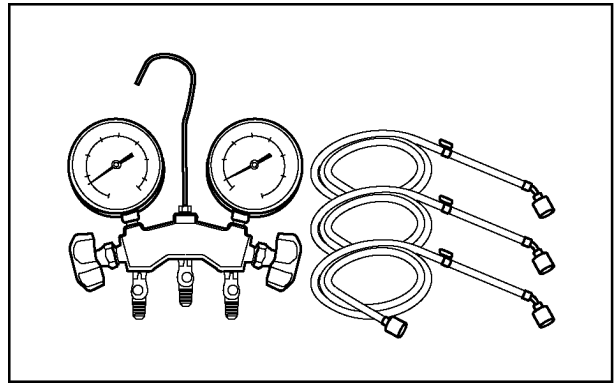


Fig. 4 Manifold gauge charging hose

Table 8 Difference between R410A and conventional charging hoses

		Conventional hoses	R410A hoses
Pressure resistance	Working pressure	3.4 MPa (35 kgf/cm ³)	5.1 MPa (52 kgf/cm ³)
	Bursting pressure	17.2 MPa (175 kgf/cm ³)	27.4 MPa (280 kgf/cm ³)
Material		NBR rubber	HNBR rubber Nylon coating inside

6. Vacuum pump adaptor

- When using a vacuum pump for R410A, it is necessary to install an electromagnetic valve to prevent the vacuum pump oil from flowing back into the charging hose. The vacuum pump adaptor is installed for that purpose. If the vacuum pump oil (mineral oil) becomes mixed with R410A, it will damage the unit.

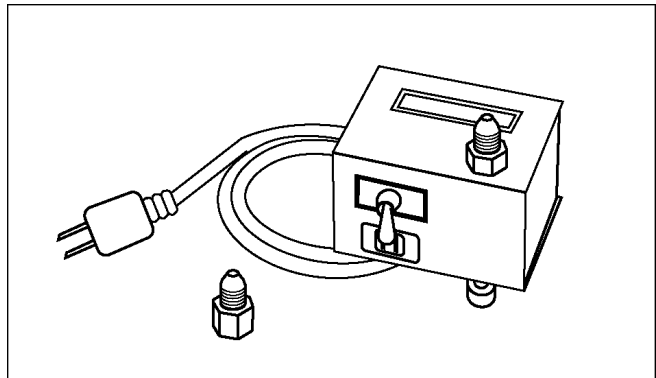


Fig. 5 Vacuum pump adaptor

7. Electric gas leak detector for HFC refrigerant

- The leak detector and halide torch that were used with CFC and HCFC cannot be used with R410A (because there is no chlorine in the refrigerant).
- The present R134a leak detector can be used, but the detection sensitivity will be lower (setting the sensitivity for R134a at 1, the level for R410A will drop to 0.6).
- For detecting small amounts of gas leakage, use the electric gas leak detector for HFC refrigerant. (Detection sensitivity with R410A is about 23 g/year.)

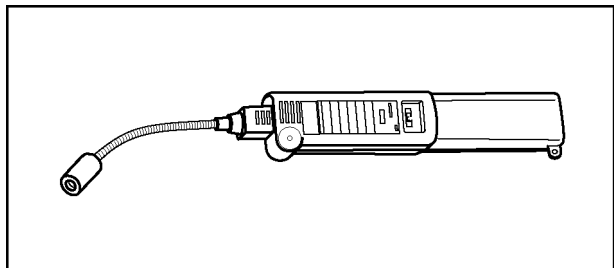


Fig. 6 Electric gas leak detector for HFC refrigerant

8. Electronic scale for refrigerant charging

- Because of the high pressure and fast vaporizing speed of R410A, the refrigerant cannot be held in a liquid phase inside the charging cylinder when charging is done using the charging cylinder method, causing bubbles to form in the measurement scale glass and making it difficult to see the reading. (Naturally, the conventional R22 charging cylinder cannot be used because of the differences in the pressure resistance, scale gradation, connecting port size, etc.)
- The electronic scale has been strengthened by using a structure in which the weight detector for the refrigerant cylinder is held by four supports. It is also equipped with two connection ports, one for R22 (7/16 UNF, 20 threads) and one for R410A (1/2 UNF, 20 threads), so it can also be used for conventional refrigerant charging.
- There are two types of electronic scales, one for 10-kg cylinders and one for 20-kg cylinders. (The 10-kg cylinder is recommended.)

Refrigerant charging is done manually by opening and closing the valve.

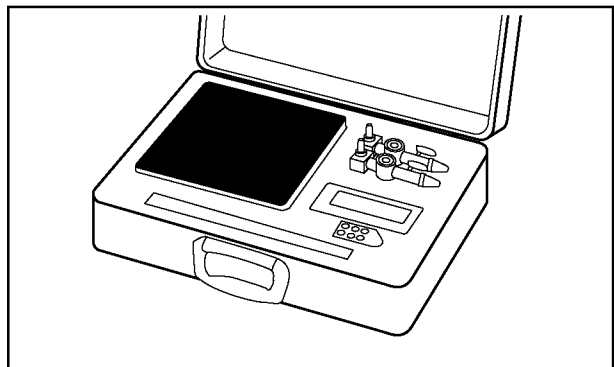


Fig. 7 Electronic scale for refrigerant charging

9. Refrigerant cylinders

- The R410A cylinders are labeled with the refrigerant name, and the coating color of the cylinder protector is pink, which is the color stipulated by ARI of the U.S.
- Cylinders equipped with a siphon tube are available to allow the cylinder to stand upright for liquid refrigerant charging.

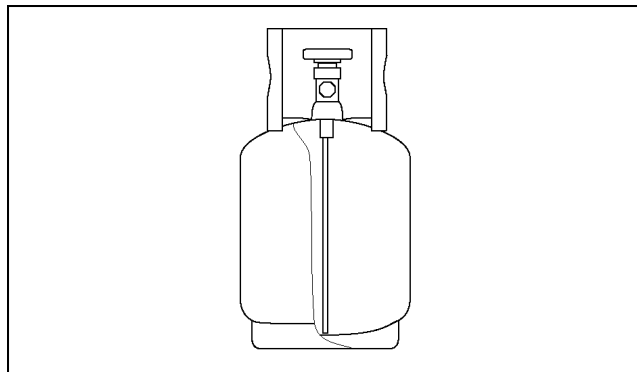


Fig. 8 Refrigerant cylinders

10. Charging orifice and packing for refrigerant cylinders

- The charging orifice must match the size of the charging hose fitting (1/2 UNF, 20 threads).
- The packing must also be made of an HFC-resistant material.

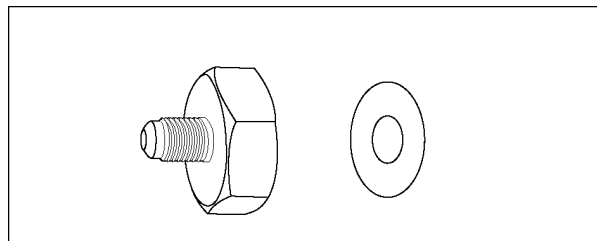


Fig. 9 Charging orifice and packing

10.2.3. R410A Tools Which Are Usable for R22 Models

Table 9 R410A tools which are usable for R22 models

	R410A tools	Usable for R22 models
(1)	Copper tube gauge for clearance adjustment	OK
(2)	Flaring tool (clutch type)	OK
(3)	Manifold gauge	NG
(4)	Charging hose	NG
(5)	Vacuum pump adaptor	OK
(6)	Electric gas leak detector for HFC refrigerant	NG
(7)	Electronic scale for refrigerant charging	OK
(8)	Refrigerant cylinder	NG
(9)	Charging orifice and packing for refrigerant cylinder	NG

10.3. REFRIGERANT PIPING WORK

10.3.1. Piping Materials

It is recommended that you use copper and copper alloy jointless pipes with a maximum oil adherence of 40 mg/10 m. Do not use pipes that are crushed, deformed, or discolored (especially the inside surface). If these inferior pipes are used, impurities may clog the expansion valves or capillaries.

Because the pressure of ACs using R410A is higher than those using R22, it is essential that you select materials that are appropriate for these standards.

The thickness of the copper tubing used for R410A is shown in Table 10. Please be aware that tubing with a thickness of only 0.7 mm is also available on the market, but this should never be used.

Table 10 Copper tube thickness (mm)

Soft pipe		Thickness (mm)	
Nominal diameter	Outside diameter (mm)	R410A	(Reference)R22
1/4	6.35	0.80	0.80
3/8	9.52	0.80	0.80
1/2	12.7	0.80	0.80

10.3.2. Processing and Connecting Piping Materials

When working with refrigerant piping, the following points must be carefully observed: no moisture or dust must be allowed to enter the piping, and there must be no refrigerant leaks.

1. Procedure and precautions for flaring work

a. Cut the pipe

Use a pipe cutter, and cut slowly so the pipe will not be deformed.

b. Remove burrs and clean shavings from the cut surface

If the shape of the pipe end is poor after removing burrs, or if shavings adhere to the flared area, it may lead to refrigerant leaks. To prevent this, turn the cut surface downward and remove burrs, then clean the surface, carefully.

c. Insert the flare nut (be sure to use the same nut that is used on the AC unit)

d. Flaring

Check the clamp bar and the cleanliness of the copper pipe. Be sure to use the clamp bar to do the flaring with accuracy. Use either an R410A flaring tool, or a conventional flaring tool. Flaring tools come in different sizes, so be sure to check the size before using. When using a conventional flaring tool, use the copper pipe gauge for clearance adjustment, etc., to ensure the correct A dimension (see Fig. 10)

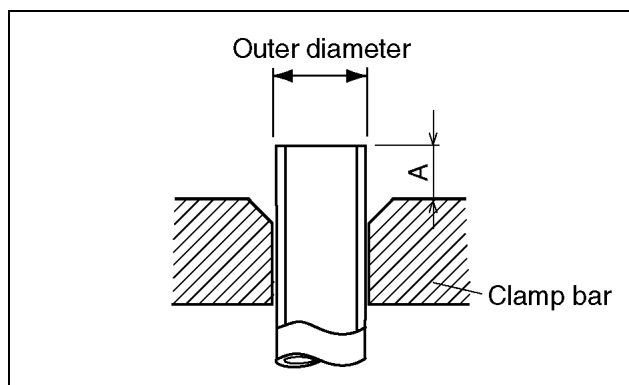


Fig. 10 Flaring dimensions

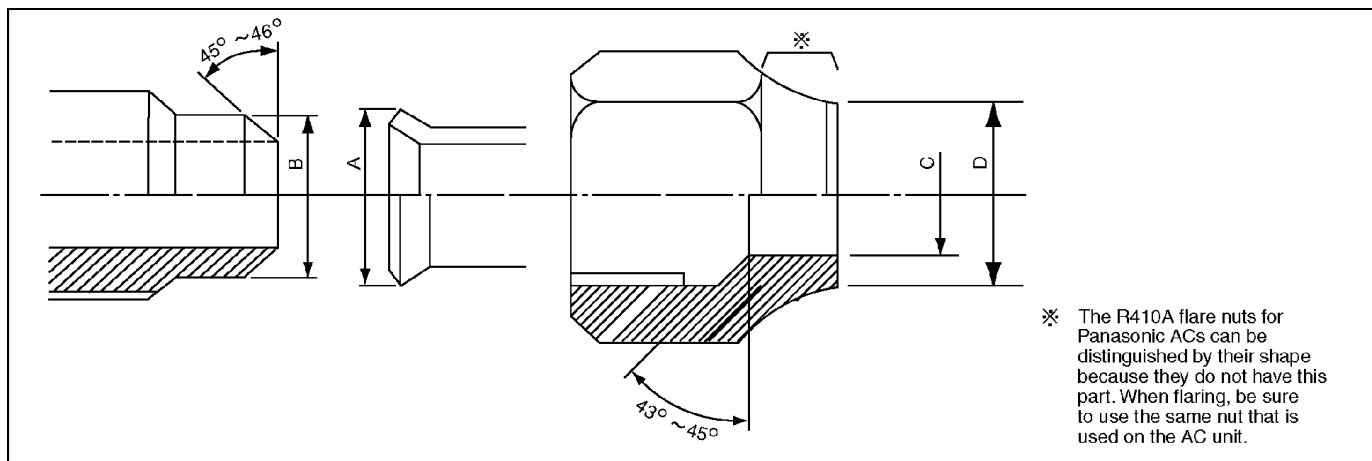


Fig. 11 Relation between the flare nut structure and flaring tool end

Table 11 R410A flaring dimensions

Nominal diameter	Outside diameter (mm)	Wall thickness (mm)	A(mm)		
			R410A flaring tool, clutch type	Conventional flaring tool	
				Clutch type	Wing-nut type
1/4	6.35	0.8	0 - 0.5	1.0 - 1.5	1.5 - 2.0
3/8	9.52	0.8	0 - 0.5	1.0 - 1.5	1.5 - 2.0
1/2	12.70	0.8	0 - 0.5	1.0 - 1.5	2.0 - 2.5

Table 12 R22 flaring dimensions

Nominal diameter	Outside diameter (mm)	Wall thickness (mm)	A(mm)		
			R410A flaring tool, clutch type	Conventional flaring tool	
				Clutch type	Wing-nut type
1/4	6.35	0.8	0 - 0.5	0.5 - 1.0	1.0 - 1.5
3/8	9.52	0.8	0 - 0.5	0.5 - 1.0	1.0 - 1.5
1/2	12.70	0.8	0 - 0.5	0.5 - 1.0	1.5 - 2.0

Table 13 R410A flare and flare nut dimensions Unit:mm

Nominal diameter	Outside diameter (mm)	Wall thickness (mm)	A +0, -0.4	B dimension	C dimension	D dimension	Flare nut width
1/4	6.35	0.8	9.1	9.2	6.5	13	17
3/8	9.52	0.8	13.2	13.5	9.7	20	22
1/2	12.70	0.8	16.6	16.0	12.9	23	26

Table 14 R22 flare and flare nut dimensions Unit:mm

Nominal diameter	Outside diameter (mm)	Wall thickness (mm)	A +0, -0.4	B dimension	C dimension	D dimension	Flare nut width
1/4	6.35	0.8	9.0	9.2	6.5	13	17
3/8	9.52	0.8	13.0	13.5	9.7	20	22
1/2	12.70	0.8	16.2	16.0	12.9	20	24

2. Procedure and precautions for flare connection

- Check to make sure there are no scratches, dust, etc., on the flare and union.
- Align the flared surface with the axial center of the union.
- Use a torque wrench, and tighten to the specified torque. The tightening torque for R410A is the same as the conventional torque value for R22. Be careful, because if the torque is too weak, it may lead to a gas leak. If it is too strong, it may split the flare nut or make it impossible to remove the flare nut.

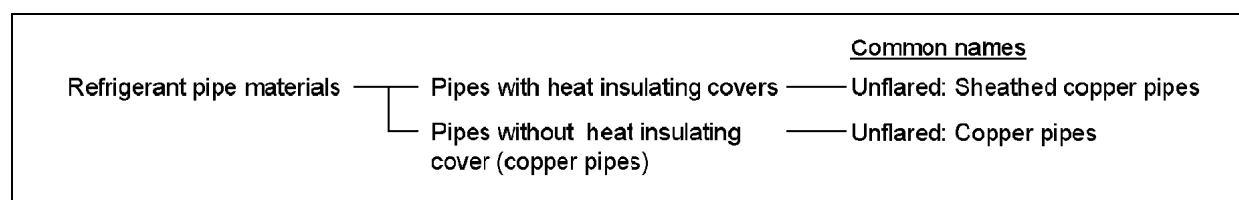
Table 15 R410A tightening torque

Nominal diameter	Outside diameter (mm)	Tightening torque N·m (kgf·cm)	Torque wrench tightening torque N·m (kgf·cm)
1/4	6.35	14 - 18 (140 - 180)	18 (180)
3/8	9.52	33 - 42 (330 - 420)	42 (420)
1/2	12.70	55 (550)	55 (550)

10.3.3. Storing and Managing Piping Materials

1. Types of piping and their storage

The following is a general classification of the refrigerant pipe materials used for ACs.



Because the gas pressure of R410A is approximately 1.6 times as high as that of R22, copper pipes with the thickness shown in Table 10, and with minimal impurities must be used. Care must also be taken during storage to ensure that pipes are not crushed, deformed, or scratched, and that no dust, moisture or other substance enters the pipe interior. When storing sheathed copper pipes or plain copper pipes, seal the openings by pinching or taping them securely.

2. Markings and management

a. Sheathed copper pipes and copper-element pipes

When using these pipes, check to make sure that they are the stipulated thickness. For flare nuts, be sure to use the same nut that is used on the AC unit.

b. Copper pipes

Use only copper pipes with the thickness given in table 10, and with minimal impurities. Because the surface of the pipe is exposed, you should take special care, and also take measures such as marking the pipes to make sure they are easily distinguished from other piping materials, to prevent mistaken use.

3. Precautions during refrigerant piping work

Take the following precautions on-site when connecting pipes. (Keep in mind that the need to control the entry of moisture and dust is even more important than in conventional piping.)

- Keep the open ends of all pipes sealed until connection with the AC equipment is complete.

- b. Take special care when doing piping work on rainy days. The entering of moisture will degrade the refrigerating machine oil, and lead to malfunctions in the equipment.
- c. Complete all pipe connections in as short a time as possible. If the pipe must be left standing for a long time after removing the seal, it must be thoroughly purged with nitrogen, or dried with a vacuum pump.

10.4. INSTALLATION, TRANSFERRING SERVICING

10.4.1. Inspecting Gas Leaks with a Vacuum Pump for New Installations (Using New Refrigerant Piping))

1. From the viewpoint of protecting the global environment, please do not release refrigerant into the atmosphere.
 - a. Connect the projecting side (pin-pushing side) of the charging hose for the manifold gauge to the service port of the 3-way valve. (1)
 - b. Fully open the handle Lo of the manifold gauge and run the vacuum pump. (2) (If the needle of the low-pressure gauge instantly reaches vacuum, re-check step a.)
 - c. Continue the vacuum process for at least 15 minutes, then check to make sure the low-pressure gauge has reached -0.1 MPa (-76 cmHg). Once the vacuum process has finished, fully close the handle Lo of the manifold gauge and stop the vacuum pump operation, then remove the charging hose that is connected to the vacuum pump adaptor. (Leave the unit in that condition for 1-2 minutes, and make sure that the needle of the manifold gauge does not return.) (2) and (3)
 - d. Turn the valve stem of the 2-way valve 90° counter-clockwise to open it, then, after 10 seconds, close it and inspect for a gas leak (4)
 - e. Remove the charging hose from the 3-way valve service port, then open both the 2-way valve and 3-way valve. (1) (4) (Turn the valve stem in the counter-clockwise direction until it gently makes contact. Do not turn it forcefully.)
 - f. Tighten the service port cap with a torque wrench (18 N·m (1.8 kgf·m)). (5) Then tighten the 2-way valve and 3-way valve caps with a torque wrench (42 N·m (4.2 kgf·m)) or (55 N·m (5.5 kgf·m)). (6)
 - g. After attaching each of the caps, inspect for a gas leak around the cap area. (5) (6)

Precautions

- Be sure to read the instructions for the vacuum pump, vacuum pump adaptor and manifold gauge prior to use, and follow the instructions carefully.
- Make sure that the vacuum pump is filled with oil up to the designated line on the oil gauge.
- The gas pressure back flow prevention valve on the charging hose is generally open during use. When you are removing the charging hose from the service port, it will come off more easily if you close this valve.

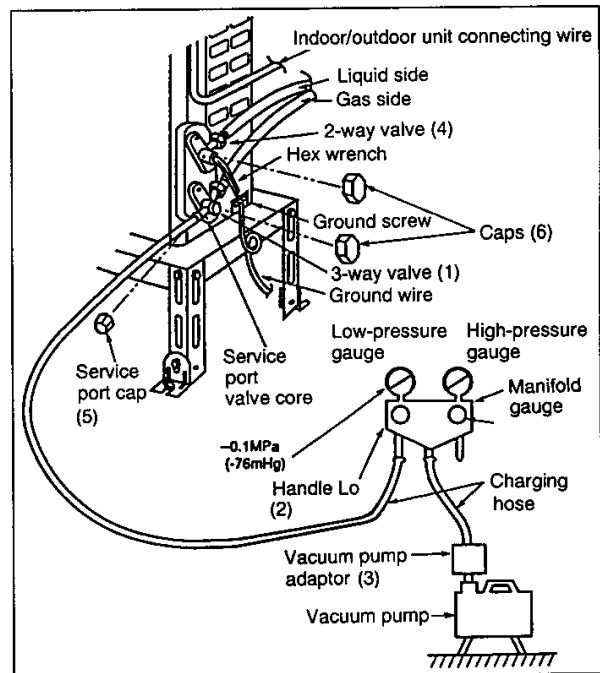


Fig. 12 Vacuum pump air purging configuration

10.4.2. Transferring (Using New Refrigerant Piping)

1. Removing the unit
 - a. Collecting the refrigerant into the outdoor unit by pumping down
 The refrigerant can be collected into the outdoor unit (pumping down) by pressing the TEST RUN button, even when the temperature of the room is low.
 - Check to make sure that the valve stems of the 2-way valve and 3-way valve have been opened by turning them counter-clockwise. (Remove the valve stem caps and check to see that the valve stems are fully opened position. Always use a hex wrench (with 4-mm opposing sides) to operate the valve stems.)

- Press the TEST RUN button on the indoor unit, and allow preliminary operation for 5-6 minutes. (TEST RUN mode)
- After stopping the operation, let the unit sit for about 3 minutes, then close the 2-way valve by turning the valve stem in the clockwise direction.
- Press the TEST RUN button on the indoor unit again, and after 2-3 minutes of operation, turn the valve stem of the 3-way valve quickly in the clockwise direction to close it, then stop the operation.
- Tighten the caps of the 2-way valve and 3-way valve to the stipulated torque.
- Remove the connection pipes (liquid side and gas side).

b. Removing the indoor and outdoor units

- Disconnect the pipes and connecting electric cables from between the indoor and outdoor units
- Put capped flare nuts onto all of the pipe connections of the indoor and outdoor units, to make sure no dust or other foreign matter enters.
- Remove the indoor and outdoor units.

2. Installing the unit

Install the unit using new refrigerant piping. Follow the instructions in section 4.1 to evacuate the pipes connecting the indoor and outdoor units, and the pipes of the indoor unit, and check for gas leaks.

10.4.3. AC Units Replacement (Using Existing Refrigerant Piping)

When replacing an R410A AC unit with another R410A AC unit, you should re-flare the refrigerant piping. Even though the replacement AC unit uses the R410A, problems occur when, for example, either the AC unit maker or the refrigerating machine oil is different.

When replacing an R22 AC unit with an R410A AC unit, the following checks and cleaning procedures are necessary but are difficult to do because of the chemical characteristics of the refrigerating machine oil (as described in items c) and d) of section 10.1.1.(2)). In this case, you should use new refrigerant piping rather than the existing piping.

1. Piping check

Because of the different pressure characteristics of R22 and R410A, the design pressure for the equipment is 1.6 times different. The wall thickness of the piping must comply with that shown in Table 10, but this is not easy to check. Also, even if the thickness is correct, there may be flattened or bent portions midway through the piping due to sharp curves. Buried sections of the piping also cannot be checked.

2. Pipe cleaning

A large quantity of refrigerating machine oil (mineral oil) adheres to existing pipes due to the refrigeration cycle circulation. If the pipes are used just as they are for the R410A cycle, the capacity will be lowered due to the incompatibility of this oil with the R410A, or irregularities may occur in the refrigeration cycle. For this reason, the piping must be thoroughly cleaned, but this is difficult with the present technology.

10.4.4. Refrigerant Compatibility (Using R410A Refrigerant in R22 ACs and Vice Versa)

Do not operate an existing R22 AC with the new R410A refrigerant. Doing so would result in improper functioning of the equipment or malfunction, and might lead to a major accident such as an explosion in the refrigeration cycle. Similarly, do not operate an R410A AC with R22 refrigerant. The chemical reaction between the refrigerating machine oil used in R410A ACs and the chlorine that is contained in R22 would cause the refrigerating machine oil to degrade and lead to malfunction.

10.4.5. Recharging Refrigerant During Servicing

When recharging is necessary, insert the specified amount of new refrigerant in accordance with the following procedure.

1. Connect the charging hose to the service port of the outdoor unit.
2. Connect the charging hose to the vacuum pump adaptor. At this time, fully open the 2-way valve and 3-way valve.
3. Fully open the handle Lo of the manifold gauge, turn on the power of the vacuum pump and continue the vacuum process for at least one hour.
4. Confirm that the low pressure gauge shows a reading of -0.1 Mpa (-76 cmHg), then fully close the handle Lo, and turn off the vacuum pump. Wait for 1-2 minutes, then check to make sure that the needle of the Low pressure gauge has not returned. See Fig.13 for the remaining steps of this procedure.
5. Set the refrigerant cylinder onto the electronic scale, then connect the hose to the cylinder and to the connection port for the electronic scale. (1)(2)

Precaution:

Be sure to set up the cylinder for liquid charging. If you use a cylinder equipped with a siphon tube, you can charge the liquid without having to turn the cylinder around

6. Remove the charging hose of the manifold gauge from the vacuum pump adaptor, and connect it to the connection port of the electronic scale. (2)(3)

7. Open the valve of the refrigerant cylinder, then open the charging valve slightly and close it. Next, press the check valve of the manifold gauge and purge the air. (2)(4) (Watch the liquid refrigerant closely at this point.)
8. After adjusting the electronic scale to zero, open the charging valve, then open the valve Lo of the manifold gauge and charge with the liquid refrigerant. (2)(5) (Be sure to read the operating instructions for the electronic scale.)
9. If you cannot charge the stipulated amount, operate the unit in the cooling mode while charging a little of the liquid at a time (about 150 g/time as a guideline). If the charging amount is insufficient from one operation, wait about one minute, then use the same procedure to do the liquid charging again.

Precaution:

Never use the gas side to allow a larger amount of liquid refrigerant to be charged while operating the unit.

10. Close the charging valve, and after charging the liquid refrigerant inside the charging hose, fully close the valve Lo of the manifold gauge, and stop the operation of the unit. (2)(5)
11. Quickly remove the charging hose from the service port. (6) If you stop midway through, the refrigerant that is in the cycle will be discharged.
12. After putting on the caps for the service port and operating valve, inspect around the caps for a gas leak. (6)(7)

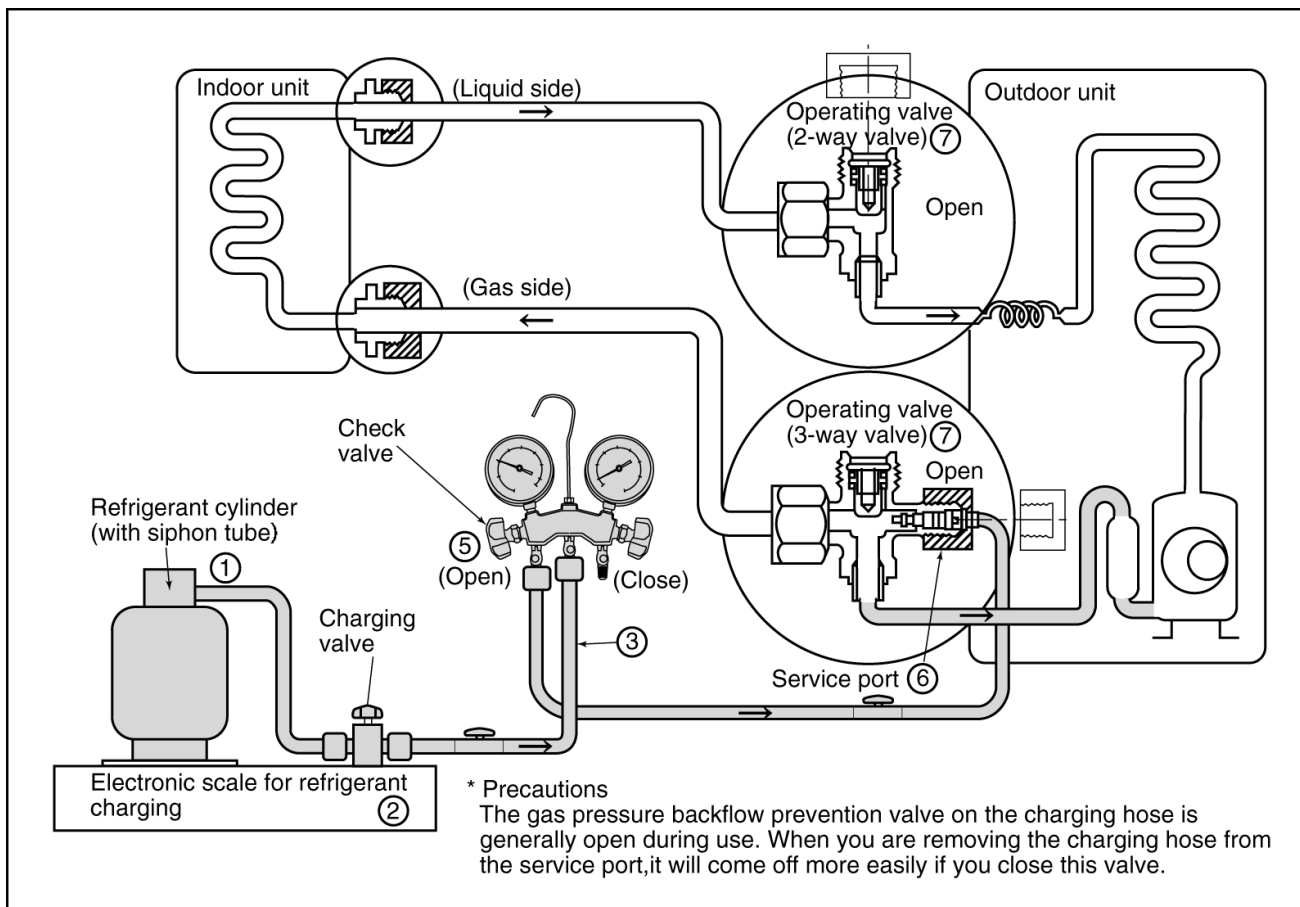


Fig. 13 Re-charging refrigerant

10.4.6. Brazing

As brazing requires sophisticated techniques and experiences, it must be performed by a qualified person.

In order to prevent the oxide film from occurring in the pipe interior during brazing, it is effective to proceed with brazing while letting dry nitrogen gas (N_2) flow.

<Brazing Method for Preventing Oxidation>

1. Attach a reducing valve to the nitrogen gas cylinder.
2. Attach a reducing valve to the nitrogen gas cylinder.
3. Apply a seal onto the clearance between the piping and inserted pipe for the nitrogen gas in order to prevent the nitrogen gas from flowing backward.
4. When the nitrogen gas is flowing, be sure to keep the piping end open.
5. Adjust the flow rate of nitrogen gas so that it is lower than $0.05 \text{ m}^3/\text{h}$, or 0.02 MPa (0.2 kgf/cm^2) by means of the reducing valve.

6. After taking the steps above, keep the nitrogen gas flowing until the piping cools down to a certain extent (i.e. temperature at which pipes are touchable with finger).
7. Completely remove the flux after brazing.

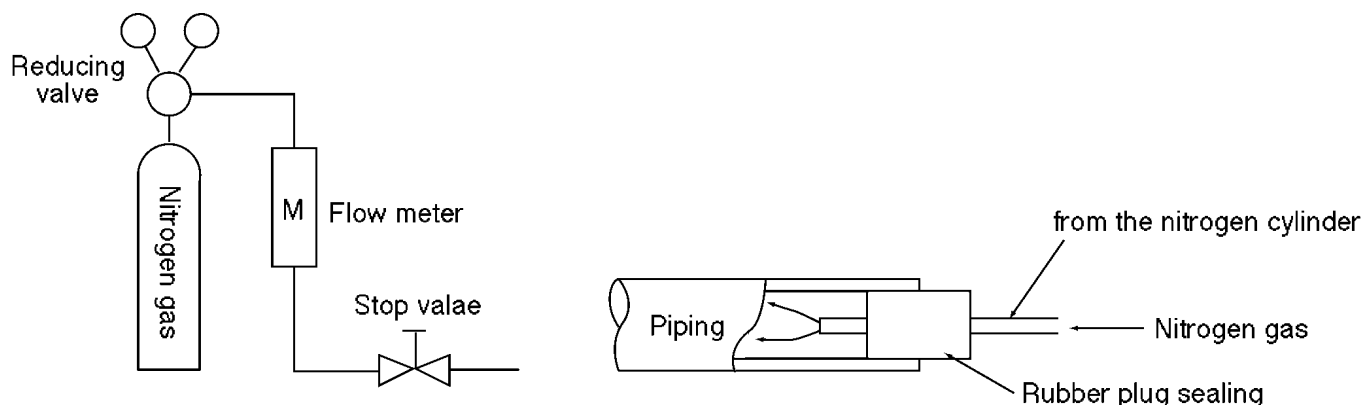


Fig. 14 Prevention of Oxidation during Brazing

* Cautions during brazing

1. General Cautions

- a. The brazing strength should be high as required.
- b. After operation, airtightness should be kept under pressurized condition.
- c. During brazing do not allow component materials to become damaged due to overheating.
- d. The refrigerant pipe work should not become blocked with scale or flux.
- e. The brazed part should not restrict the flow in the refrigerant circuit.
- f. No corrosion should occur from the brazed part.

2. Prevention of Overheating

Due to heating, the interior and exterior surfaces of treated metal may oxidize. Especially, when the interior of the refrigerant circuit oxidizes due to overheating, scale occurs and stays in the circuit as dust, thus exerting a fatally adverse effect. So, make brazing at adequate brazing temperature and with a minimum of heating area.

3. Overheating Protection

In order to prevent components near the brazed part from overheating damage or quality deterioration due to flame or heat, take adequate steps for protection such as (1) by shielding with a metal plate, (2) by using a wet cloth, and (3) by means of heat absorbent.

4. Movement during Brazing

Eliminate all vibration during brazing to protect brazed joints from cracking and breakage.

5. Oxidation Preventative

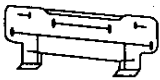
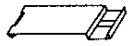
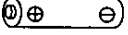
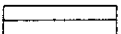
In order to improve the brazing efficiency, various types of antioxidant are available on the market. However, the constituents of these are widely varied, and some are anticipated to corrode the piping materials, or adversely affect HFC refrigerant, lubricating oil, etc. Exercise care when using an oxidation preventive.

10.4.7. Servicing Tips

The drier must also be replaced whenever replacing the refrigerant cycle parts. Replace the refrigerant cycle parts first before replacing the drier. The drier is supplied in a vacuum pack. Perform brazing immediately after opening the vacuum pack, and then start the vacuum within two hours. In addition, the drier also needs to be replaced when the refrigerant has leaked completely.

11 Installation Information

11.1. ATTACHED ACCESSORIES

No.	Accessories part	Qty.	No.	Accessories part	Qty.
1	Installation plate 	1	6	Drain elbow 	1
2	Installation plate fixing screw 	6	7	Clamping cover of piping 	1
3	Remote control 	1	8	Vinyl tape 	3
4	Battery 	2	9	Vinyl tape 	1
5	Air purifying filter 	2			

11.2. SELECT THE BEST LOCATION

11.2.1. INDOOR UNIT

- There should not be any heat source or steam near the unit.
- There should not be any obstacles blocking the air circulation.
- A place where air circulation in the room is good.
- A place where drainage can be easily done.
- A place where noise prevention is taken into consideration.
- Do not install the unit near the door way.
- Ensure the spaces indicated by arrows from the wall, ceiling or other obstacles.
- Indoor unit of this room air conditioner shall be installed on the wall in a height of at least 2.3m.

11.2.2. OUTDOOR UNIT

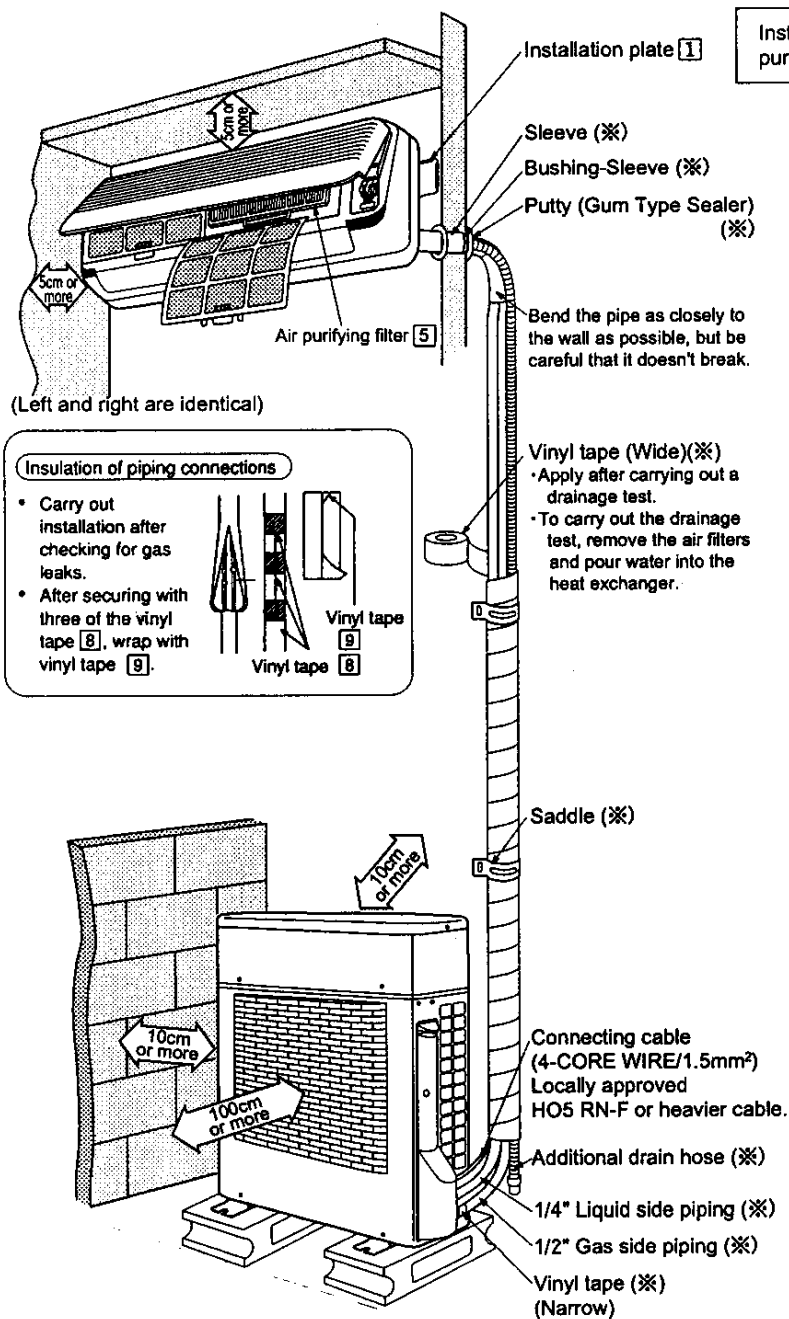
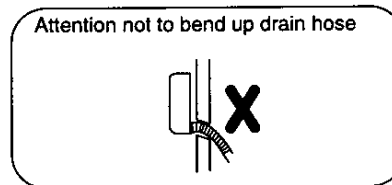
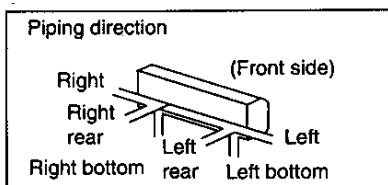
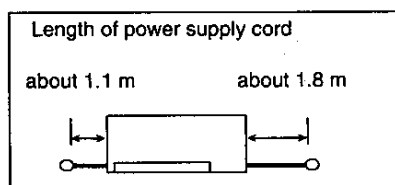
- If an awning is built over the unit to prevent direct sunlight or rain, be careful that heat radiation from the condenser is not obstructed.
- There should not be any animal or plant which could be affected by hot discharged air.
- Keep the spaces indicated by arrows from wall, ceiling, fence or other obstacles.
- Do not place any obstacles which may cause a short circuit of the discharged air.

11.2.3. PIPING LENGTH

Piping size		Common Length (m)	Max. Elevation (m)	Max. Piping Length (m)
Gas	Liquid			
1/2"	1/4"	7	5	10

The additional gas charge is unnecessary up to the Max. piping length.

11.3. INDOOR/OUTDOOR UNIT INSTALLATION DIAGRAM



This illustration is for explanation purposes only.
The indoor unit will actually face a different way.
At least three directions of air discharge and inlet shall be opened to prevent any stagnant air.

12 Servicing Information

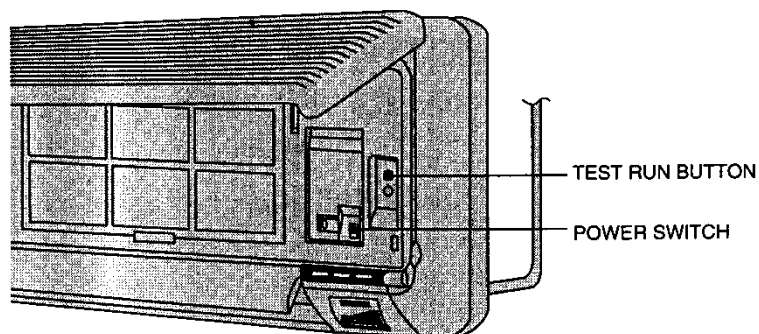
12.1. TROUBLESHOOTING

12.1.1. RATED FREQUENCY OPERATION

During troubleshooting and servicing, rated compressor operating frequency must be obtained in order to check the specification and technical data. Below are the methods used to obtain rated compressor operating specification.

1. Cooling (73Hz)

- a. Press the Test Run button on the indoor unit. The air conditioner starts operation at Cooling rated frequency.



- b. Short the service terminal (CN-S) of the outdoor printed circuit board. The air conditioner starts operation at Cooling rated frequency.

2. Heating (92Hz)

Keep pressing the Test Run button, switch off and on the Power Switch, then release the Test Run button. The air conditioner starts operation at Heating rated frequency.

12.1.2. TROUBLESHOOTING AIR CONDITIONER

Refrigeration cycle system

In order to diagnose malfunctions, make sure that there are no electrical problems before inspecting the refrigeration cycle. Such problems include insufficient insulation, problem with the power source, malfunction of a compressor or a fan.

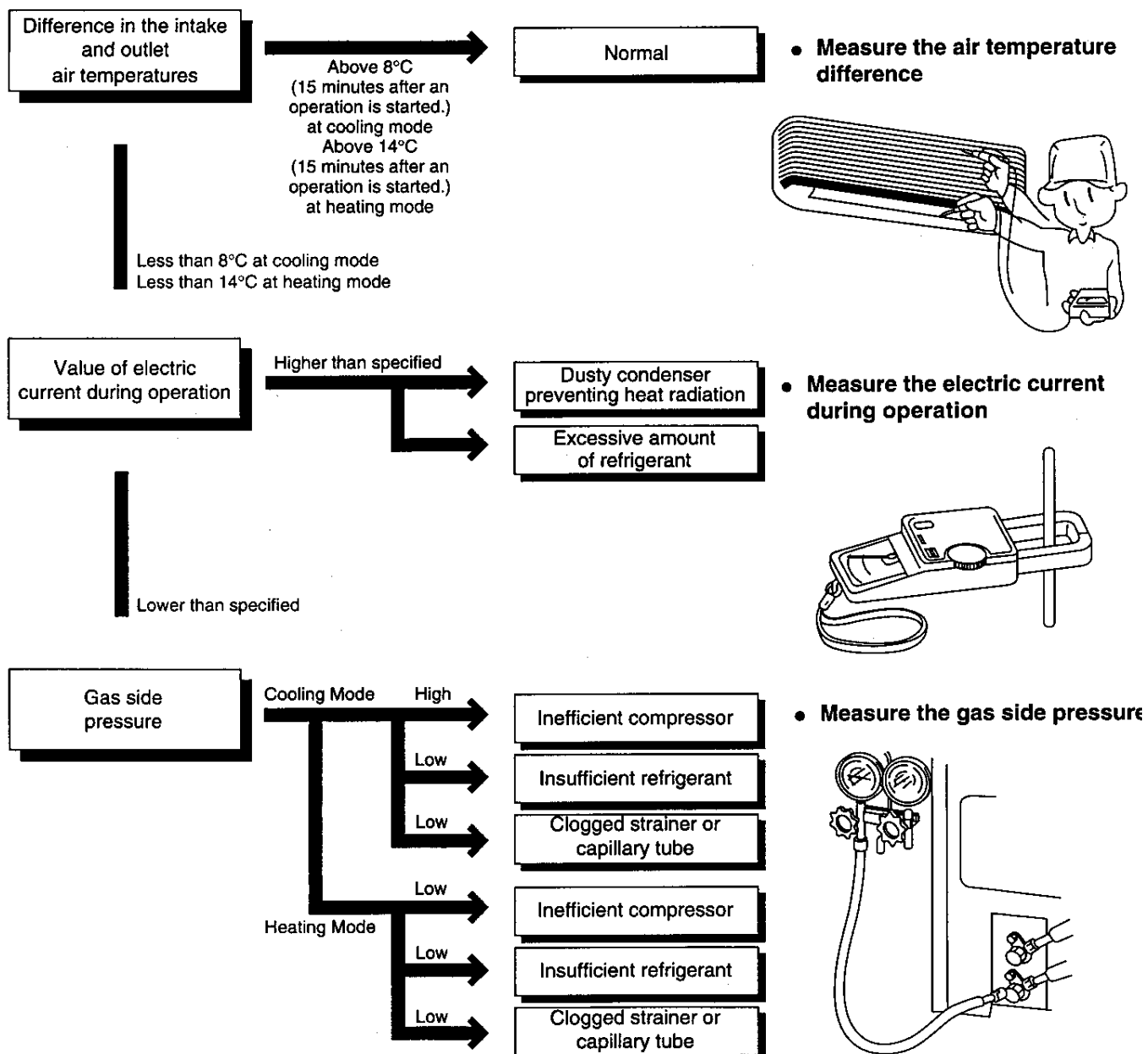
The normal outlet air temperature and pressure of the refrigeration cycle depends on various conditions; the standard values for them are shown in the following table.

	Gas pressure MPa (kg/cm ² G)	Outlet air temperature (°C)
Cooling Mode	0.9~1.2 (9~12)	12~16
Heating Mode	2.3~2.9 (23~29)	36~48

Normal Pressure and Outlet Air Temperature (Standard)

Condition:

- Indoor fan speed; High
- Outdoor temperature is 35 °C at cooling mode and 7 °C at heating mode.
- Compressor operates at rated frequency.



Relationship between the condition of the air conditioner and pressure and electric current

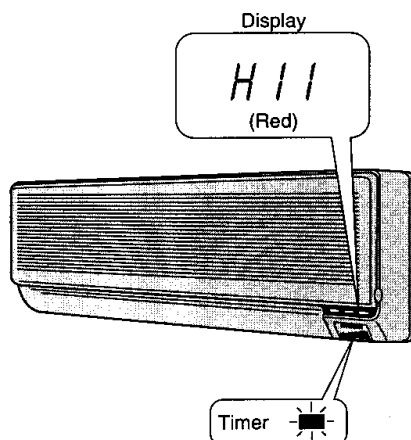
Condition of the air conditioner	Cooling Mode			Heating Mode		
	Low Pressure	High Pressure	Electric current during operation	Low Pressure	High Pressure	Electric current during operation
Insufficient refrigerant (gas leakage)	↘	↘	↘	↘	↘	↘
Clogged capillary tube or Strainer	↘	↘	↘	↗	↗	↗
Short circuit in the indoor unit	↘	↘	↘	↗	↗	↗
Heat radiation deficiency of the outdoor unit	↗	↗	↗	↘	↘	↘
Inefficient compression	↗	↗	↗	↘	↘	↘

- Carry out the measurements of pressure, electric current, and temperature fifteen minutes after an operation is stated.

12.2. SELF DIAGNOSIS DISPLAY

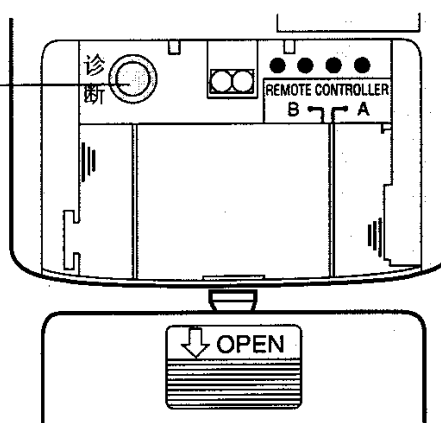
The diagnostic display can be seen on the receiver of the Front Grille.

- When an abnormality occurs, the unit automatically stops, and the TIMER LED blinks to indicate a malfunction. At the same time, the type of abnormality will be indicated on the receiver as shown in the diagram below. Providing this information reduces the time spent in diagnosing procedures.



- The diagnostic display vanishes when the power is turned off.
- When power is re-supplied and the Diagnostic Button on the Remote Control is pressed, the type of the previous abnormality and the protection control works will be displayed on the receiver for approximately 10 seconds.

Diagnostic Button
To be used by the service technician only.



- By starting cooling operation using TEST RUN button and press the Diagnostic Button at the remote control, the previous abnormalities are deleted.
- Depending on the type of abnormality, you may be able to override the abnormality and use temporary operation (for abnormalities indicated by "●" mark in the table below).

Use the remote control to select cooling or heating operation mode and press OFF/ON button. At this moment, four short beeps "bip.bip.bip.bip" will sound and TIMER LED will blink.

Diagnosis display	Abnormality/Protection control	Abnormality Judgement	Temporary operation	Primary location to verify
H11	Indoor/outdoor abnormal communication	1 min after starting operation	-	- Internal/external cable connections - Indoor/Outdoor PCB
H14	Indoor intake air temperature sensor abnormality		-	Intake air temperature sensor (defective or disconnected)
H15	Outdoor compressor temperature sensor abnormality		-	Compressor temperature sensor (defective or disconnected)
H16	Outdoor Current Transformer open circuit		-	- Outdoor PCB - Power transistor module
H19	Indoor fan motor mechanism lock		-	- Indoor PCB - Fan motor
H23	Indoor heat exchanger temperature sensor abnormality		● (Cooling only)	Heat exchanger temperature sensor (defective or disconnected)
H27	Outdoor air temperature sensor abnormality		●	Outdoor temperature sensor (defective or disconnected)
H28	Outdoor heat exchanger temperature sensor abnormality		●	Outdoor heat exchanger temperature sensor (defective or disconnected)
H98	Indoor high pressure protection		-	- Air filter dirty - Air circulation short circuit
H99	Indoor heat exchanger anti-freezing protection		-	- Insufficient refrigerant - Air filter dirty
F11	Cooling/Heating cycle changeover abnormality	4 times occurrence within 40 minutes	-	4-way valve - V-coil
F91	Refrigeration cycle abnormality	2 times occurrence within 30 minutes	-	No refrigerant (3-way valve is closed)
F93	Compressor motor rotation abnormality	2 times occurrence within 30 minutes	-	- Compressor - Outdoor P.C.Board
F95	Over heating protections	3 times occurrence within 30 minutes	-	Pipe sensor
F96	Outdoor power transistor module overheating protection	4 times occurrence within 30 minutes	-	- Excess refrigerant - Improper heat radiation - Power transistor
F97	Outdoor compressor overheating protection	4 times occurrence within 20 minutes	-	- Insufficient refrigerant - Compressor
F98	Total running current protection	3 times occurrence within 30 minutes	-	- Excess refrigerant - Improper heat radiation
F99	Outdoor Direct Current (DC) peak detection	2 times occurrence continuously	-	- Outdoor PCB - Power transistor - Compressor

1. Current Transformer Defective

When the Current Transformer (CT) is an open circuit, total running current is less than 1.51 A and the indicated frequency is 73Hz or above. After 3 minutes of operation, the abnormality signal is sent from outdoor to indoor and [H16] is displayed.

2. 4 Way Valve Defective

a. Heating Operation (except Deice)

After 4 minutes of operation, the indoor heat exchanger temperature is lower than 5°C. The operation stops and restarts after 3 minutes. If this phenomenon occurs for 4 times within 40 minutes, [F11] is displayed.

b. Cooling Operation

After 4 minutes of operation, indoor heat exchanger temperature is higher than 45°C. The operation stops and restarts after 3 minutes. If this phenomenon occurs for 4 times within 40 minutes, [F11] is displayed.

The abnormality judgement is not carried out, in the following conditions:

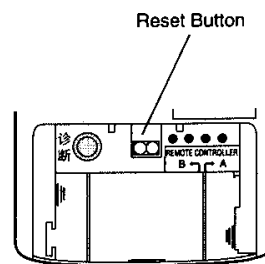
- deice operation
- 2 minutes after deice operation
- hot start
- 2 minutes after hot start
- 3 minutes after heating and cooling/soft dry mode changeover

12.3. REMOTE CONTROL

12.3.1. Remote Control reset

When the batteries are inserted for the first time, or the batteries are replaced, all the indications will blink and the remote control might not work.

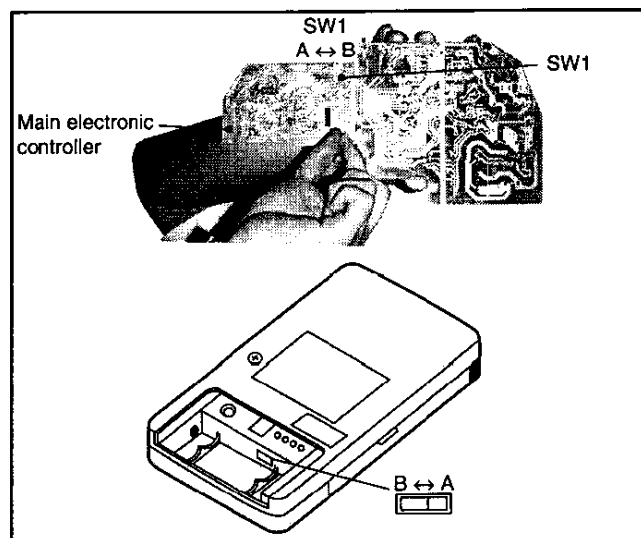
If this happens, remove the back cover of the remote control and you will find a resetting terminal, and by shorting it with a minus screwdriver, it will return to normal.



12.3.2. Changing the wireless remote control transmission code

When two indoor units are installed in the same room, in order to prevent operating errors caused by using two remote controls, set up the remote control [B ←→ A] switch (SW1).

The unit is set to A when it is shipped.

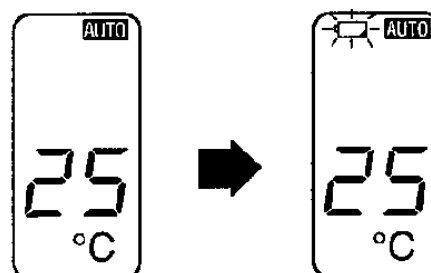


By adding a jumper wire to the remote control side and shorting the Jx at the indoor printed circuit board, it is possible to select 4 types of transmission codes including one at time of delivery condition (1).

	Remote control		Indoor printed circuit board		Note
	Switch SW B ←→ A	J-B	Switch SW1	Jx	
1	A	---	A	---	At product delivery
2	B	---	B	---	
3	A	Jumper wire	A	Shorted	
4	B	Jumper wire	B	Shorted	

12.3.3. Remote Control Batteries

The batteries can be used for one year approximately. When the batteries become close to 2.6V, the battery mark  flashes. Replace the batteries immediately. Otherwise, the remote control display will not appear.



12.4. DISASSEMBLY OF PARTS

12.4.1. Inspection points for the Indoor Electronic Controller

The Electronic Controller, a signal Receiver, a power monitor, as indicator and a self diagnosis indicator display can be seen by removing the Front Grille and Control Board Cover, as shown in the Fig. 1.

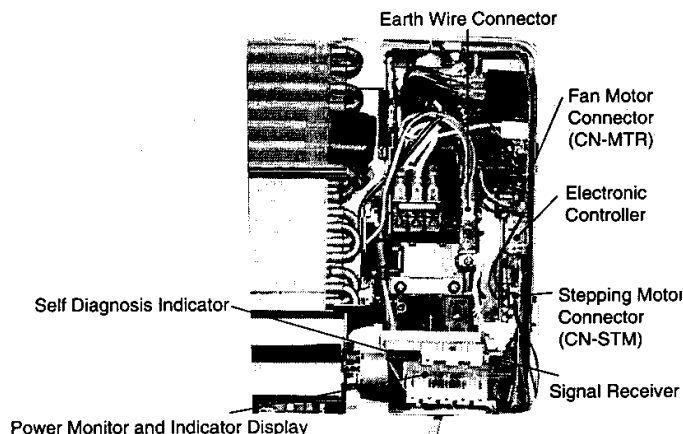


Fig. 1

12.4.2. Indoor Fan Motor removal procedure

1. Remove the connector CN-MTR (GREEN) of Fan Motor and connector CN-STM (WHITE) of stepping motor from the electronic controller. Release the earth wire (YELLOW-GREEN) from the control board and sensors from its holders. (Refer Fig. 1)

2. Remove the Control Board

The Control Board can be removed by releasing the top, left and right tabs shown in Fig. 2, 3, 4.

Releasing the 2 right tabs by pressing down the top tab and pushing up the bottom tab.

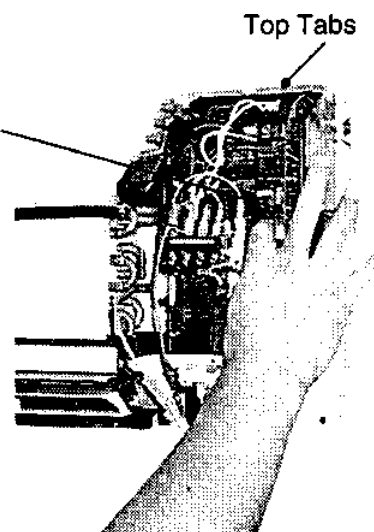


Fig. 2

Releasing the 2 left tabs by pressing down the top tab and pushing up the bottom tabs

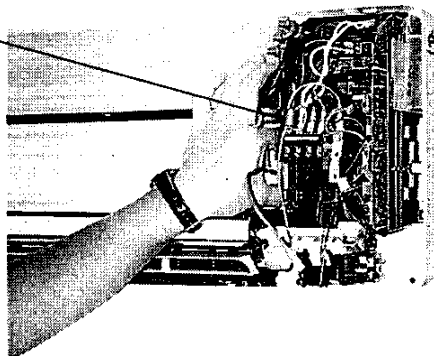


Fig. 3

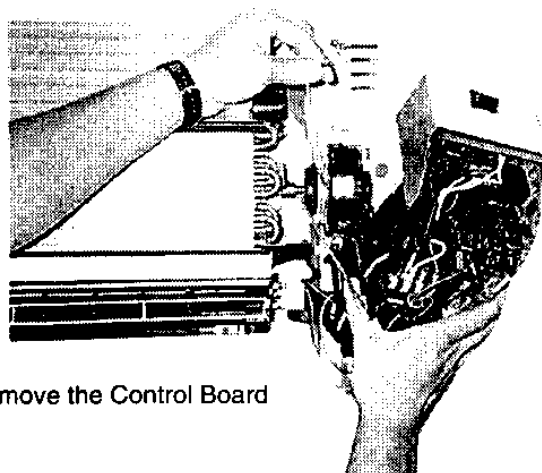


Fig. 4

3. Remove the Fan Motor

Loosen the Fan Motor securing screw at the junction with Cross Flow Fan. (Fig. 5)

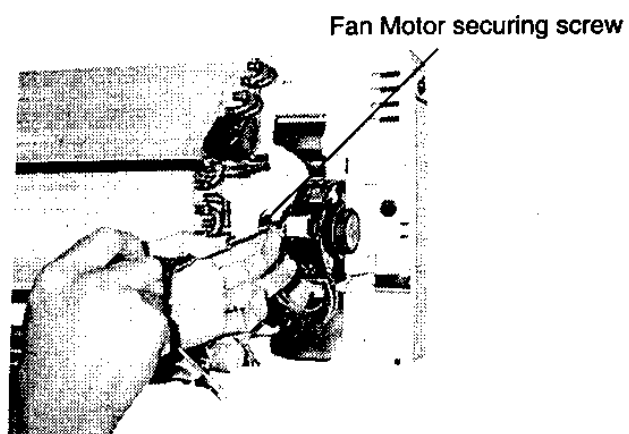


Fig. 5

Remove the particular piece and the Fan Motor can be taken off as shown in Fig. 6 and 7.

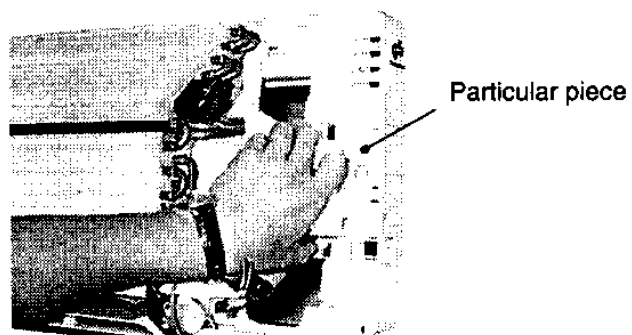
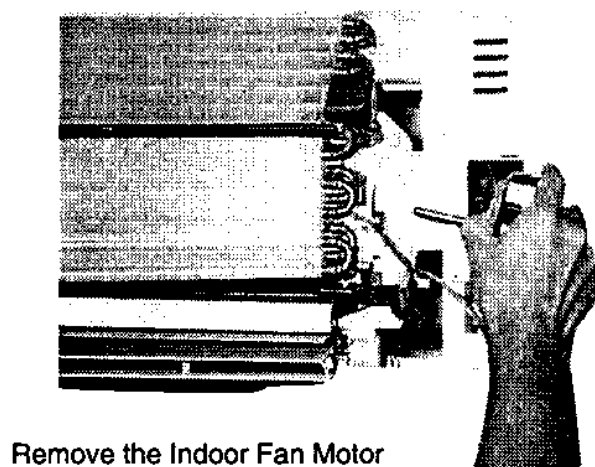


Fig. 6



Remove the Indoor Fan Motor

Fig. 7

4. To fix the Indoor Fan Motor, ensure that the Fan Motor securing screw is positioned at the rear end and the Fan Motor lead wire is positioned parallel to the Fan Motor. (Fig. 8)

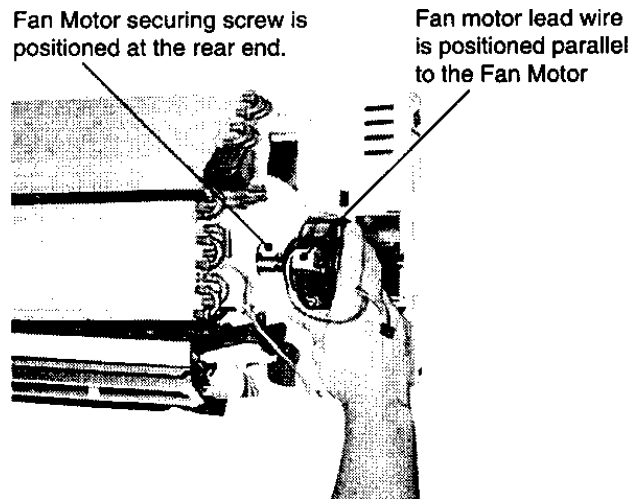


Fig. 8

12.4.3. Cross Flow Fan removal procedure

1. Remove the Indoor Fan Motor.
(Refer to the removal procedure of the Indoor Fan Motor.) (Fig. 9)

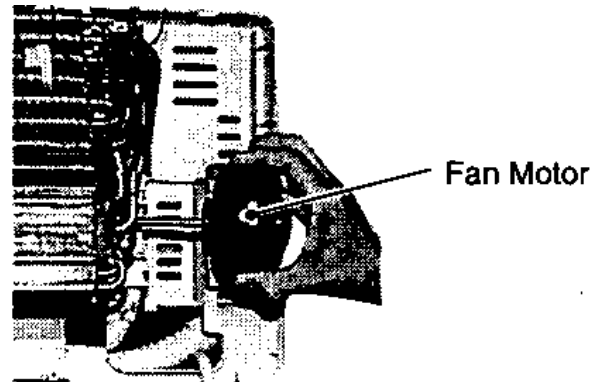


Fig. 9

2. Remove the Air Discharge Grille by taking off the screws that hold the Air Discharge Grille and then pull the Air Discharge Grille in a down and forward direction. (Fig. 10)

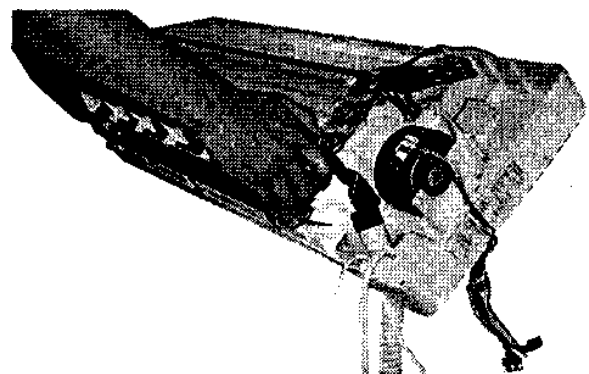


Fig. 10

3. Pull off the Bearing at the left of the Cross Flow Fan. (Fig. 11)

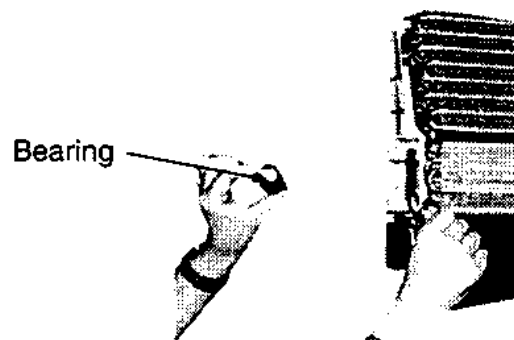


Fig. 11

4. Take off the mounting tab on the left side of the Heat Exchanger, pull the Heat Exchanger forward (left side) and remove the Cross Flow Fan. (Fig. 12)

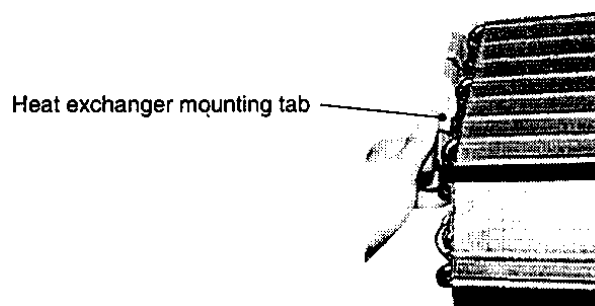


Fig. 12

12.4.4. Removal procedure of Display Electronic Controller

Release the Display Electronic Controller from the tab as shown in Fig. 13.

Move the Display Electronic Controller to the right and pull it towards you.

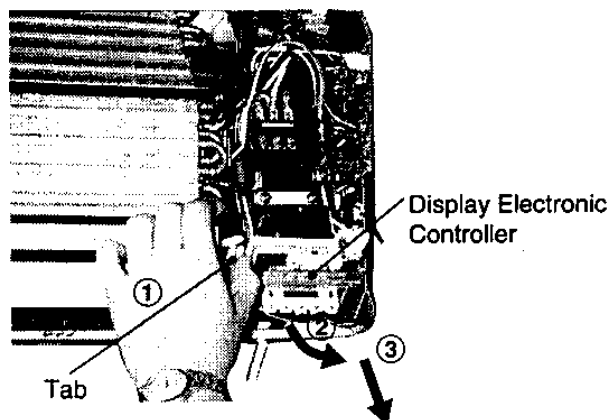


Fig. 13

12.4.5. Removal procedure of Main and Power Electronic Controller

1. Release all connectors which are connected to the electronic controller.
2. Pull the electronic controller towards you as shown in Fig. 14.

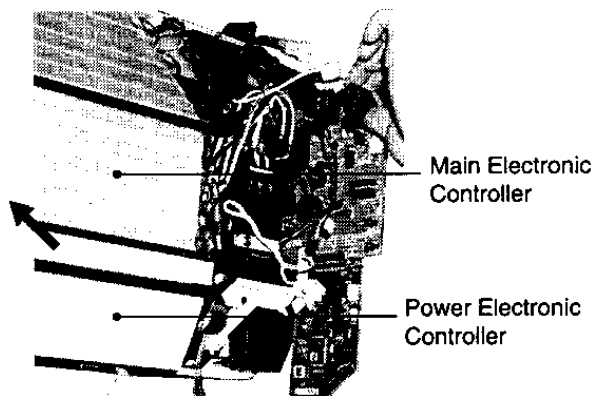


Fig. 14

3. Remove the connector as shown in Fig. 15.

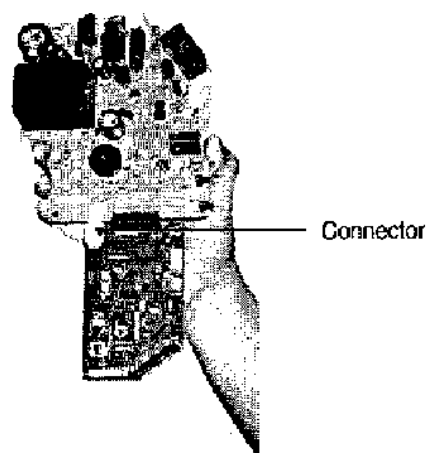


Fig. 15

4. Bend the 2 electronic controller in 'L' shape as shown in Fig. 16 and release the main electronic controller from the power electronic PCB.

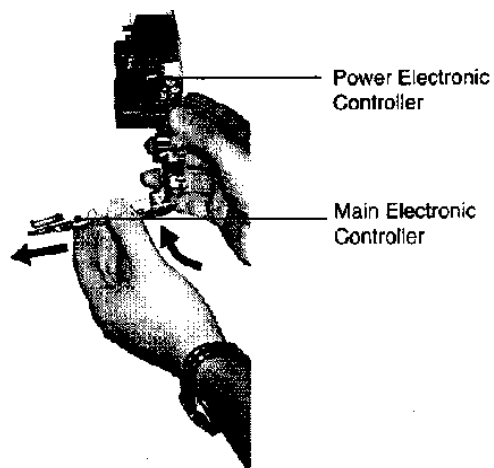


Fig. 16

12.4.6. Inspection Method for the Outdoor Unit

1. Remove the control box. (Two screws)
2. Remove the particular plate.
3. Remove the control box cover.
4. Remove the front panel. (Four screws) (Fig.17)

WARNING:

- Be sure to return the wiring to its original position.
- There are many high voltage components within the heat sink cover so never touch the interior during operation. Wait at least two minutes after power has been turned OFF.

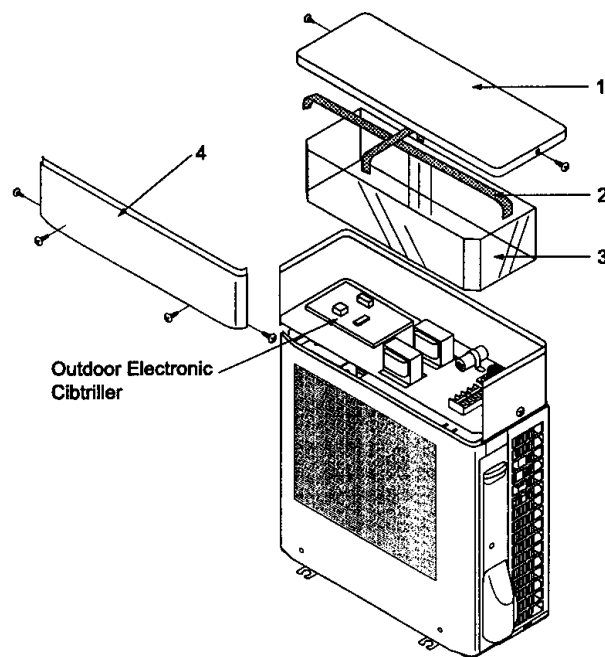


Fig. 17

5. Remove the control box. (Two screws)
6. Remove the control box cover.
7. Remove the Power transistor module. (Two screws) (Fig.18)

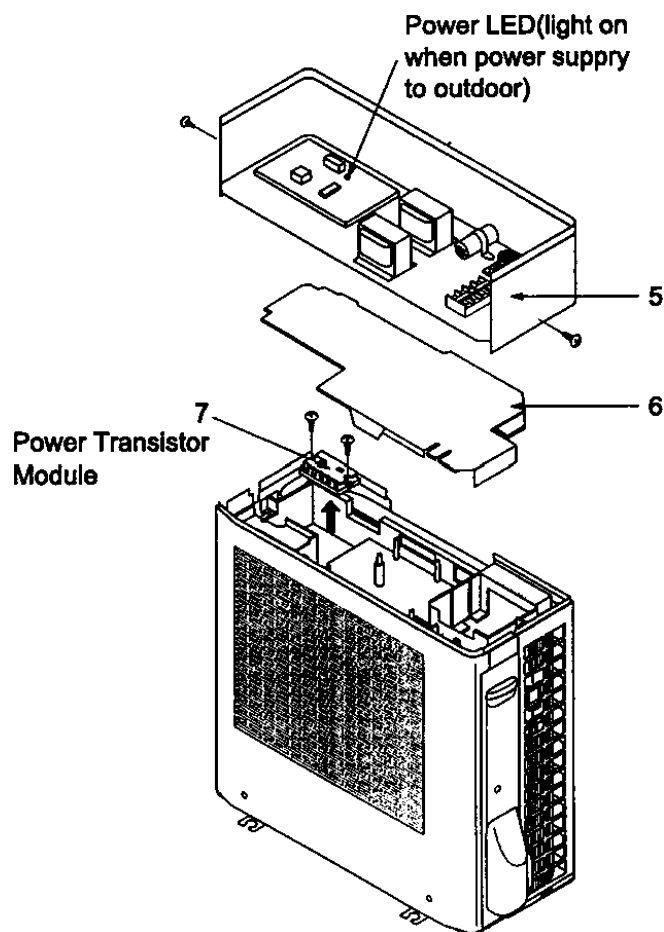


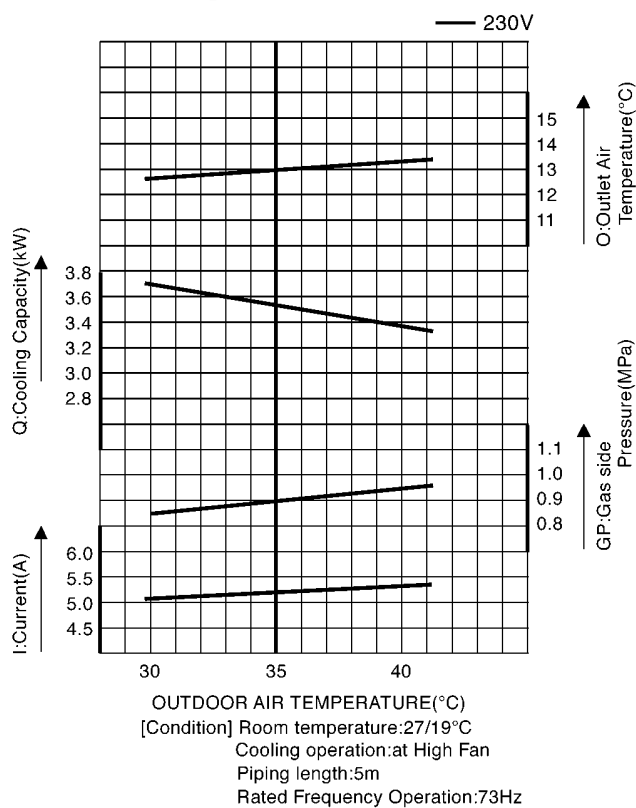
Fig. 18

13 Technical Data

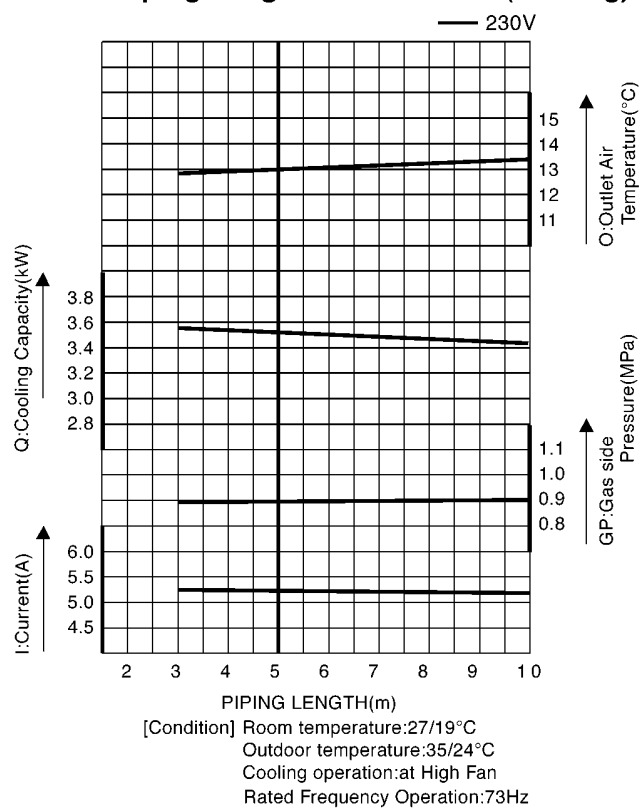
13.1. OPERATION CHARACTERISTICS

CS-VG120KE / CU-VG120KE

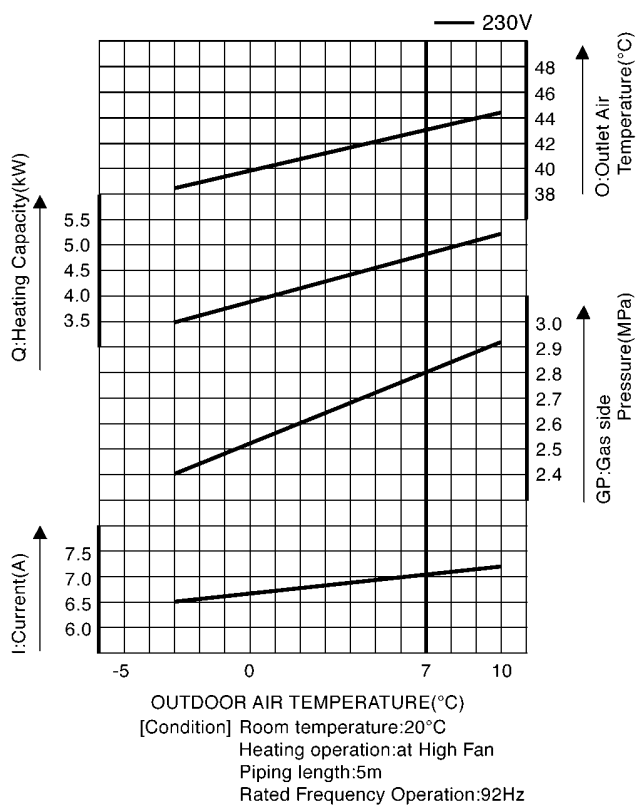
• Cooling Characteristic



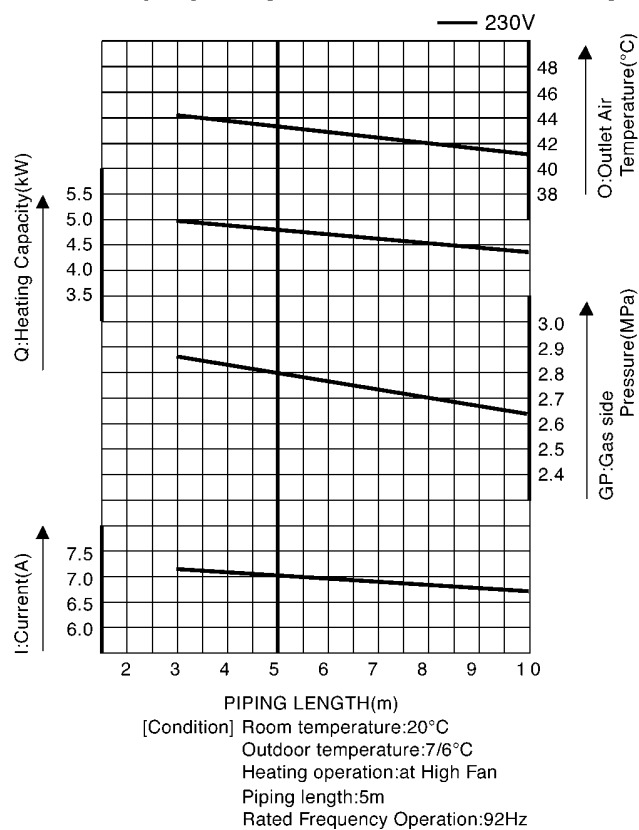
• Piping Length Characteristic(Cooling)



• Heating Characteristic



• Piping Length Characteristic(Heating)



14 Electronic Circuit Diagram

14.1. HOW TO USE ELECTRONIC CIRCUIT DIAGRAM

Before the circuit diagram, read the following carefully.

Voltage measurement

voltage has been measured with a digital tester when the indoor fan is set at high fan speed under the following conditions without setting the timer.

Use them for servicing.

Voltage indication is in Red at cooling and all operations.

Voltage indication is in (Red) at heating.

	Intake air temperature	Temperature setting	Discharge air temperature	Pipe temperature
Cooling	27 °C	16 °C	17 °C	15 °C
Heating	20 °C	30 °C	40 °C	50 °C

Indications for capacitor

a. Unit μ μ F P pF

Indications for resistance

a. K $K\Omega$ M $M\Omega$
W watt Not indicated 1/4 W

Diode without indication MA165

※ Circuit Diagram is subject to change without notice for further development.

14.2. TIMER TABLE

14.2.1. Indoor

Name	Time	Test Mode (When test point Short-circulated)
4 way valve abnormality	4 min.	2.4 sec.
Outdoor air temp. for Hz No. detection	30 min.	0
Anti-dew formation control	20 min.	0
Anti-freezing Control	6 min.	0
Theorem OFF delay	3 min.	0
Low pressure control (gas leakage) compressor OFF time	3 min.	0
Time delay safety control	2 min. 58 sec.	0
Odor timer status shift time	10 sec.	0
	90 sec.	
	20 sec.	
Intake air temp. sampling time	2 min.	0
Hz lock time	30 sec.	0
Self diagnosis display time	10 sec.	0
Auto mode judgement sampling time	20 sec.	0
24 hours ON / OFF timer		1 hour → 1 min.
Heating SSHi fan shift	120 min.	12 sec.
Cooling SHi fan speed shift	30 min.	3 sec.
Hot start forced completion	4 min.	0
Anti-fog discharge control	30 min.	3 sec.
	90 min.	9sec.
	180 min.	18 sec.
	420 min.	42 sec.
Auto mode judgement interval	30 min.	3 sec.
Sleep mode time shift		1 hour → 6 sec.
After Hot start / Deice	2 min.	12 sec.

14.2.2. Outdoor

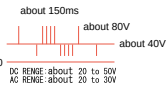
Name	Time	Test Mode (When test point Short-circulated)
DC PEAK	30 sec.	3 sec.
Deice detection	120 min.	24 sec.
	80 min.	18 sec.
	40 min.	8 sec.
	40 min.	8 sec.
Deice forced completion	11 min.	66 sec.
Time delay safety control	3 min.	0
Hz lock time	30 sec.	0
Outdoor fan delay operation control	30 sec.	0
4 way valve delay operation control	3 min.	0

14.3. CS-VG120KE (INDOOR UNIT)

CS-VG120KE
CU-VG120KE

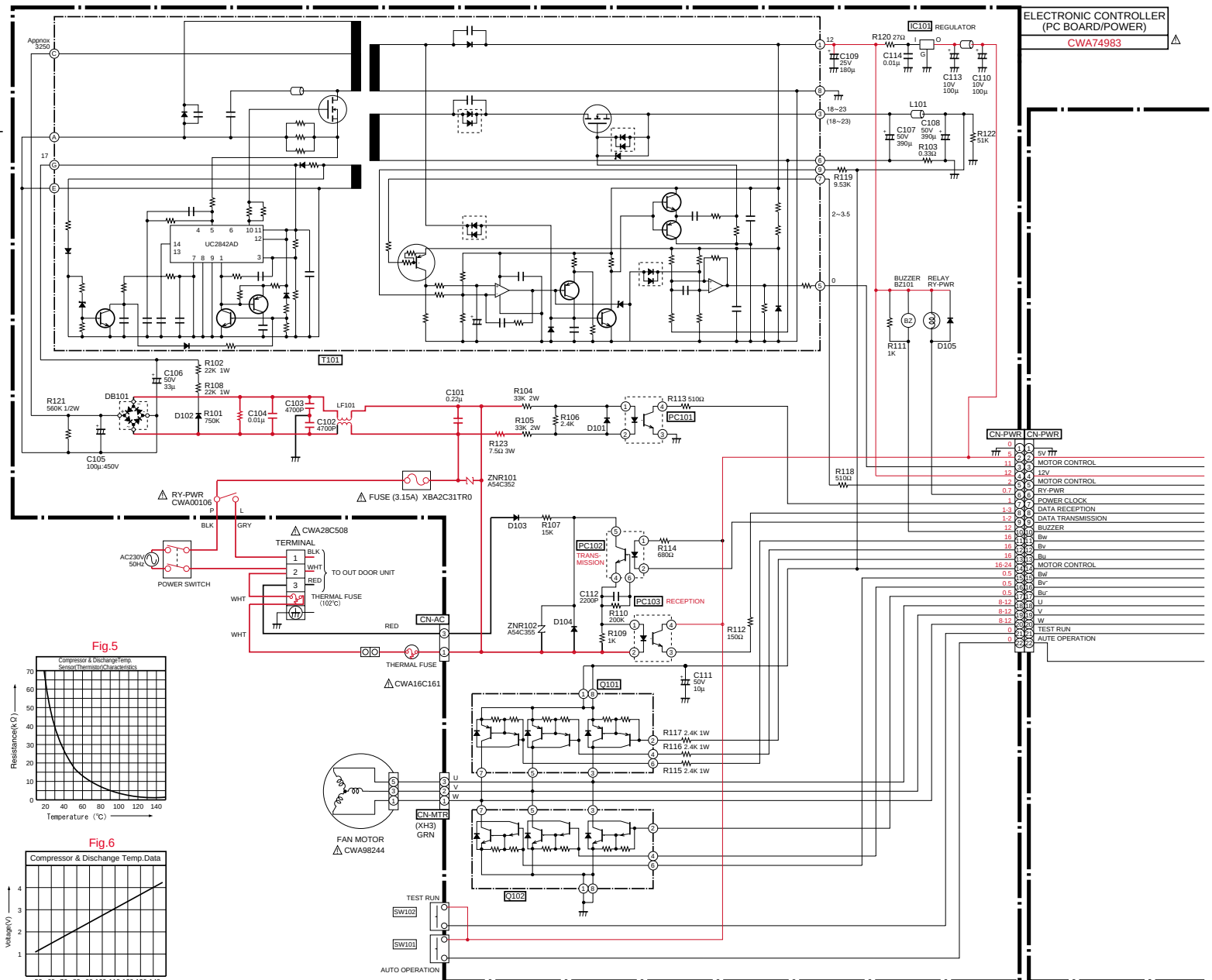
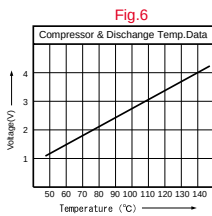
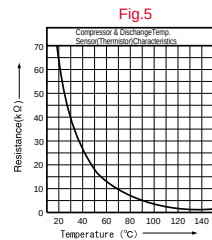
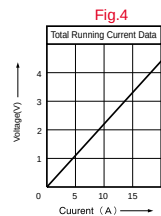
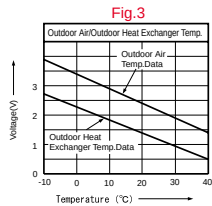
INDOOR UNIT

TERMINAL 2-3 VOLTAGE



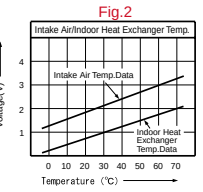
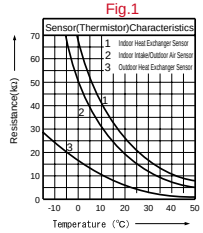
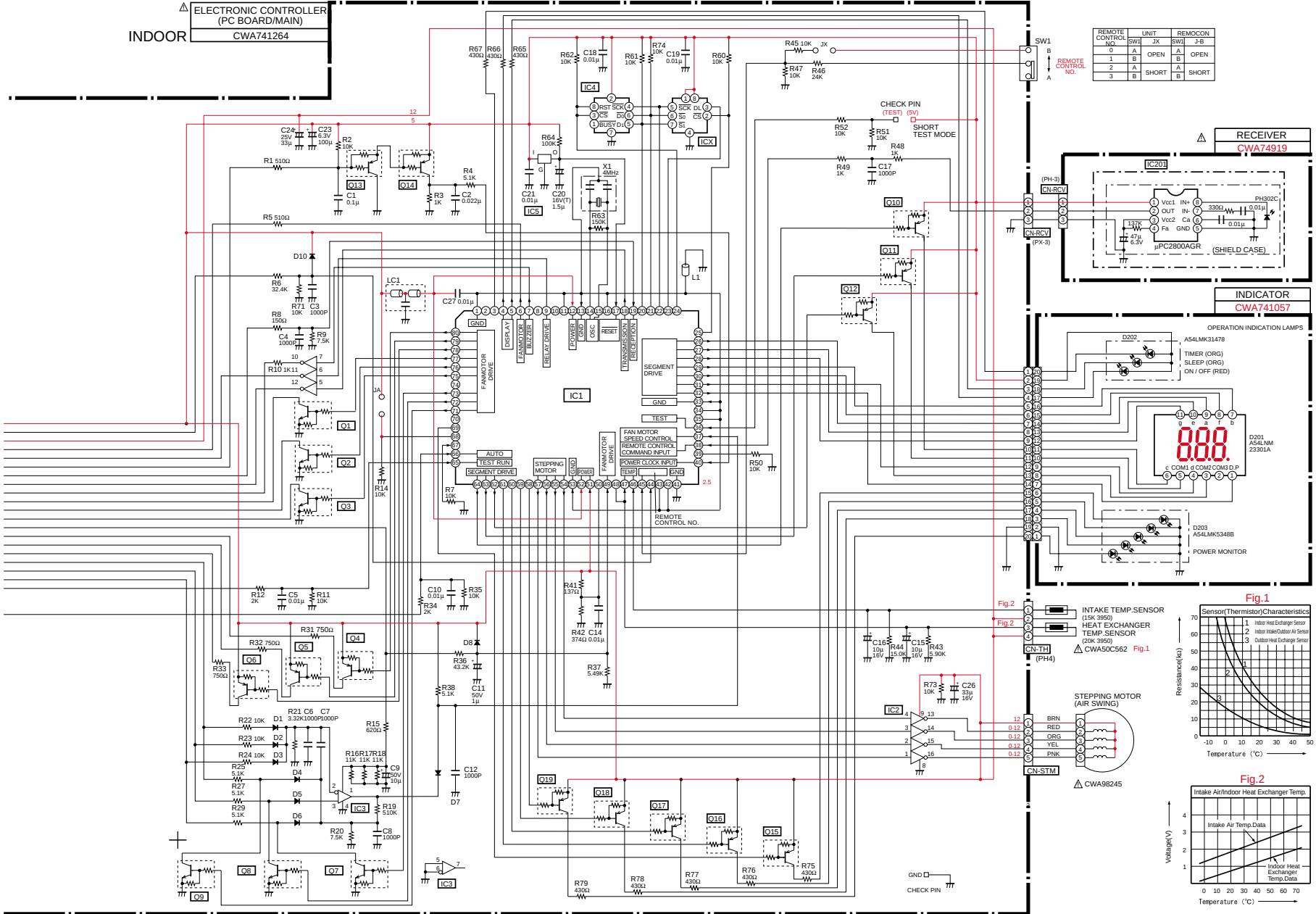
FAN MOTOR

Fan level	UV VV	Between AC VOLTAGE	U-GND V-GND W-GND DC VOLTAGE
15	18V	12V	11
14	14	10	10
13	13	9	9
12	12	8	8
STOP	0	0	0

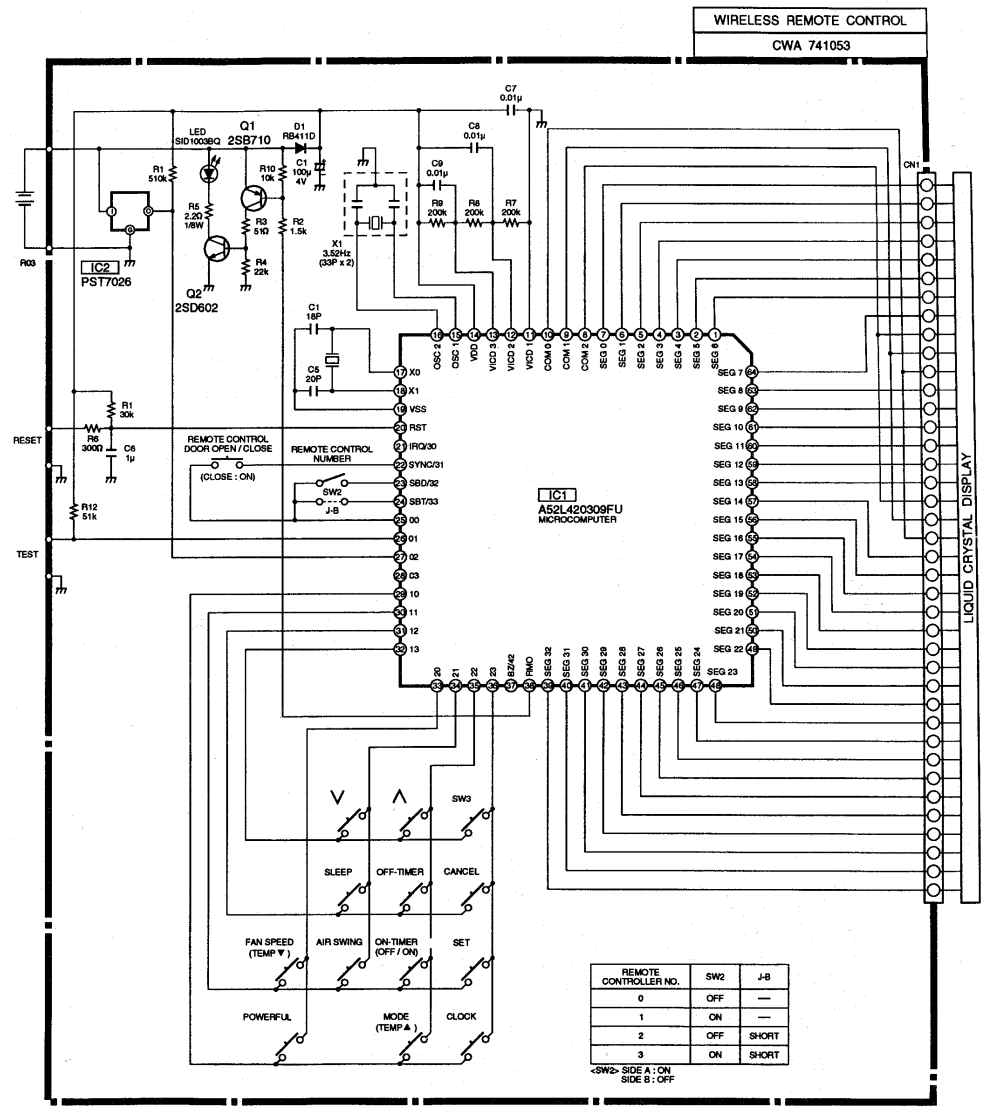


INDOOR
ELECTRONIC CONTROLLER
(PC BOARD/MAIN)
CWA741264

REMOTE CONTROL NO.	UNIT SW1	JX	SW2	REMOCON J-B
0	A	OPEN	A	OPEN
1	B	OPEN	B	OPEN
2	A	A	A	SHORT
3	B	SHORT	B	SHORT



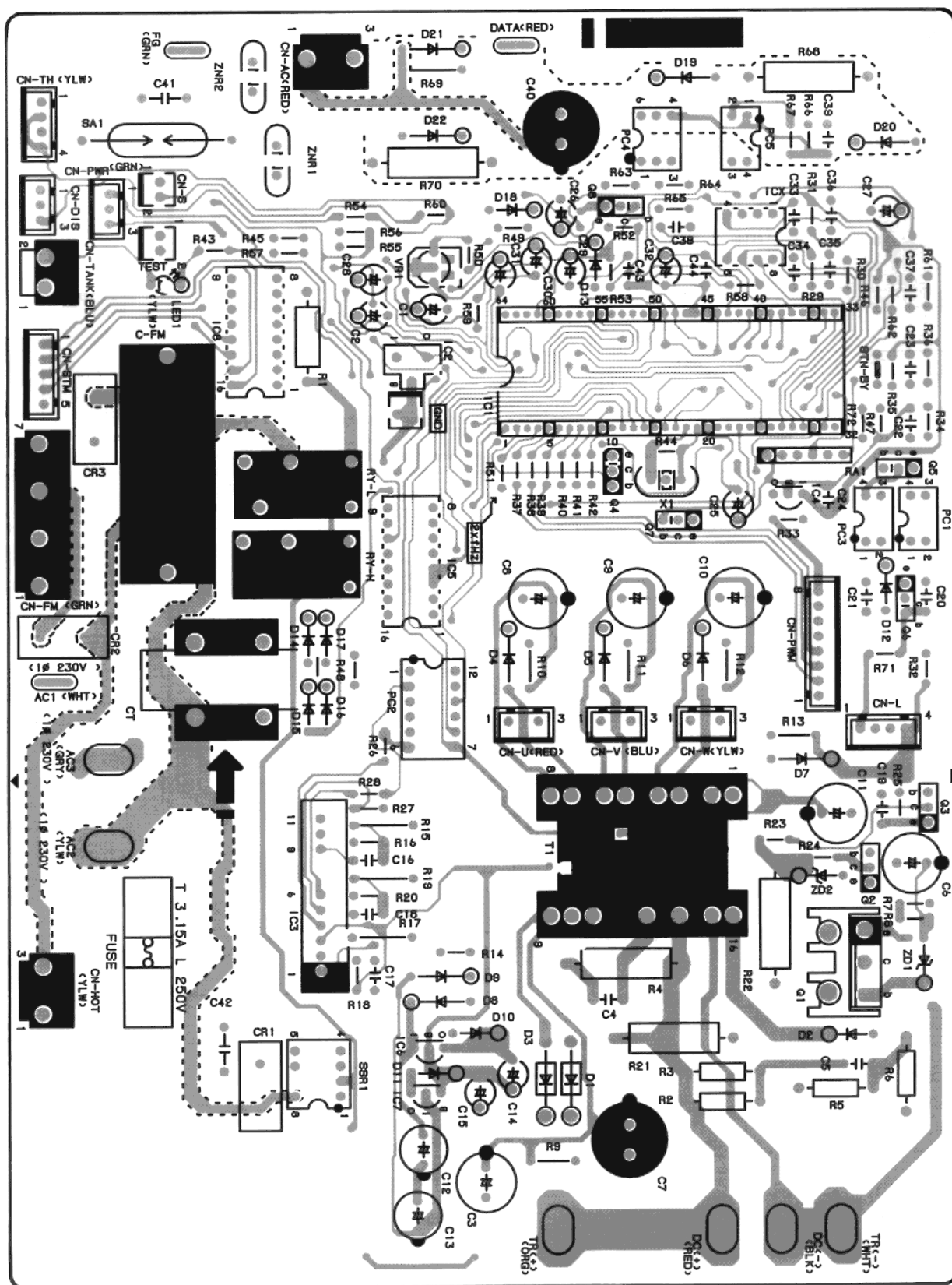
● REMOTE CONTROLLER



15 Printed Circuit Board

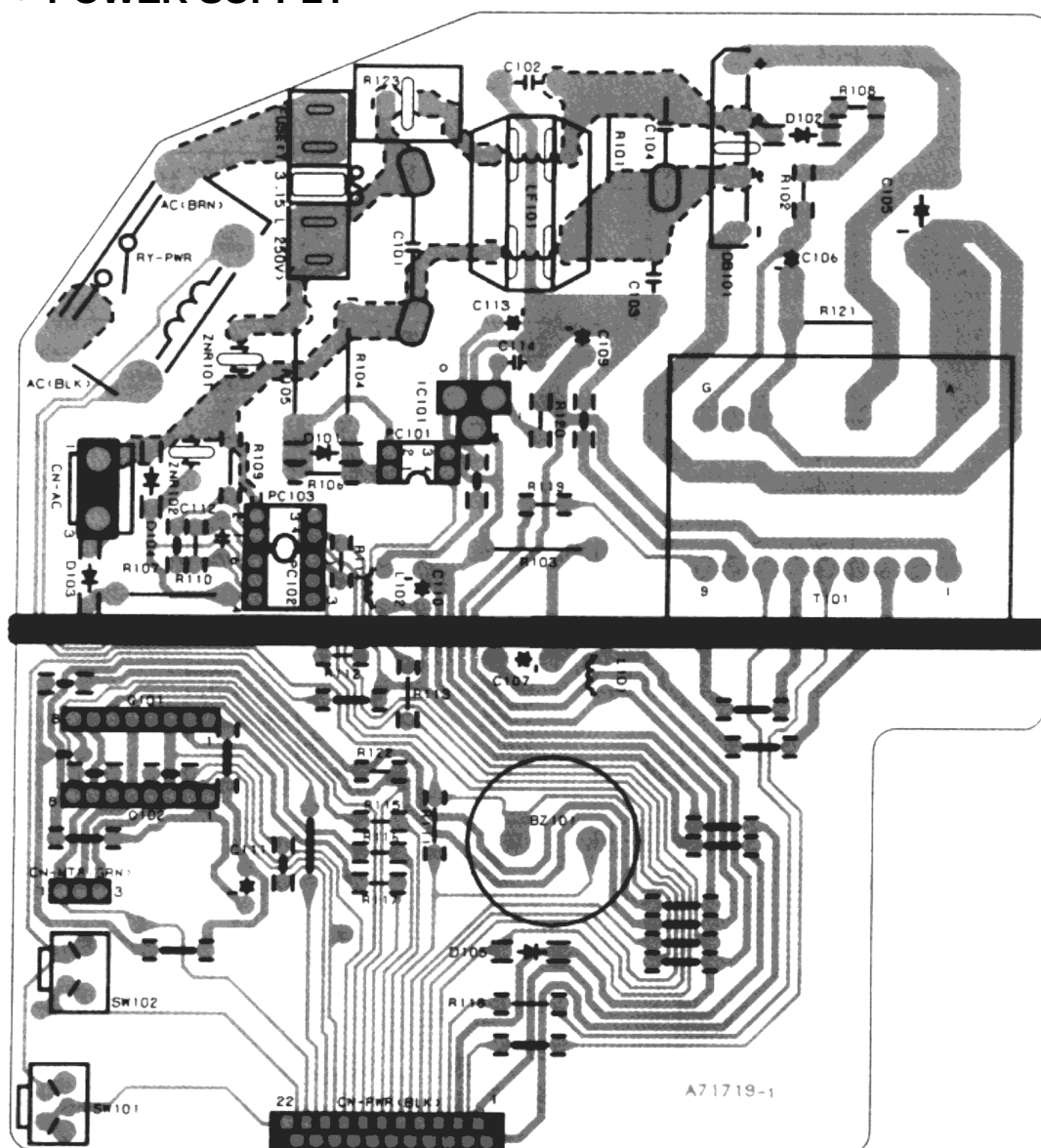
15.1. CS-VG120KE (INDOOR UNIT)

(PARTS SIDE)

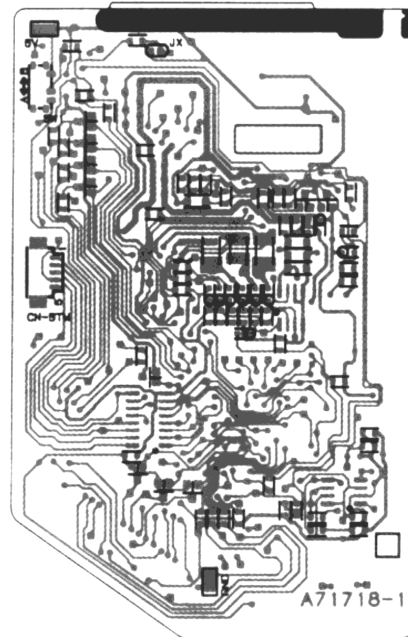


15.2. CU-VG120KE (OUTDOOR UNIT)

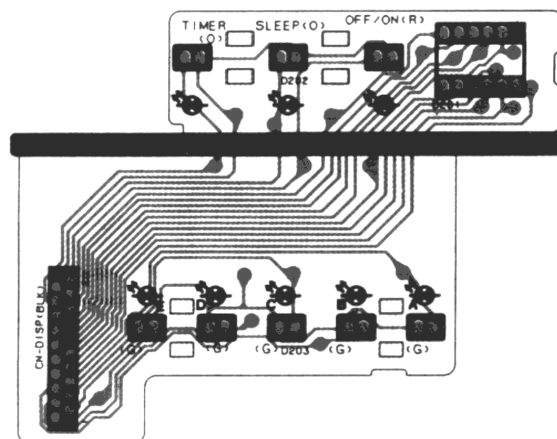
● POWER SUPPLY



● MAIN



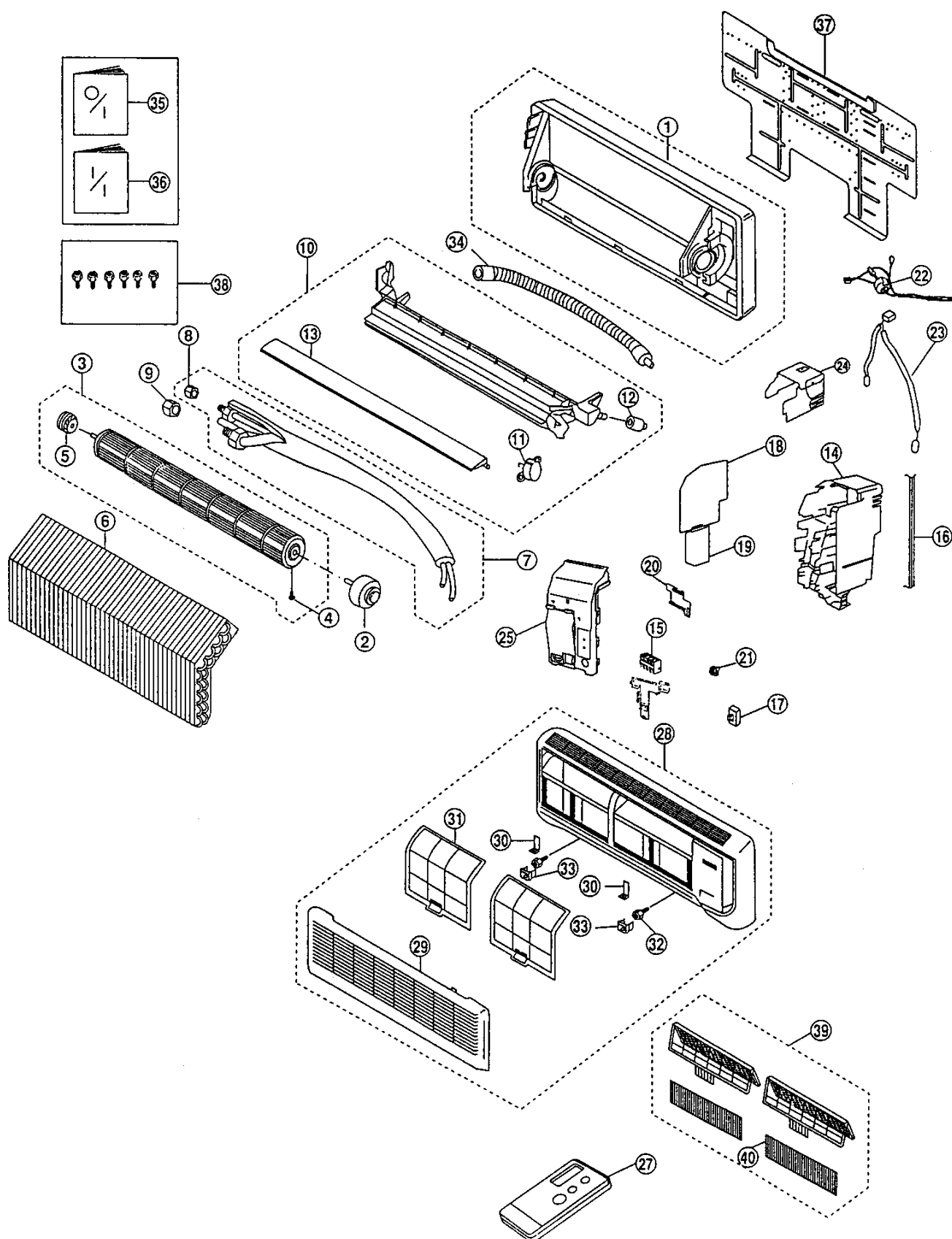
● INDICATOR



16 Exploded View & Replacement Parts List

16.1. CS-VG120KE

16.1.1. Exploded View



16.1.2. Replacement Parts List

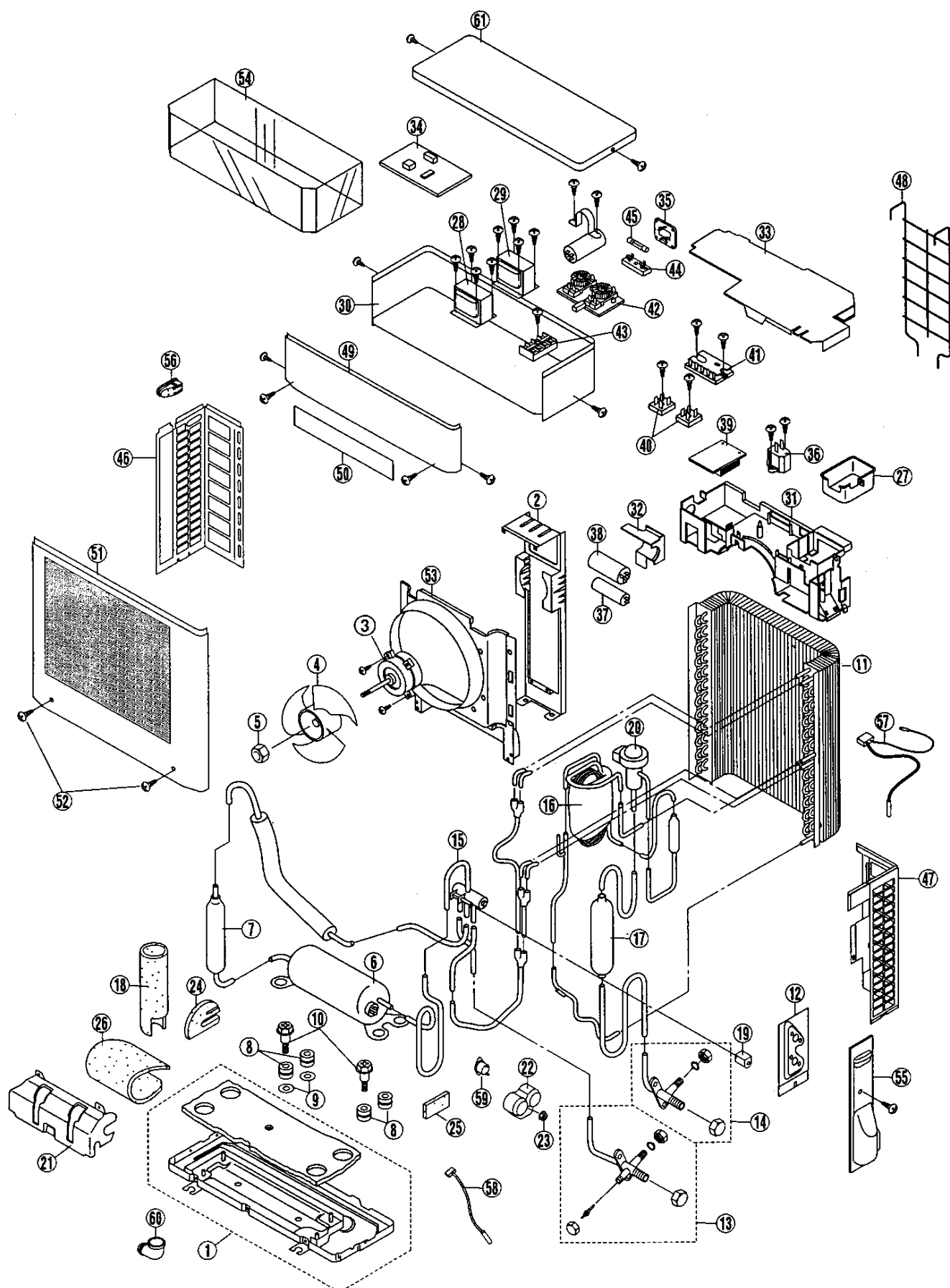
No.	CS-VG120KE	DESCRIPTION & NAME	Q'TY	Remarks	
1	CWD50C202	CHASSY COMPLETE	1	M	
2	CWA98244	FAN MOTOR	1	M	●
3	CWH02C053	CROSS-FLOW FAN COMPLETE	1	M	
4	CWH4580304	SCREW-CROSSFLOW FAN	1	M	
5	CWH64K007	BEARING ASS'Y	1	M	
6	CWB30C211	EVAPORATOR	1	J	
7	CWT01C2103	TUBE ASS'Y COMPLETE	1	J	
8	CWT25086	FLARE NUT (1/4")	1	J	●
9	CWT25096	FLARE NUT (1/2")	1	J	●
10	CWE20C534	DISCHARGE GRILLE COMPLETE	1	M	
11	CWA98245	MOTOR-AIR SWING	1	M	●
12	CWH52C003	CAP-DRAIN TRAY	1	M	
13	CWE24394	VANE	1	M	
14	CWH10908	CONTROL BOARD	1	M	
15	CWA28C508	TERMINAL BOARD COMPLETE	1	M	●
16	CWA20C678	POWER SUPPLY CORD	1	M	
17	CWA04088	SLIDE SWITCH	1	M	●
18	CWA742140	PC BOARD WITH COMPONENT (POWER)	1	J	●
19	CWA742133	PC BOARD WITH COMPONENT (MAIN)	1	J	●
20	CWA741057	PC BOARD WITH COMPONENT (INDICATOR)	1	M	●
21	CWA74919	PC BOARD WITH COMPONENT (RECEIVER)	1	M	●
22	CWA16C161	FUSE COMPLETE	1	M	●
23	CWA50C562	SENSOR COMPLETE	1	M	●
24	CWH13406	CONTROL BOARD TOP COVER	1	M	
25	CWH13C284	CONTROL BOARD FRONT COVER	1	M	
27	CWA75C614	REMOTE CONTROL COMPLETE	1	M	●
28	CWE11C880	FRONT GRILLE COMPLETE	1	J	
29	CWE22C287	INTAKE GRILLE COMPLETE	1	M	
30	CWD93C070	PARTICULAR PIECE	2	M	
31	CWD00215	AIR FILTER	2	M	
32	XTN4+16C	SCREW-FRONT GRILLE	2	M	
33	CWH52230	CAP FRONT GRILLE	2	M	
34	CWH5880580	DRAIN HOSE	1	M	
35	CWF563064	OPERATING INSTRUCTIONS	1	J	
36	CWF612056	INSTALLATION INSTRUCTIONS	1	J	
37	CWH36122	INSTLLATION PLATE	1	M	
38	CWH82C194	BAG COMPLETE-INSTALLATION SCREW	1	M	
39	CWD00C111	AIR PURIFYING FILTER COMPLETE	1	M	
40	CWD00220	AIR PURIFYING FILTER	2	M	●

(Note)

- Spare parts are supplied from ;
"M" is from MACC, Malaysia (Vender code : 086).
"J" is from Japan.
- "●"marked parts are recommended to be kept in stock.

16.2. CU-VG120KE

16.2.1. Exploded View



16.2.2. Replacement Parts List

Ref. No.	CU-VG120KE	DESCRIPTION & NAME	Q'TY	Remarks	
1	CWD50K620AX	CHASSY COMPLETE	1	J	
2	CWD54189	FAN MOTOR BRACKET	1	J	
3	CWA92269	MOTOR	1	J	●
4	CWH03K021X	PROPELLER FAN	1	J	
5	CWH56053	NUT	1	M	
6	CWB092028	COMPRESSOR	1	J	●
7	CWB13038	ACCUMLATOR	1	J	
8	CWH50130	ANTI-VIBRATION BUSHING	4	J	
9	CWB81036	PACKING	2	J	
10	CWH55370	BOLT	2	J	
11	CWB32C278	CONDENSER	1	J	
12	CWH35131A	HOLDER-COUPPLING	1	J	
13	CWB011010	3-WAY VALVE	1	J	●
14	CWB02293	2-WAY VALVE	1	J	●
15	CWB00048	4-WAY VALVE	1	J	●
16	CWT01C2081	TUBE ASS'Y	1	J	●
17	CWB10036	DRYER	1	J	●
18	CWG30715	SOUND PROOF MATERIAL	1	J	
19	CWA43C657	4-WAY VALVE COIL	1	J	●
20	CWA43C656	2-WAY VALVE COIL	1	J	●
21	CWH15C086	SOUND-PROOF BOARD	1	J	
22	CWH17013	TERMINAL COVER	1	J	
23	CWH7080300	NUT	1	M	
24	CWG30606	SOUND PROOF MATERIAL	1	J	
25	CWG30687	SOUND PROOF MATERIAL	1	J	
26	CWG30714	SOUND PROOF MATERIAL	1	J	
27	CWD76196	BOX SHAPED PIECE	1	J	
28	CWA42134	REACTOR	1	M	
29	CWA42135	REACTOR	1	M	
30	CWH10K414A	CONTROL BOARD	1	J	
31	CWH10872	CONTROL BOARD	1	J	
32	CWD66242	BOX SHAPED PLATE	1	J	
33	CWH13431	CONTROL BOARD COVER	1	J	
34	CWA742134	PC BOARD WITH COMPONENT	1	J	●
35	CWA74184	PC BOARD WITH COMPONENT	1	J	●
36	CWA47038	RESISTOR FOR ELECTRONIC	1	J	●
37	CWA31137	SH CAPACITOR	1	M	●
38	CWA30199	CAPACITOR	1	M	●
39	CWA58180	HEATSINK	1	J	
40	A54S25VB60	DIODE	2	M	●
41	CWA55C127	TRANSISTOR	1	J	●
42	CWA49C196	NOISE FILTER	1	M	
43	CWA28K224	TERMINAL BOARD COMPLETE	1	M	
45	XBACW066	FUSE	1	M	●
46	CWE04125AX	CABINET SIDE PLATE	1	J	
47	CWE04126AX	CABINET SIDE PLATE	1	J	
48	CWD04120A	WIRE NET	1	J	
49	CWE06C116A	CABINET FRONT PLATE	1	J	
50	CWE37652	BUDGE	1	J	
51	CWE11C877	FRONT GRILLE	1	J	
52	CWH55250	SCREW-FRONT GRILLE	2	J	
53	CWD31125	AIR GUIDER P.F.	1	J	
54	CWH13432	CONTROL BOARD COVER	1	J	
55	CWH13C254	CONTROL BOARD COVER	1	J	
56	CWE16037C	HANDLE	1	M	
57	CWA50C580	THERMISTOR	1	J	●
58	CWA50C581	THERMISTOR	1	J	●
59	CWA67C1752	LEAD WIRE COMPLETE	1	J	●
60	CWH5850080	DRAIN ELBOW	1	M	
61	CWE03098A	CABINET TOP PLATE	1	J	

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17 Electronic Parts List

(Model:CWA742140) POWER PCB

SYMBOL	PART NO.	DESCRIPTION & NAME	REMARKS
BZ101	A48004	BUZZER	M
D101	A54MA165T5	DIODE	M
D102	A54RA15-10V3	DIODE	M
D103,D104	A54RA15-06V3	DIODE	M
D105	A54RA15-01V3	DIODE	M
DB101	A54D3SBA60F1	DIODE	M
FUSE	XBA2C31TRO	FUSE	M
IC101	A52C095	INTEGRATED CIRCUIT	M
PC101,PC103	A52LP621-1B4	PHOTO COUPLER	M
PC102	A52PS2633E	PHOTO COUPLER	M
Q101	A523AA13	TRANSISTOR	M
Q102	A52STA03A	TRANSISTOR	M
RY-PWR	A00106	RELEY	M
SW101,SW102	A01059	SWITCH	M
T101	A40328	TRANSFORMER	M
ZNR101	A54C352	ZNR	M
ZNR102	A54C355	ZNR	M

(Model:CWA742133) MAIN PCB

SYMBOL	PART NO.	DESCRIPTION & NAME	REMARKS
D1 ~ D8,D10	A541SS355T	DIODE	M
IC1	A5278366F027	INTEGRATED CIRCUIT	J
IC2	A52A2003GR2	INTEGRATED CIRCUIT	M
IC3	A52MPC393G22	INTEGRATED CIRCUIT	M
IC4	A52XL9011BF2	INTEGRATED CIRCUIT	M
IC5	A52PST600DR	INTEGRATED CIRCUIT	M
CIX	A53803A	INTEGRATED CIRCUIT	J
Q1 ~ Q3,Q7 ~ Q9	A55DC114EKT	TRANSISTOR	M
Q4 ~ Q6,Q10 ~ Q12,Q13 ~ Q19	A55DA143XKTX	TRANSISTOR	M
Q13	A55DC143XKTX	TRANSISTOR	M
SW1	A04085	SWITCH	M
X1	A45CTS4MG02T	RESONATOR	M

(Model:CWA742134) OUTDOOR PCB

SYMBOL	PART NO.	DESCRIPTION & NAME	REMARKS
CT	A40178	CURRENT TRANSFORMER	M
C-FM	A31719	CAPACITOR (1.8μF/370V)	J
D1,D3	A54RA44-02V	DIODE	M
D10 ~ D18	A54MA165TA5	DIODE	M
D2	A541SS136T	DIODE	M
D4 ~ D9	A54RB22-04V3	DIODE	J
D19 ~ D21	A54RA15-06V3	DIODE	M
D22	A54RA15-10V3	DIODE	M
IC1	A5278362CW25	INTEGRATED CIRCUIT	J
IC2	A52C050	INTEGRATED CIRCUIT	M
IC3	A52BX8594	INTEGRATED CIRCUIT	J
IC4	A52M51951AST	INTEGRATED CIRCUIT	M
IC5,IC8	A52MPA2003C	INTEGRATED CIRCUIT	M
IC6	A52AN78L10TA	INTEGRATED CIRCUIT	J
IC7	A52AN79L10TA	INTEGRATED CIRCUIT	J
ICX	A53804A	INTEGRATED CIRCUIT	J
LED1	A5448YPXTA3E	LED	J
FUSE	XBA2C31TRO	FUSE	M
PC1,PC3	A52LP621-1G4	PHOTO COUPLER	M
PC2	A52LP621-3G4	PHOTO COUPLER	J
PC4	A52PS2633E	PHOTO COUPLER	M
PC5	A52LP621-1B4	PHOTO COUPLER	M
Q1	A55C075	TRANSISTOR	J
Q2,Q3,Q5	A55C1740STPQ	TRANSISTOR	M
Q4	A55C17402TPQ	TRANSISTOR	M
Q6	A55UN411FTA	TRANSISTOR	M
Q7	A55UN4116QTA	TRANSISTOR	J
Q8	A55UN421FTA	TRANSISTOR	M
RY-H	A00161	RELEY	M
RY-L	A00160	RELEY	M
T1	A40350	TRANSFORMER	J
VR1	A44EVMEASB22	VARIABLE RESISTOR	M
X1	A45ST4MGW04T	RESONATOR	J
ZD1	A54D5.6EB3TB	TENAR DIODE	J
ZD2	A54D20EL4TB	TENAR DIODE	M

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MEMO :

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